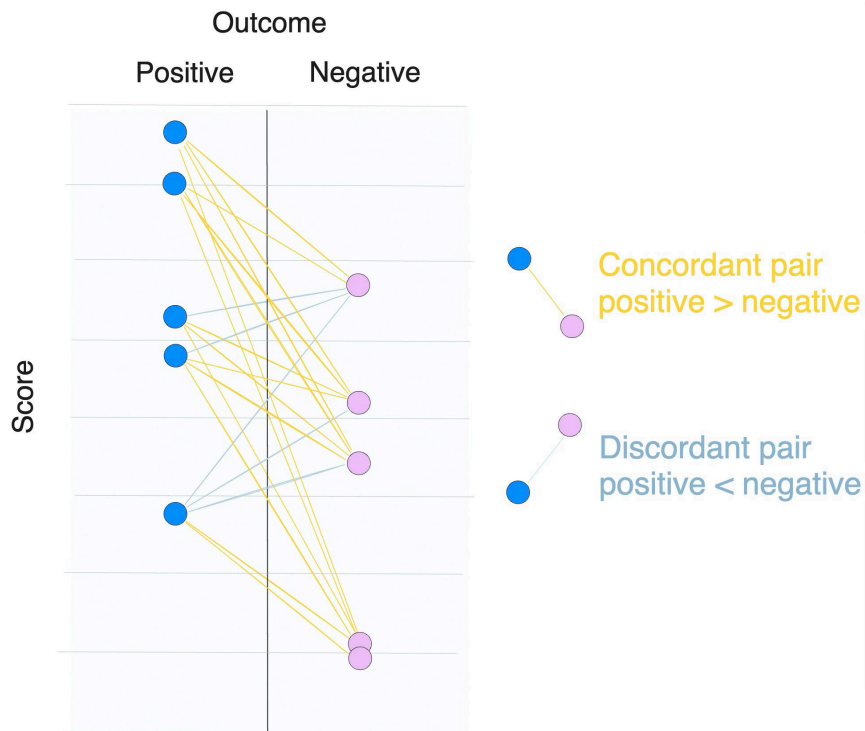




Metrics for Credit and ML Models

A practitioner binder: discrimination, calibration, equity, and trust



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Beyond accuracy: Measures for assessing machine learning models, pitfalls and guidelines

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Abstract

Pattern recognition predictive models have become an important tool for analysis of neuroimaging data and answering important questions from clinical and cognitive neuroscience. Regardless of the application, the most commonly used method to quantify model performance is to calculate prediction accuracy, i.e. the proportion of correctly classified samples. While simple and intuitive, other performance measures are often more appropriate with respect to many common goals of neuroimaging pattern recognition studies. In this paper, we will review alternative performance measures and focus on their interpretation and practical aspects of model evaluation. Specifically, we will focus on 4 families of performance measures: 1) categorical performance measures such as accuracy, 2) rank based performance measures such as the area under the curve, 3) probabilistic performance measures based on quadratic error such as Brier score, and 4) probabilistic performance measures based on information criteria such as logarithmic score. We will examine their statistical properties in various settings using simulated data and real neuroimaging data derived from public datasets. Results showed that accuracy had the worst performance with respect to statistical power, detecting model improvement, selecting informative features and reliability of results. Therefore in most cases, it should not be used to make statistical inference about model performance. Accuracy should also be avoided for evaluating utility of clinical models, because it does not take into account clinically relevant information, such as relative cost of false-positive and false-negative misclassification or calibration of probabilistic predictions. We recommend alternative evaluation criteria with respect to the goals of a specific machine learning model.

Introduction

Machine learning predictive models have become an integral method for many areas of clinical and cognitive neuroscience, including classification of patients with brain disorders from healthy controls, treatment response prediction or, in a cognitive neuroscience setting, identifying brain areas containing information about experimental conditions. They allow making potentially clinically important predictions and testing hypotheses about brain function that would not be possible using traditional mass univariate methods (i.e., effects distributed across multiple variables). Regardless of the application, it is important to evaluate the quality of predictions on new, previously unseen data.

A common method to estimate the quality of model predictions is to use cross-validation and calculate the average prediction performance across test samples (Varoquaux et al., 2017) or balanced variants that account for different class frequencies (Brodersen et al., 2010). Selection of appropriate performance measures is a widely studied topic in other areas of science, such as weather forecasting (Mason, 2008), medicine (Steyerberg et al., 2010) or finance (Hamerle et al., 2003). However, in the neuroimaging field, this has received surprisingly little attention. For example, many introductory review and tutorial articles focused on machine learning in

neuroimaging (Haynes and Rees, 2006; Pereira et al., 2009; Varoquaux et al., 2017) do not discuss performance measures other than accuracy (i.e., simple proportion of correctly classified samples). Accuracy is also the most frequently reported performance measure in neuroimaging studies employing machine learning, even though in many cases alternative performance measures may be more appropriate for the specific problem at hand. However, no performance measure is perfect and suitable for all situations and different performance measures capture different aspects of model predictions. Thus, a thoughtful choice needs to be made in order to evaluate the model predictions based on what is important in each specific situation. Different performance measures also have different statistical properties which need to be taken into account. For example, how reliable or reproducible are the results, or what is the power of detecting a statistically significant effect?

In this paper, we provide a didactic overview of various performance measures focusing on their interpretation and practical aspects of their usage. We divide the measures into four families based on what aspects of a model prediction they evaluate; (1) measures evaluating categorical predictions, (2) ranks of predictions, (3) probabilistic predictions with respect to a squared error, and (4) probabilistic predictions with respect to information criteria. This includes accuracy, the area under the receiver operating characteristic curve, the Brier score, and the logarithmic score respectively, as most prominent members of each family. Next, we perform an extensive empirical evaluation of statistical properties of these measures, focusing on power to detect a statistically significant effect, power to detect a model improvement, evaluation of stability of the feature selection process, and evaluation of the reliability of results. We show that accuracy performs the worst with respect to all examined statistical properties. Last, we discuss appropriate evaluation criteria with respect to various goals of the specific machine learning model.

Tutorial on performance measures

This section is a didactic overview of 4 main types of performance measures.

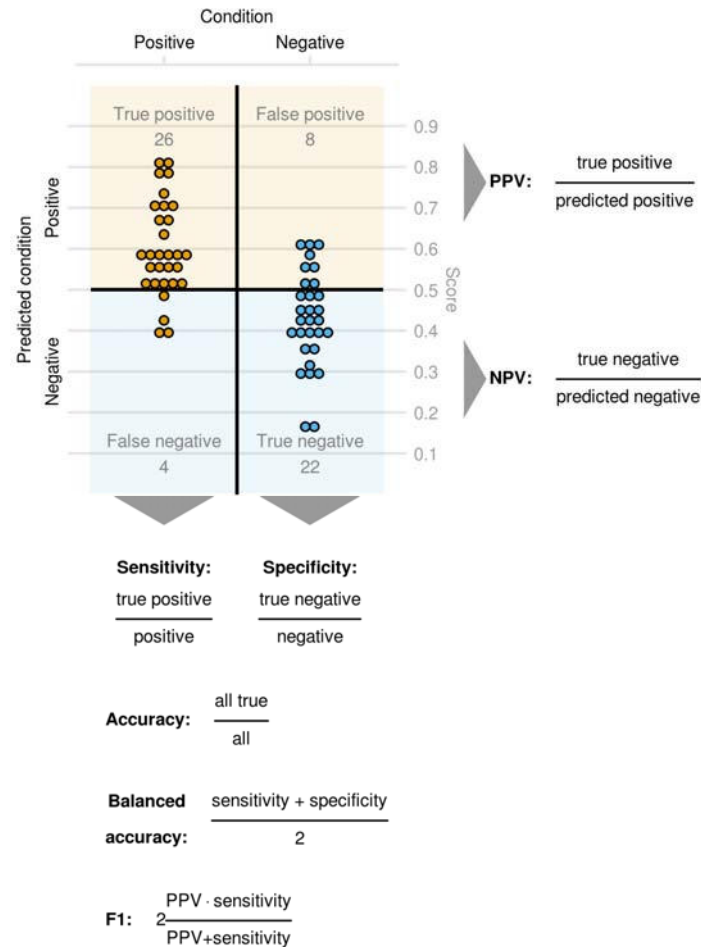


Figure 1: Definition of categorical predictions and their relationship to a confusion matrix. Note that the predicted score (e.g. predicted probability from a logistic regression, here depicted by the position of the predictions on the y-axis) needs to be dichotomized into categorical predictions, thus the magnitude of a miss-classification is not taken into account. Balanced accuracy equals accuracy when the class frequencies are equal, otherwise misclassification of the minority class is weighted higher, therefore, the chance level stays at 0.5. PPV: positive predictive value. NPV: Negative predictive value.

Categorical measures

The most commonly used performance measures are based on an evaluation of categorical predictions. These can be derived from a confusion matrix where predicted labels are displayed in rows and observed labels are displayed in columns (Figure 1). The most commonly used measure is accuracy, which is a simple proportion of all samples classified correctly. This can be misleading in imbalanced classes, since for example if the disease prevalence is 1% it is trivial to obtain an accuracy of 99% just by always reporting no disease. For this reason, balanced accuracy is often used instead (Brodersen et al., 2010). Balanced accuracy is simply the arithmetic mean of

sensitivity and specificity, thus it is equal to accuracy when the class frequencies are the same (Figure 1), otherwise, because sensitivity and specificity are contributing equally, it weighs false positive and false negative miss classifications according to class frequencies.

Sensitivity (true positive rate, recall) is the proportion of positive samples classified as positive, or in other words, it is the probability that a sample from a positive class will be classified as positive. Specificity (true negative rate) is the counterpart to sensitivity measuring the proportion of negative samples correctly classified as negative. Sensitivity and specificity do not take class frequencies (and class imbalances) or disease prevalence into account and they do not capture what is important in many practical applications. For example, in a clinical setting, we don't know if the patient has a disease or not (otherwise, there would be no need for testing). Instead, what is known are the results of the test (positive or negative) and we would like to know, given the result of the test, what is the probability that the patient will have a disease or not. This is measured by positive predictive value (PPV) and negative predictive value (NPV) for positive and negative test results, respectively. It is easy to misinterpret sensitivity as PPV, the difference is subtle but crucial. To illustrate, say we have the following confusion matrix:

		Condition	
		Positive	Negative
Predicted Condition	Positive	10	90
	Negative	0	900

This gives us 100% sensitivity ($10/10$), 91% specificity ($900/990$), 91% accuracy ($((10+900)/1000)$ and 96% balanced accuracy ($((10/10 + 900/990)/2)$). According to these measures, the model performs well. However, in this example the disease prevalence is 1% ($10/1000$), so only 10% of patients with a positive test result are truly positive, the remaining 90% are misclassified (i.e., cell with positive predicted condition but negative actual condition in confusion matrix). Therefore, the test may actually be useless in practice. Another measure used in cases with imbalanced classes is the F1 score, which is the harmonic mean of PPV and sensitivity (so if one of the elements is close to 0, the whole score will be close to 0) (Figure 1). Since the harmonic and not the arithmetic mean is used, both sensitivity and PPV need to be high in order for the score to be high. Thus, this score emphasizes the balance between sensitivity and PPV and it assumes that both are equally important.

All categorical measures suffer from two main problems: first, they depend on an arbitrarily selected classification threshold. Each data point counts either as a correct or incorrect classification, without taking the magnitude of the error into account. If the decision threshold is set to 50%, then if a model predicts that a disease probability in a healthy subject is 49% percent and thus classifies this subject as healthy, this is indistinguishable from a disease probability of 1% in another healthy subject (also classified as healthy). Imagine two models, one predicts that a healthy subjects has a disease with a probability of 99% and other with a probability of 51%. The latter model is obviously better since the error is smaller, but if these predictions are thresholded at traditional 50% percent, the accuracy will not be able to detect the improvement. Due to this insensitivity, compared to measures that do not require dichotomization of predictions, using accuracy necessarily leads to a significant loss of statistical power. Second, false positive and false negative misclassification are weighted as being equally bad or according to class frequencies (balanced accuracy), which is often inappropriate. Rather, misclassification costs are asymmetric and depend on consequences of such misclassification. For example, in a clinical context, misclassifying a healthy subject as diseased is worse when this misclassification will lead to an

unnecessary open brain surgery, than when it will lead to a prescription of medications with no side effects.

One way to combat unequal misclassification cost is to select the decision threshold according to cost-benefit analysis, such that the action is made when the expected benefit of the action outweighs the expected harm. This is then evaluated by cost-weighted accuracy, where true positives and false positives are weighted according to their relative cost. Although this allows to evaluate categorical predictions according to their utility and not arbitrarily, the dichotomization still creates problems for many practical applications. For example, in a clinical setting, the categorized predictions hide potentially useful clinical information from decision-makers (be it clinician, patient, an insurance company, etc.). If a prediction is only the presence versus the absence of a disease, a decision maker cannot take into account if a probability of the disease is 1%, 15% or 40%. This effectively moves the decision from stakeholders to a data analyst assuming that the chosen decision threshold is appropriate and constant for every situation and every patient, thus not allowing to take clinicians' opinions or patients' preferences into account.

Rank based measures

Another important family of performance measures are those based on ranks of predictions. Here, predictions are not categorical, but all predictions are ranked from lowest to highest with respect to the probability of an outcome.

The most common measure from this family is the area under the receiver operating characteristic curve (AUC). It measures a separation between two distributions, with a maximum score of 1 meaning perfect separation or no overlap between distributions and 0.5 being chance level when the distributions cannot be separated. In the case of model evaluation, the two distributions are distributions of predicted values (e.g. probability of a disease) for each target group. AUC is identical or closely related to multiple more or less known concepts, some of which we will review below.

The most common way of interpreting the AUC is through the receiver operating characteristic curve (ROC) (Figure 2A). This is a plot showing how a proportion of false positive and false negative misclassifications (i.e. sensitivity and specificity) changes as a function of the decision threshold. AUC is then the area under this curve. The curve itself is useful because it visualizes sensitivity and specificity across all thresholds not only for one threshold as in the case with the confusion matrix. Thus it allows choosing a decision threshold with an appropriate balance between sensitivity and specificity. We can see that if a model has no predictive power, then regardless of the threshold, the proportion of false positives and false negatives will always sum to 1, therefore the ROC curve will be a straight line across the diagonal and thus the area under this curve will be 0.5.

AUC can also be seen as quantifying to which extent two distributions overlap. We can rearrange the ROC plot, in such a way that the threshold value is on the x-axis and two curves are shown, one for the false-negative rate (1-sensitivity) and one for the true negative rate, both on the y-axis. The area between the diagonal and the ROC curve in the ROC plot (Figure 2A) is now the area between these two curves. These two curves are cumulative distributions of subjects from each class. The area between these curves represents the non-overlapping areas of two distributions (Figure 2B).

AUC is identical to C-index or a concordance probability in the case of a binary outcome (Hanley and McNeil, 1982). This is the probability that a randomly chosen data point forms a positive class is ranked higher than a randomly chosen example forms the negative class. E.g. if we have two patients, one with disease and one without, AUC is the probability that the model will correctly rank patients with a disease to have a higher risk of the disease than patients without the disease. This

can also be interpreted as a proportion of all pairs of subjects in the dataset, where a subject with the disease is ranked higher than a subject without the disease (Figure 3B).

AUC is also a rescaling of several rank-based correlation measures, including Sommers d_{xy} (Newson, 2002). It is also a normalized Mann-Whitney U statistic (Mason and Graham, 2002), which is used in the Mann-Whitney U test or Wilcoxon rank-sum test, a popular nonparametric alternative for a t-test. The latter connection is especially important because it means that by testing the statistical significance of a difference between two groups using the Mann-Whitney U test, one is also testing the statistical significance of AUC and vice versa (Figure 3A).

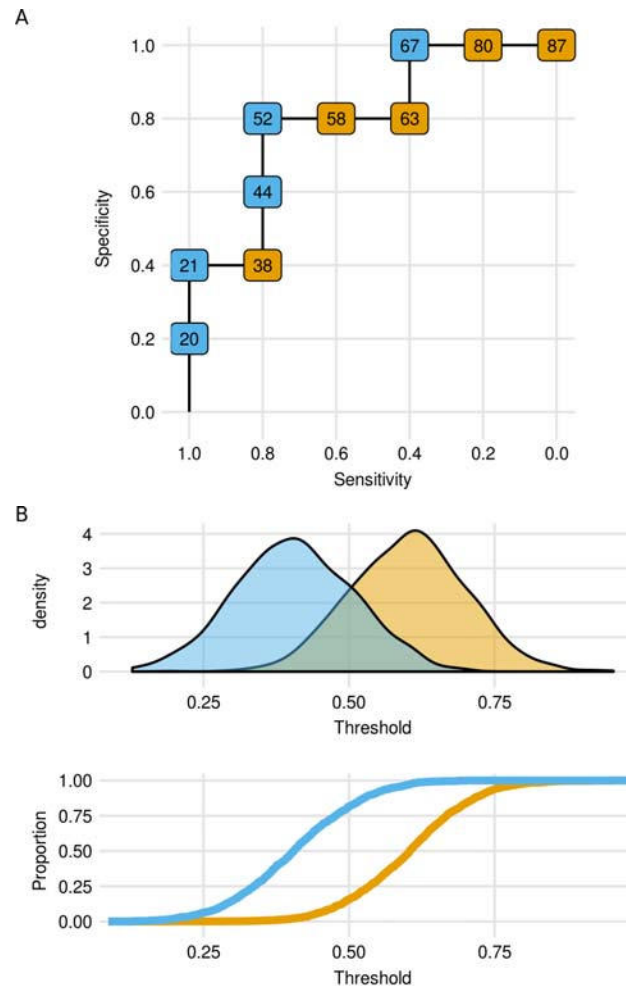


Figure 2. A: ROC curve for the same data as in Figure 3. The points on the curve represent threshold values that divide continuous prediction into two classes (represented as orange and blue colors). B: AUC as an overlap between two distributions. Two distributions (top) are transformed into cumulative distribution (bottom). The non-overlapping area of these two distributions is the area between two cumulative distribution curves, which equals to the area between the diagonal to the ROC curve or AUC - 0.5.

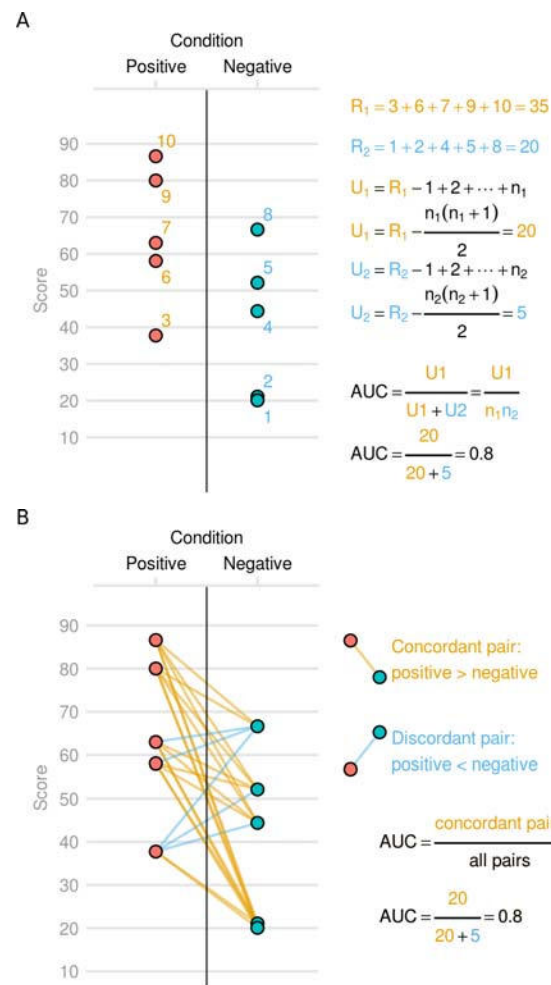


Figure 3. Interpretation and construction of AUC from the figure 2A as an **A) sum of ranks, i.e. Mann-Whitney U statistics**. Predictions are ranked from lowest to highest, and the sum of ranks R_1 for the positive class is computed (or R_2 for negative class). From the sum of ranks, Mann-Whitney U statistic is computed by subtracting the sum of ranks of one group from the sum of ranks of all predictions. AUC is the normalized version of the Mann-Whitney U , by dividing U statistics by the maximum possible value of the U statistics and **B) concordance measure**. AUC can be interpreted as a proportion of pairs of subjects where a subject from the positive class is ranked higher than a subject from the negative class, or where a randomly selected subject from the positive class is ranked higher than a randomly selected subject from the negative class.

Quadratic error-based measures

Performance measures based on the quadratic error (together with information-based measures) quantify the performance of probabilistic predictions directly, without any intermediate transformation of predictions to categorical predictions or ranks. This makes them the most sensitive measures to capture signal in the data or model improvement, although it requires that the model predictions are in the form of probabilities.

Brier score (Brier, 1950) is the most prominent example from this category. It is a mean squared error between predicted values and observed values, where predictions are coded as 0 and 1, for the case of the binary classifier. The Brier score can be straightforwardly generalised to multi-class classification using 'one-hot' dummy coding.

$$Brier = \frac{1}{n} \sum_{i=1}^n (y_i - p_i)^2$$

For example, if a model predicts that a patient with a disease has a disease with a probability of 0.8, the squared error will be $(1-0.8)^2=0.04$ and the Brier score is the average of error of all predictions across all subjects. Brier score has values between 0 and 1 (smaller the better), with 0.25 for a chance level predictions in a case where there is an equal number of subjects in both groups (because $0.5^2=0.25$). Compared to AUC, it takes into account specifically predicted probabilities of an outcome, not only their ranks. The score is improved when the predictions are well calibrated, so when the predicted probabilities correspond to observed frequencies of misclassification (e.g. in subjects with the predicted probability of a disease of 0.8, 80% of these subjects will have the disease).

One of the difficulties with employing the Brier score in practice is that – unlike accuracy and AUC – it lacks a simple intuitive interpretation. In order to make it more intuitive, it can be rescaled to form a pseudo R^2 measure analogous to variance explained used in evaluate regression models.

The Scaled Brier score is defined as

$$Brier_{scaled} = 1 - \frac{Brier}{Brier_{max}}$$

where the $Brier_{max}$ is the maximum score that can be obtained with a non-informative model

$$Brier_{max} = \frac{1}{n} \sum_{i=1}^n (p_i) \times \left(1 - \frac{1}{n} \sum_{i=1}^n (p_i)\right)$$

Thus a non-informative model will have a score of 0 and a perfect model score of 1, regardless of class frequencies.

One intuitively interpretable measure is a discrimination slope, also known as Tjur's pseudo R^2 (Tjur, 2009).

$$Tjur's\ pseudo\ R^2 = \frac{1}{n_a} \sum_{i=1}^{n_a} (p_{ai}) - \frac{1}{n_b} \sum_{i=1}^{n_b} (p_{bi})$$

This is simply the difference between the mean of predicted probabilities of two classes, which can also be easily visualized (Figure 4).

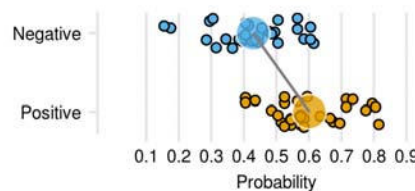


Figure 4: Tjur's pseudo R^2 . Interpretation of Tjur's pseudo R^2 is the difference between mean predicted probability of the positive group and the negative group.

Information criteria

Information theory provides a natural framework to evaluate the quality of model predictions according to how much information about the outcome is contained in the probabilistic predictions. The most important information-theoretical measure is the logarithmic score, defined as

$$\text{Logarithmic score} = \ln(p_{\text{target}})$$

where p_{target} is the predicted probability of the observed target. If the target is coded as 0 and 1, and the score is averaged across all predictions, this becomes

$$\text{Logarithmic score} = \frac{1}{n} \sum_{i=1}^n y \times \ln(p_i) + (1 - y_i) \times \ln(1 - p_i)$$

For example, a model predicts that a patient with a disease has a disease with a probability of 0.8, the logarithmic score will be $\ln(0.8) = -0.22$. It has many interpretations and strong connections to other important mathematical concepts. It is a log-likelihood of observed outcomes given model predictions, it also equals to Kullback–Leibler or relative entropy between observed target values and predicted probabilities. It quantifies an average information loss (in bits) about the target given that we have only imperfect, probabilistic predictions. It also quantifies how surprising the observed targets are given the probabilistic predictions. In a certain sense, the logarithmic score is an optimal score. It has been shown that in betting or investing, the expected long term gains are directly proportional to the amount of information the gambler has about the outcome. For example, if two gamblers are betting against each other repeatedly on an outcome of football games, the gambler whose predictions are better according to the logarithmic score, will on average multiply her wealth, even if her predictions might be worse according to Brier score or AUC (Kelly, 1956; Roulston and Smith, 2000).

In practice, the logarithmic score and Brier score usually produce similar results and the difference is evident only with severe misclassification. The logarithmic score can grow to infinity when the predicted probability of an outcome is close to zero.

This might be considered undesirable since a single extreme wrong prediction can severely affect the summary score of an otherwise well-performing model. On the other hand, it can be argued that this is a desirable property of a score and predictions close to 0 and 1 should be discouraged. Intuitively, the logarithmic score measures surprise, if an event that is supposed to be absolutely impossible ($p=0$) happens, we rightly ought to be infinitely surprised. Similarly, if we know that an event is absolutely certain ($p=1$), then the right approach (mathematically) would be to bet all our money on this event, thus it is desirable that the wrong prediction is maximally penalized.

Similarly to the Brier score, the logarithmic score can be scaled to make it more intuitively interpretable. One popular way is to use Nagelkerke pseudo R^2 (Nagelkerke, 1991) which is defined as

$$R^2_{\text{Nagelkerke}} = \frac{1 - \left(\frac{L(P_{\text{null}})}{L(P)} \right)^{2/N}}{1 - L(P_{\text{null}})^{2/N}}$$

Where $L(P)$ is a logarithmic score of model and $L(P_{\text{null}})$ is a logarithmic score of chance level predictions (i.e. predicting only class frequencies).

Empirical evaluation

Here we perform several empirical evaluations of statistical properties of the selected performance measures, one from each family described above, namely, accuracy, AUC, Brier score and logarithmic score. We examine: (i) the statistical power of detecting a statistically significant result,

(ii) the power to detect a statistically significant improvement in model performance, (iii) the feature selection stability, and (iv) the reliability of cross-validation results. We employ real and simulated datasets.

Statistical power of detecting a statistically significant result

Statistical tests: In this section, we evaluate how often a given method is likely to detect a statistically significant effect (i.e., a classifier that exceeds chance levels at a given significance level, nominally $p < 0.05$). In this case, the focus is on statistical significance, but not on the absolute level of performance. We used the same datasets using the different performance measures described above. We tested the power of a model to make a statistically significant prediction on an independent test set. Statistical significance of the accuracy measure was obtained using a binomial test. To obtain statistical significance of AUC, we performed Mann-Whitney U test. Since AUC is equivalent to Wilcoxon-Mann-Whitney U statistics, performing Mann-Whitney-U test on model predictions equals to performing a statistical significance test of AUC. P-values for Brier score and log score were obtained using a permutation test, where the real labels were shuffled for a maximum of 10,000 times and the specific scores were computed for each shuffle, thus obtaining an empirical null distribution of scores where the specific p-value corresponds to a percentile of the observed test statistic in this distribution. For permutation tests, we employed early stopping criteria according to (Gandy, 2009), that stopped the permutation when the chance of making wrong decision to reject the null hypothesis was lower than 0.001. Performing the permutation test for accuracy and AUC is not necessary, because their distributions are known and can be computed exactly. These tests are only valid because we are testing the statistical significance in an independent test set. However, they would produce overly optimistic results in a cross-validation setting. This is because data points between folds are no longer completely independent and a permutation test where a model is refitted in each permutation should be used instead (Noirhomme et al., 2014; Varoquaux et al., 2017).

Experiments:

First, we examined statistical power on simulated distributions of model predictions, without any machine learning modeling. Similar to the first experiment, we repeatedly sampled from two one-dimensional Gaussian distributions 1 SD apart representing the distribution of model predictions of two classes. These predictions were transformed into 0-1 range using a logistic function and into categorical predictions by thresholding at 0.5 threshold. We performed 1000 simulations for sample sizes 20, 80, 140, 200, and recorded proportion of times each statistical test obtained a statistically significant result ($p < 0.05$).

Second, we examined statistical power of a support vector machine (SVM) to discriminate between two groups in the simulated dataset. The simulated dataset consisted of 6 independent variables, 3 signal variables and 3 noise variables. Signal variables were each randomly sampled from a Gaussian distribution with $SD=1$ and $mean=0$ for group 0 and $mean=0.3$ for group 1. Three noise variables were each randomly sampled from a Gaussian distribution with $mean=0$ and $SD=1$. We repeatedly sampled training and test set from this dataset of size 40, 80, 120, 140, and fit a support vector machine classifier in the training set and evaluated the statistical significance of the predictions in the test set. We used C-SVM implementation of an SVM with a linear kernel from a package kernlab (Karatzoglou et al., 2004), with a C parameter fixed at 1. SVM predictions were transformed into probabilities using Platt scaling (Platt, 1999), as implemented in kernlab.

Third, we examined statistical power on real neuroimaging datasets, including OASIS cross-sectional (Marcus et al., 2010) and ABIDE datasets (Craddock et al., 2009). We used already preprocessed OASIS VBM data as provided by the OASIS project using nilearn dataset fetching

functionality (Abraham et al., 2014) The preprocessing consisted of brain extraction, segmentation of white matter and gray matter, and normalization to standard space using DARTEL. Details of the preprocessing can be found elsewhere (Marcus et al., 2010). Here we used gray matter and white matter data separately in order to predict biological sex and diagnostic status (presence or absence of dementia). To reduce the computational load, we reduced the dimensionality of the WM and GM datasets to 100 principal components each. Furthermore, we used already preprocessed ABIDE dataset as provided by the preprocessed-connectome-project (Craddock et al., 2009) to predict biological sex. These consisted of ROI average cortical thickness data obtained using ANTs pipeline (Das et al., 2009), defined using sulcus landmarks according to the Desikan-Killiany-Tourville (DKT) protocol (Klein and Tourville, 2012).

Additionally, we included common non-imaging machine learning benchmark datasets obtained from using mlbench and kernlab libraries originally from UCI machine learning repository (Dua and Graff, 2017). These included Pima Indians diabetes, sonar, musk, and spam.

Finally, in order to compare statistical power to detect statistically significant above chance prediction according to specific performance measures, it is important to show that the higher power is due to higher sensitivity of the performance measure and not because the used statistical test is overly optimistic. To do this, we repeated all experiments including simulated and real datasets, but with permuting true labels before each simulation to destroy any relationship between the data and target outcome.

Results: In all simulated and real datasets, when the null hypothesis was true (i.e. performing experiments on data with shuffled labels), statistical significance $p < 0.05$ was obtained approximately 5% of times for significance tests of AUC, Brier score and logarithmic score, as expected (supplementary figure 1). The binomial test was often overly conservative, i.e., $p < 0.05$ was obtained less than 5% of times. This is a known behavior of binomial test in small samples caused by a limited number of values in the null distribution (Fig 4 shows the example of $n=20$ sample). This conservativeness is worse in small samples and it disappears when the sample size is sufficiently large (i.e. $N=5000$). This behavior is not limited to a binomial test, it also happens if the p-value of accuracy is calculated using permutation test, which is just a random approximation of the exact binomial test, and the low resolution of the null distribution of accuracy results is present even when this distribution is obtained using permutations.

For all experiments and all sample sizes, tests using accuracy had the lowest power. Brier score, logarithmic score, and AUC performed approximately the same. At the sample size where AUC, Brier score, the logarithmic score obtained the common goal of 80% power, accuracy obtained only 60% power in all datasets (see figure 5). Results for all datasets separately are in supplementary figure 2

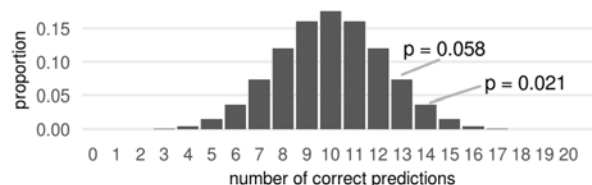


Fig 5: Null distribution of correct predictions for $n=20$ illustrating why significance test for categorical predictions is conservative in small samples at specific significance levels. Since the predictions are categorical, the null distributions consist only of a limited number of values. It is not possible to obtain $p=0.05$, in order to obtain $p<0.05$ it is necessary to get at least 14 correct predictions, corresponding to $p=0.021$. Although this makes the test conservative at the $p = 0.05$

level, it is not conservative at $p = 0.058$ or $p=0.021$ levels obtained by getting 13 or 14 correct predictions respectively.

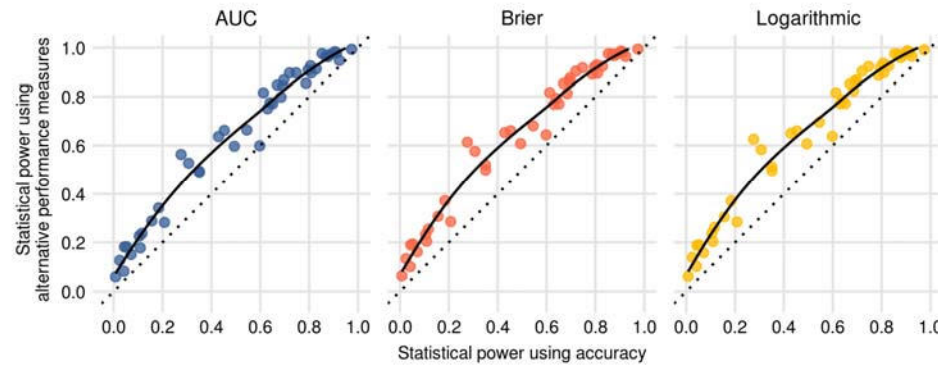


Figure 6. Comparison of statistical power of accuracy and alternative performance measures. We calculated statistical power to find significant above-chance performance in the test set ($p < 0.05$) across multiple simulated and real datasets with varying sample sizes. Each dot represents a proportion of statistically significant results across 1000 draws from a specific dataset and sample size. This figure shows that all alternative measures show greater power than accuracy for detecting a significant effect.

Power to detect a statistically significant improvement in model performance

In the previous section, we evaluated power to detect an effect that is significantly different from zero (i.e. exceeding ‘chance’ level), however, in many cases, it is important to detect a difference between two different classification models, potentially trained using different algorithms or different features. For example, we might want to know if a whole brain-machine learning model predicts remission of a depressive episode better than simple clinical data. Or we might want to compare the performance of two different methods e.g. support vector machines and deep learning. In both of these cases, it is important to evaluate the difference statistically, otherwise the apparent superiority of one model might be due to chance and translate to better predictions in a population.

Statistical tests: We obtained statistical significance of a difference in accuracy between two models using McNemar test using an exact binomial distribution. To test differences between two models using Brier score and logarithmic score, we used a permutation sign flip test testing the hypothesis that the difference in errors between two models is centered at 0. If the predictions were categorical, the sign-flip test would approximate the results of the exact McNemar test. There is no equivalent test for AUC because errors depend on ranks and thus cannot be computed for individual data points. Instead, we have used DeLong’s non-parametric test of differences between two AUC (DeLong et al., 1988).

Experiments: We compared the performance of two models on the same simulated and real datasets as in the previous section. Each time, we compared the performance of a model that was trained on the whole training set, with a model trained using only a subsample of the training set of size between 10-90% of the full training set. The model trained with fewer data points should eventually perform worse than a model using all available data. For each sample size of the restricted model, we have calculated the proportion of times a statistical test found a statistically significant difference in the performance of these two models.

Results: The difference between the power of different performance measures was higher than in the testing against the null hypothesis of no effect in the previous section. Accuracy had the lowest power, followed by AUC and Brier and logarithmic score performed approximately the same.

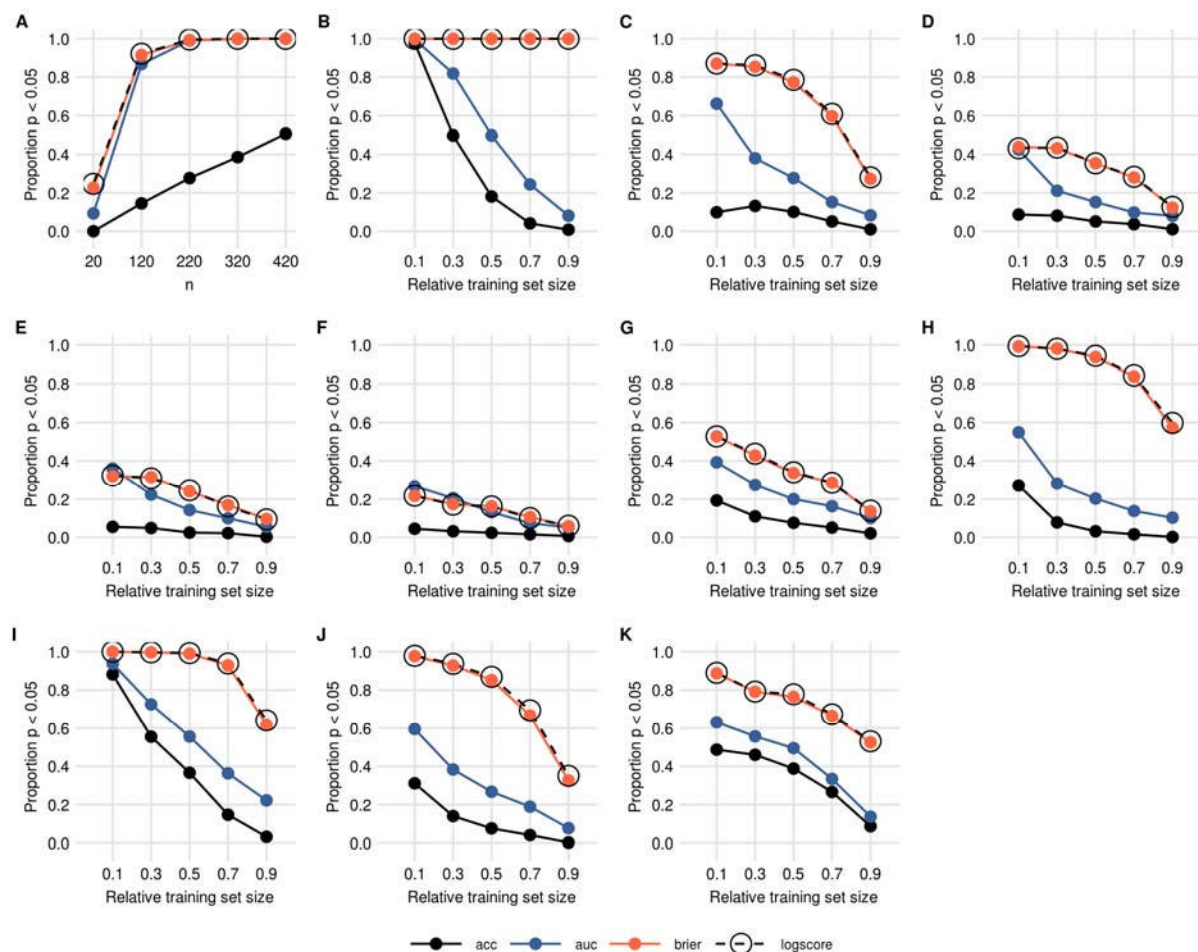


Figure 7 Statistical power of comparing 2 competing models. One model was trained using the whole training set and the second model was trained on fewer subjects, using only a proportion of the training set. Y-axis shows the proportion of statistically significant results obtain by comparing the performance of two models, from 1000 random draws from each dataset. A: simulated predictions, the performance of the two models was fixed, but we manipulated the sample size. B: SVM fitted on simulated data, C: OASIS gray matter gender prediction, OASIS white matter gender predictions, E: OASIS gray matter diagnosis prediction, F: OASIS white matter diagnosis prediction, G: ABIDE cortical thickness gender prediction, H: Pima Indians diabetes benchmark dataset, I: Sonar benchmark dataset, J: Musk benchmark dataset K: Spam benchmark dataset.

Evaluation of stability of the feature selection process

Feature selection is an important part of machine learning with the goal of selecting a subset of features that are important for prediction. It is usually done in order to make models more interpretable and improve their performance. Different feature selection criteria lead to a different set of selected features. Here we evaluated which specific performance measure (accuracy, AUC, brier score, logarithmic score), leads to better feature selection results when its improvement is used as a criterion for feature selection. We performed a greedy forward stepwise feature selection, starting with 0 features and subsequently adding additional features into the model that improve the model performance the most according to a specific performance measure. This is a noisy feature selection process but preferably we would want informative features to be selected on average more often than non-informative features. We constructed stability paths according to Meinshausen Buhlman (Meinshausen and Bühlmann, 2010), the feature selection procedure was performed repeatedly on a random subsample of a dataset and the probability of selecting a specific feature

was computed for any number of selected features in the final model. This was performed on 2 benchmark machine learning datasets 1, Pima Indians diabetes dataset and 2, spam prediction datasets. To each dataset, we added non-informative features by randomly permuting some of the original features, thus destroying any information they can have about the outcome. We compared how often original informative features are selected compared to non-informative features.

Results: Signal features (i.e. features that had not been permuted) were selected most often if the feature selection was performed according to logarithmic score and Brier score, followed by AUC and accuracy. For example for spam data, only one signal feature was stably selected using accuracy.

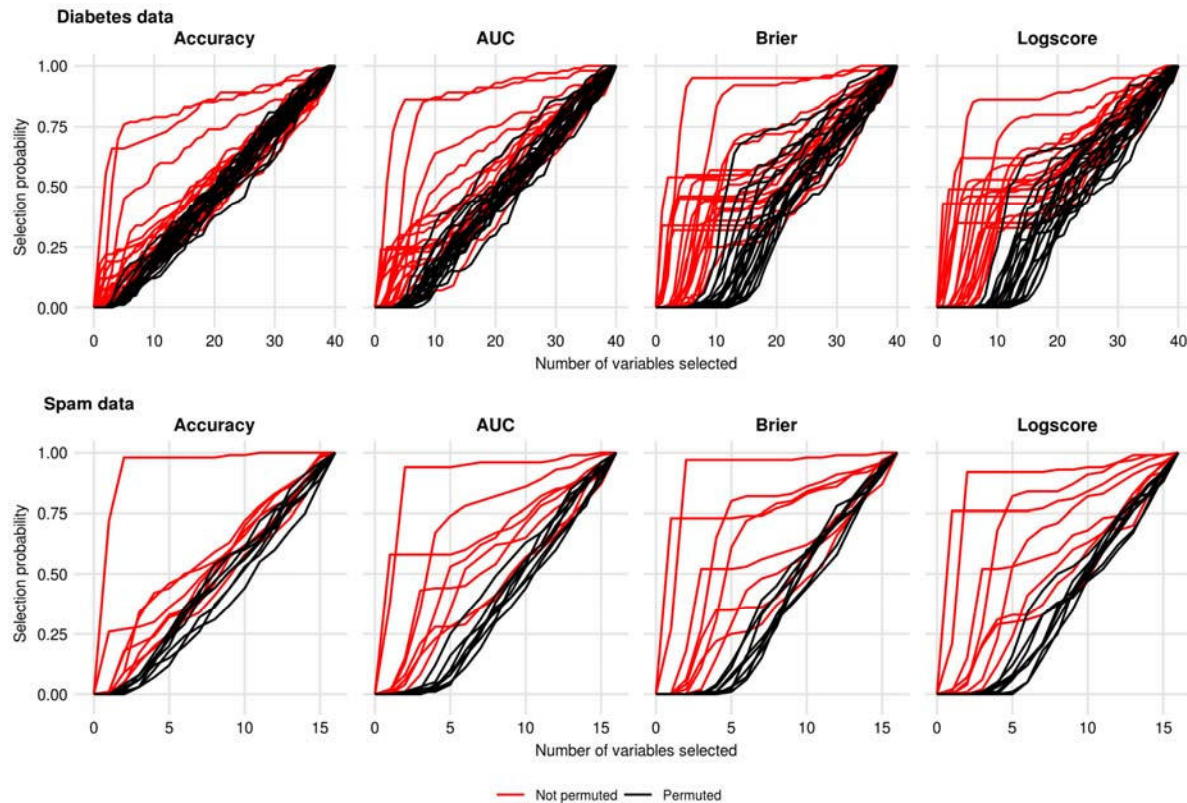


Figure 8: stability of feature selections. We can see stability paths of individual features if they are selected according to specific performance measures in a greedy forward stepwise feature selection procedure. Signal features (red) were selected sooner using logarithmic score and brier score compared to accuracy and AUC.

Reliability of the cross-validation measure

Previously (Varoquaux et al., 2017) showed that cross-validation estimates of accuracy have high variance and high error with respect to the accuracy obtained on a large hold-out validation set, especially when the sample size is low. Here we compared the relationship between cross-validation performance and hold-out performance for accuracy, AUC, Brier score and logarithmic score. We performed a procedure similar to Varoquaux 2017. We selected 6 large sample datasets from UCI machine learning repository in order to have at least 1000 samples in the hold-out set. Further, we manipulated datasets by randomly flipping labels to 0-20% data points, thus creating many datasets with different true performance. In each of these datasets, we estimated model performance using

10-fold cross-validation and compared it to out of sample performance on the validation sample of size 1000. We repeated this procedure for different sizes of the training set 50, 100, 150, 200, and 250.

Results: In almost all comparisons, reliability of the cross-validation results (as measured by Spearman correlation between cross-validation performance and hold-out performance) was higher for logarithmic score and Brier score, compared to accuracy and AUC. This was true when the results for different datasets were combined together (Figure 7A) or separated (Figure 7B).

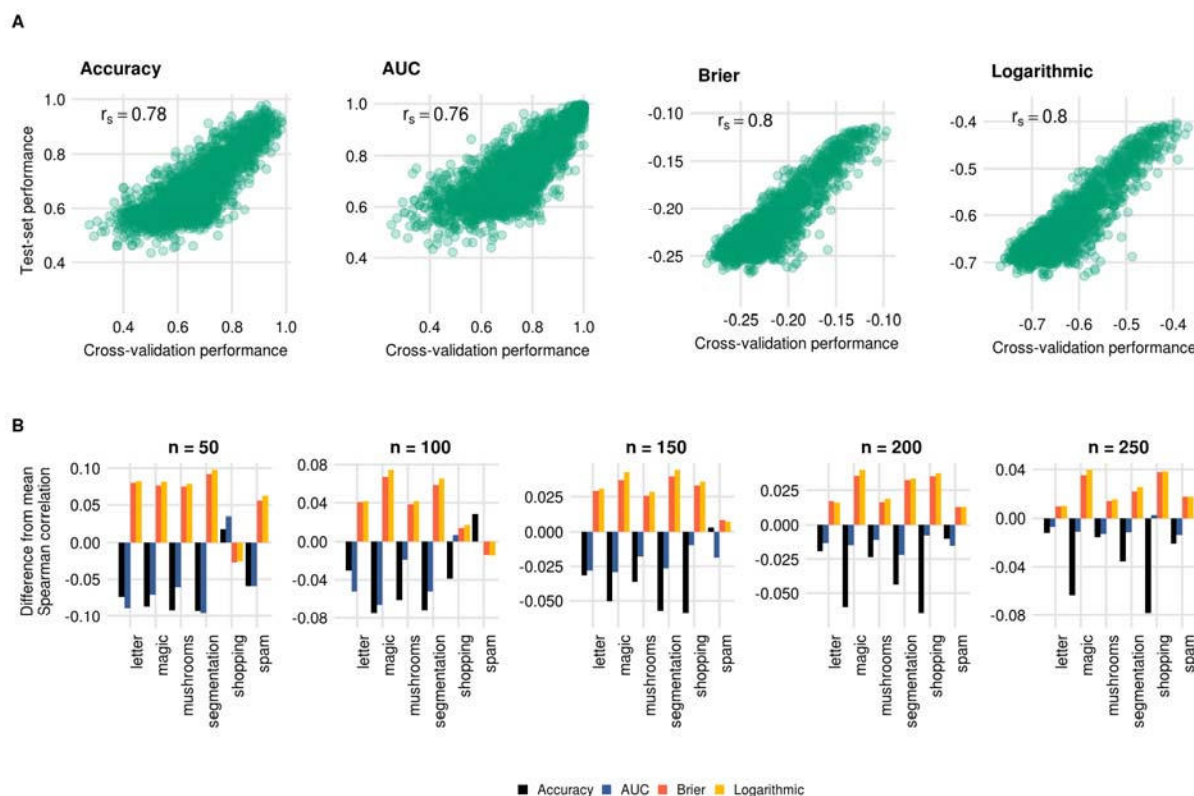


Figure 7: Comparison between performance in the cross-validation and on the holdout set. A: comparison across 6 datasets for a training set of size 50. B comparing Spearman correlation between cross-validation performance and hold-out performance for each sample size and dataset separately.

Data and code availability

All data and code necessary to reproduce the experiments is at <https://github.com/dinga92/beyond-acc>

Discussion and recommendations

The choice of a performance measure should be motivated by the goals of the specific prediction model. In this report, we reviewed exemplars of 4 families of performance measures evaluating different aspects of model predictions. Namely, categorical predictions, ranks of predictions and probabilistic predictions according to quadratic error and information content.

Often, such as in brain decoding and encoding, the primary goal of machine learning models is not to make predictions but to learn something about how information is represented in the brain. Two common goals of encoding or decoding studies are to establish if a region contains any information at all about the stimulus or behavior, and to compare the relative amount of information about stimulus between multiple ROIs (Naselaris et al., 2011).

In this case, the performance measure should be chosen to maximize the statistical power of a test and reliability of results. Accuracy, although the most commonly used performance measure, performed the worst in all statistical aspects we have examined in this study compared to alternative measures (i.e. AUC, Brier score, logarithmic score). It has the lowest statistical power to find significant results and to detect a model improvement, it leads to unstable feature selection and the results are least likely to replicate across samples from the same population. For these reasons, the accuracy should not be used to make a statistical inference, although if needed it can be reported as an additional descriptive statistic.

The loss of power when using accuracy is significant. Compared to alternative measures, the power to detect statistically significant results dropped from 80% to 60% or from 60% to 40% on average across simulated and real datasets. The reason for this is twofold. First, accuracy does not take the magnitude of an error into account, thus it is a very crude and insensitive measure of model performance. The model improvement can only be detected at the decision threshold, thus leading to severely suboptimal inference. Second, since accuracy evaluates only categorical predictions, it can only take a limited number of values, thus a null distribution is tabulated which leads to conservative p-values (Fig 4). The smaller the sample size, the worse this effect is.

The loss of power is even more prominent when the goal is to find statistically significant improvement on an already well-performing model. In this situation, both accuracy and AUC perform significantly worse than probabilistic measures (log score, Brier score). This effect is especially prominent when comparing already well performing models. When the discrimination between classes is already large, there is only a small chance that a model improvement would result in changing of ranks of predictions (for the change in AUC) or predictions crossing the decision threshold (for the change in accuracy). Even potentially important model improvements can be missed if the model is assessed based on accuracy. Situations are easily constructed where model performance improves significantly, but without ever changing the proportion of correctly classified samples. Or in other words, statistical power to detect an improvement in model performance according to accuracy can be effectively zero.

The crudeness of accuracy leads to another important problem, and that is a loss of reliability and reproducibility of the results, as shown in our comparison of cross-validation results. If we replicate the same analysis on a different sample from the same population, the results using accuracy will be less similar to each other than results using alternative measures. This, together with a significant loss of power, leads to less replicable results. It is important to note that there is no upside to these problems in the form of for example higher confidence in conservative statistically significant results using accuracy. As it was pointed out before (Button et al., 2013; Ioannidis, 2005; Loken and Gelman, 2017) low power necessary leads to overestimation of the found effect, low reproducibility of the results, and higher chance that the observed statistically significant effect size does not reflect the true effect size. (Varoquaux et al., 2017) pointed out that the error bars for accuracy under cross-validation are large, and that the results are much more variable than is usually appreciated. From our results, it is clear that a large portion of this variability is attributable to using accuracy as a performance measure, and that unfavorable statistical properties of cross-validation can be improved simply by using alternative performance measures.

In many situations, the statistical properties of a specific performance measure are not an important aspect of model evaluation. In a clinical context, the utility of model predictions for a patient or a

clinician is far more important than statistical power. Model evaluation based on categorical predictions is in this situation inappropriate because categorical predictions hide potentially clinically important information from decision-makers, and they assume that the optimal decision threshold is known and identical regardless of patient or situation. In clinical settings, it is generally recommended (Moons et al., 2015) to evaluate models based on their discrimination, usually measured by AUC and calibration which measures how well the predicted probabilities matches observed frequencies. In situations where the optimal threshold is fixed and known or when the decisions need to be made fully automatically, without any additional human intervention, it is appropriate to evaluate model performance with respect to its categorical predictions. However, misclassification should be weighted according to relative the cost of false positive and false negative misclassification in order to properly evaluate the utility of the model. Accuracy weighs false positive and false negative misclassification equally (or according to class frequencies with balanced accuracy) which is almost never the case, thus it will lead to wrong decisions.

To present and visualize the model performance, multiple options exist. Good visualization should be intuitive and informative. Confusion matrices, although common, do not show all available information because they show only categorical predictions. This can be improved by plotting the whole distribution of predicted probabilities per target class in the form of histograms, raincloud plots, or dot plots as in figure 4. This directly shows how well the model separates target classes, it might reveal outliers or situations where the performance is driven by only a small subset of accurately classified data points. This can be accompanied by a calibration plot with predicted probabilities on the x-axis and observed frequencies on the y-axis with fitted regression curve, showing how reliable the predicted probabilities are. An additional commonly used visualization is a ROC curve. This is arguably less informative and less intuitive than plotting the predicted probability distributions directly, however, it can be used in specific situations where it is useful to visualize the range of sensitivities and specificities across different decision thresholds.

Conclusion

We extensively compared classification performance metrics from four families using simulated and real data. In all statistical properties we evaluated, accuracy performed the worst, thus it should not be used as a metric to statistically evaluate model predictions. Summary measures based on probability predictions (i.e. logarithmic score or Brier score) performed the best and thus, we recommend the use of these measures instead of accuracy. For model interpretation and presentation, various measures can be reported at the same time, together with a graphical representation of model predictions. If the model is supposed to be used in practice such as in a clinical setting, summary measures are not enough. Rather, we recommend that the model be evaluated with respect to its discrimination power, calibration, and clinical utility. Accuracy should be avoided because it weighs false positive and false negative misclassification equally, or according to class frequencies but not according to consequences for a patient.

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Model Selection with Gini Indices under Auto-Calibration

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Abstract

The Gini index does not give a strictly consistent scoring rule in general. Therefore, maximizing the Gini index may lead to wrong decisions. The main issue is that the Gini index is a rank-based score that is not calibration-sensitive. We show that the Gini index allows for strictly consistent scoring if we restrict to the class of auto-calibrated regression models.

Keywords. Regression model, binary classification, Gini index, Gini score, consistency, consistent scoring, auto-calibration, Lorenz curve, concentration curve, cumulative accuracy profile, CAP, receiver operating characteristics, ROC, area under the curve, AUC, accuracy ratio, Somers' D , forecast-dominance.

1 Introduction

The Gini index (Gini score, accuracy ratio) is a popular tool for model selection in machine learning, and there are versions of the Gini index that are used to evaluate actuarial pricing models and financial credit risk models; see Frees et al. [7, 8], Denuit et al. [3], Engelmann et al. [6] and Tasche [17]. However, in general, the Gini index does not give a (strictly) consistent scoring rule; Example 3 of Byrne [1] gives a counterexample. (Strict) consistency is an important property in model selection because it ensures that maximizing the Gini index does not lead to a wrong model choice; see Gneiting [10] and Gneiting–Raftery [11]. The Gini index can be obtained from Somers' D [16], which essentially considers Kendall's τ ; see Newson [14]. Intuitively, this tells us that the Gini index is a rank-based score that is not calibration-sensitive. The missing piece to make the Gini index a strictly consistent scoring rule is to restrict it to the class of auto-calibrated regression models, this is proved in Theorem 4.5, below; for auto-calibration we refer to Krüger–Ziegel [12], Denuit et al. [2] and Section 7.4.2 of Wüthrich–Merz [18].

Organization. In the next section, we introduce the notion of strictly consistent scoring rules. In Section 3, we discuss the Gini index as it is usually used in the machine learning community. In Section 4, we introduce and discuss the property of having an auto-calibrated regression model (forecasts), and we prove that the Gini index gives a strictly consistent scoring rule if we restrict to the class of auto-calibrated regression models. This makes the maximization of the Gini index a sensible model selection tool on the class of auto-calibrated regression models. Finally, in Section 5 we conclude.

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2 Consistent scoring rules

Let $(Y, \mathbf{X})$ be a random tuple on a sufficiently rich probability space $(\Omega, \mathcal{A}, \mathbb{P})$ with real-valued non-negative response Y having finite mean and with covariates $\mathbf{X}$. Denote by $\mathcal{F}$ the family of potential distributions of $(Y, \mathbf{X})$ being supported on $\mathcal{Y} \times \mathcal{X}$. Let $F_{Y|\mathbf{X}}$ be the conditional distribution of Y , given $\mathbf{X}$. For any model $(Y, \mathbf{X}) \sim F \in \mathcal{F}$, we consider the conditional mean functional T

$$F_{Y|\mathbf{X}} \mapsto T(F_{Y|\mathbf{X}}) = \mu^\dagger(\mathbf{X}) = \mathbb{E}[Y | \mathbf{X}],$$

where $\mathbf{X} \mapsto \mu^\dagger(\mathbf{X}) = \mathbb{E}[Y | \mathbf{X}]$ denotes the true regression function of the chosen model. The main task in regression modeling is to find this unknown true regression function $\mu^\dagger(\cdot)$ from i.i.d. data $(Y_i, \mathbf{X}_i)$, $1 \leq i \leq n$, having the same distribution as $(Y, \mathbf{X})$.

Choose a scoring function $S : \mathcal{Y} \times \mathbb{R} \rightarrow \mathbb{R}$ giving us the score $\mathbb{E}[S(Y, \hat{\mu}(\mathbf{X}))]$ for regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$ and $(Y, \mathbf{X}) \sim F \in \mathcal{F}$. A *scoring rule* is obtained by selecting the argument(s) $\hat{\mu}^*(\cdot)$ that maximize the score over the regression functions $\hat{\mu}(\cdot)$, subject to existence,

$$\hat{\mu}^*(\cdot) \in \arg \max_{\hat{\mu}(\cdot)} \mathbb{E}[S(Y, \hat{\mu}(\mathbf{X}))], \quad (2.1)$$

under the given model choice $(Y, \mathbf{X}) \sim F \in \mathcal{F}$.

A scoring rule is called *consistent* on $\mathcal{F}$ for the conditional mean functional T , if for any model $(Y, \mathbf{X}) \sim F \in \mathcal{F}$ with conditional distributions $F_{Y|\mathbf{X}}$ of Y , given $\mathbf{X}$, we have $S(Y, T(F_{Y|\mathbf{X}})) \in L^1(\mathbb{P})$, and for any regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$ with $S(Y, \hat{\mu}(\mathbf{X})) \in L^1(\mathbb{P})$ we have

$$\mathbb{E}[S(Y, T(F_{Y|\mathbf{X}}))] \geq \mathbb{E}[S(Y, \hat{\mu}(\mathbf{X}))]. \quad (2.2)$$

A scoring rule is called *strictly consistent* on $\mathcal{F}$ for the conditional mean functional T , if it is consistent on $\mathcal{F}$, and if an identity in (2.2) holds if and only if $\hat{\mu}(\mathbf{X}) = T(F_{Y|\mathbf{X}}) = \mu^\dagger(\mathbf{X})$, a.s.

Remarks 2.1 • Strict consistency implies that the true regression function $\mu^\dagger(\cdot)$ is the unique maximizer in (2.1), and it can be estimated by score maximization (assuming it is contained in the set over which we optimize, which we generally do). Empirically, we then consider for i.i.d. data $(Y_i, \mathbf{X}_i)$, $1 \leq i \leq n$,

$$\arg \max_{\hat{\mu}(\cdot)} \frac{1}{n} \sum_{i=1}^n S(Y_i, \hat{\mu}(\mathbf{X}_i)),$$

where we still need to ensure that we can exchange the limit $n \rightarrow \infty$ and the $\arg \max$ -operator to asymptotically select the true regression function $\mu^\dagger(\cdot)$ under strict consistency.

- Formula (2.2) states unconditional consistency as we average over the distribution of $\mathbf{X}$. For conditional consistency (in $\mathbf{X}$) and its relation to the unconditional version we refer to Section 2.2 in Dimitriadis et al. [5]. A point prediction version of consistency is given in Definition 1 in Gneiting [10].
- For scoring rule (2.1) we consider a maximization. By a sign switch we can turn this into a minimization problem, and in that case we rather speak about expected loss minimization.
- Typically, we restrict (2.1)-(2.2) to smaller classes of regression functions $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$. In the sequel, we will require continuity for these smaller classes, and, further below, we require the auto-calibration property. This requires that the true regression function $\mu^\dagger(\cdot)$ has this continuity, auto-calibration it will satisfy automatically, see Lemma 4.1, below.

3 The Gini index in machine learning

In the sequel we assume $\hat{\mu}(\mathbf{X})$ to have a continuous distribution $F_{\hat{\mu}(\mathbf{X})}$ for all $(Y, \mathbf{X}) \sim F \in \mathcal{F}$ and for any considered regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$. This implies $F_{\hat{\mu}(\mathbf{X})}(F_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha)) = \alpha$ for all $\alpha \in (0, 1)$, and with $F_{\hat{\mu}(\mathbf{X})}^{-1}$ denoting the left-continuous generalized inverse of $F_{\hat{\mu}(\mathbf{X})}$.

In machine learning (ML) one considers the *cumulative accuracy profile* (CAP) defined by

$$\alpha \in (0, 1) \quad \mapsto \quad \text{CAP}_{Y, \hat{\mu}(\mathbf{X})}(\alpha) = \frac{1}{\mathbb{E}[Y]} \mathbb{E} \left[Y \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) > F_{\hat{\mu}(\mathbf{X})}^{-1}(1-\alpha)\}} \right] \in [0, 1].$$

In actuarial science, the CAP is also called concentration curve (up to sign switches), see Denuit–Trufin [4]. The CAP measures a rank-based correlation between the prediction $\hat{\mu}(\mathbf{X})$ and the response Y .

The *Gini index* (*Gini score*, *Gini ratio*, *Gini coefficient*, *accuracy ratio*) in ML is defined by

$$G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}} = \frac{\int_0^1 \text{CAP}_{Y, \hat{\mu}(\mathbf{X})}(\alpha) d\alpha - 1/2}{\int_0^1 \text{CAP}_{Y, Y}(\alpha) d\alpha - 1/2}, \quad (3.1)$$

where we additionally assume that Y has an (unconditional) continuous distribution F_Y . For a geometric interpretation see Figure 1 (lhs) and formula (4.3), below.

Remarks 3.1 • The denominator in (3.1) does not use the regression function $\hat{\mu}(\cdot)$, i.e., it has no impact on model selection by maximizing the Gini index $G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}}$ over $\hat{\mu}(\cdot)$. Hence, for scoring we can focus on the term in the numerator

$$\begin{aligned} \int_0^1 \text{CAP}_{Y, \hat{\mu}(\mathbf{X})}(\alpha) d\alpha &= \frac{1}{\mathbb{E}[Y]} \mathbb{E} \left[Y \int_0^1 \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) > F_{\hat{\mu}(\mathbf{X})}^{-1}(1-\alpha)\}} d\alpha \right] \\ &= \frac{1}{\mathbb{E}[Y]} \mathbb{E} \left[Y \mathbb{P} \left[F_{\hat{\mu}(\mathbf{X})}^{-1}(U) < \hat{\mu}(\mathbf{X}) \mid \hat{\mu}(\mathbf{X}) \right] \right] \\ &= \frac{1}{\mathbb{E}[Y]} \mathbb{E} [Y F_{\hat{\mu}(\mathbf{X})}(\hat{\mu}(\mathbf{X}))], \end{aligned} \quad (3.2)$$

for an independent $(0, 1)$ -uniform random variable U and where we use continuity of $F_{\hat{\mu}(\mathbf{X})}$. This shows that the Gini index in ML is not calibration-sensitive because $F_{\hat{\mu}(\mathbf{X})}(\hat{\mu}(\mathbf{X}))$ has a $(0, 1)$ -uniform distribution, i.e., the specific distribution of $\hat{\mu}(\mathbf{X})$ does not matter, but only its correlation with Y matters.

- Since typically the true data model $(Y, \mathbf{X}) \sim F$ is not known, the Gini index in ML (3.1) is replaced by an empirical version

$$\widehat{G}_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}} = \frac{\int_0^1 \widehat{\text{CAP}}_{Y, \hat{\mu}(\mathbf{X})}(\alpha) d\alpha - 1/2}{\int_0^1 \widehat{\text{CAP}}_{Y, Y}(\alpha) d\alpha - 1/2} \leq 1, \quad (3.3)$$

where we set

$$\widehat{\text{CAP}}_{Y, \hat{\mu}(\mathbf{X})}(\alpha) = \frac{1}{\frac{1}{n} \sum_{i=1}^n Y_i} \frac{1}{n} \sum_{i=1}^n Y_i \mathbb{1}_{\{\hat{\mu}(\mathbf{X}_i) > \hat{\mu}(\mathbf{X}_{(\lceil (1-\alpha)n \rceil)})\}}, \quad (3.4)$$

for i.i.d. data $(Y_i, \mathbf{X}_i)$, $1 \leq i \leq n$, having the same distribution as $(Y, \mathbf{X})$, and for order statistics $\hat{\mu}(\mathbf{X}_{(1)}) < \hat{\mu}(\mathbf{X}_{(2)}) < \dots < \hat{\mu}(\mathbf{X}_{(n)})$; note that by assumption the distribution of $\hat{\mu}(\mathbf{X})$ is continuous which implies that all observations $\hat{\mu}(\mathbf{X}_i)$ are mutually different for $1 \leq i \leq n$, and we have a strict ordering in the order statistics.

- Let us further comment on (3.4). First, if we mirror the CAP at the diagonal we have

$$\begin{aligned}\text{CAP}_{Y,\hat{\mu}(\mathbf{X})}^{-}(\alpha) &= \frac{1}{\mathbb{E}[Y]} \mathbb{E} \left[Y \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) \leq F_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha)\}} \right] \\ &= 1 - \frac{1}{\mathbb{E}[Y]} \mathbb{E} \left[Y \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) > F_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha)\}} \right] = 1 - \text{CAP}_{Y,\hat{\mu}(\mathbf{X})}(1 - \alpha).\end{aligned}\tag{3.5}$$

For an empirical version of the mirrored CAP we replace the above expression by

$$\begin{aligned}\widehat{\text{CAP}}_{Y,\hat{\mu}(\mathbf{X})}^{-}(\alpha) &= \frac{1}{\frac{1}{n} \sum_{i=1}^n Y_i} \frac{1}{n} \sum_{i=1}^n Y_i \mathbb{1}_{\{\hat{\mu}(\mathbf{X}_i) \leq \widehat{F}_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha)\}} \\ &= 1 - \frac{1}{\frac{1}{n} \sum_{i=1}^n Y_i} \frac{1}{n} \sum_{i=1}^n Y_i \mathbb{1}_{\{\hat{\mu}(\mathbf{X}_i) > \widehat{\mu}(\mathbf{X}_{(\lceil \alpha n \rceil)})\}},\end{aligned}$$

where in the last indicator we use the empirical distribution, for $m \in \mathbb{R}$ and $\alpha \in (0, 1)$,

$$\widehat{F}_{\hat{\mu}(\mathbf{X})}(m) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{\hat{\mu}(\mathbf{X}_i) \leq m\}} \quad \text{and} \quad \widehat{F}_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha) = \widehat{\mu}(\mathbf{X}_{(\lceil \alpha n \rceil)}).$$

This justifies the choice in (3.4). Similarly, we have for the denominator in (3.3)

$$\widehat{\text{CAP}}_{Y,Y}^{-}(\alpha) = \frac{1}{\frac{1}{n} \sum_{i=1}^n Y_i} \frac{1}{n} \sum_{i=1}^n Y_i \mathbb{1}_{\{Y_i \leq Y_{(\lceil \alpha n \rceil)}\}} \stackrel{(*)}{=} \frac{1}{\frac{1}{n} \sum_{i=1}^n Y_i} \frac{1}{n} \sum_{i=1}^{\lceil \alpha n \rceil} Y_{(i)},\tag{3.6}$$

for the identity $\stackrel{(*)}{=}$ to hold for any $\alpha \in (0, 1)$, we need to assume that we have a strict ordering $Y_{(1)} < Y_{(2)} < \dots < Y_{(n)}$, i.e., that there are no ties in the observations $(Y_i)_{1 \leq i \leq n}$, which is the case because Y was assumed to have a continuous distribution F_Y . This then motivates to set

$$\begin{aligned}\widehat{\text{CAP}}_{Y,Y}(\alpha) &= 1 - \widehat{\text{CAP}}_{Y,Y}^{-}(1 - \alpha) = \frac{1}{\frac{1}{n} \sum_{i=1}^n Y_i} \frac{1}{n} \sum_{i=1}^n Y_i \mathbb{1}_{\{Y_i > Y_{(\lceil (1-\alpha)n \rceil)}\}} \\ &\stackrel{(*)}{=} \frac{1}{\frac{1}{n} \sum_{i=1}^n Y_i} \frac{1}{n} \sum_{i=\lceil (1-\alpha)n \rceil + 1}^n Y_{(i)}.\end{aligned}\tag{3.7}$$

If we have a perfect joint ordering between $(Y_i)_{1 \leq i \leq n}$ and $(\hat{\mu}(\mathbf{X}_i))_{1 \leq i \leq n}$, the upper bound in (3.3) is attained, see (3.4) and (3.7). This is the motivation for the scaling in (3.1).

In the definition of the Gini index in ML (3.1) we have assumed that Y has a continuous distribution F_Y . This is not the case for discrete responses Y . Therefore, in the discrete case we need to replace the denominator in (3.1) by a different object. For illustrative purposes we show the binary classification case in the next example.

Example 3.2 (binary classification) We consider a binary classification example with true regression function

$$\mathbf{X} \mapsto p^{\dagger}(\mathbf{X}) = \mathbb{E}[Y | \mathbf{X}] = \mathbb{P}[Y = 1 | \mathbf{X}] \in (0, 1).$$

That is, Y is conditionally Bernoulli distributed, given $\mathbf{X}$, with probability $p^\dagger(\mathbf{X}) \in (0, 1)$ and range $\mathcal{Y} = \{0, 1\}$. In this case the CAP for a regression function $\mathbf{X} \mapsto \hat{p}(\mathbf{X})$ with continuous distribution $F_{\hat{p}(\mathbf{X})}$ is for $\alpha \in (0, 1)$ given by

$$\begin{aligned} \text{CAP}_{Y, \hat{p}(\mathbf{X})}(\alpha) &= \frac{1}{\mathbb{E}[Y]} \mathbb{E} \left[Y \mathbf{1}_{\{\hat{p}(\mathbf{X}) > F_{\hat{p}(\mathbf{X})}^{-1}(1-\alpha)\}} \right] \\ &= \frac{1}{\mathbb{P}[Y = 1]} \mathbb{E} \left[\mathbf{1}_{\{Y=1, \hat{p}(\mathbf{X}) > F_{\hat{p}(\mathbf{X})}^{-1}(1-\alpha)\}} \right] \\ &= \mathbb{P} \left[\hat{p}(\mathbf{X}) > F_{\hat{p}(\mathbf{X})}^{-1}(1-\alpha) \mid Y = 1 \right] \\ &= 1 - F_{\hat{p}(\mathbf{X})|Y=1} \left(F_{\hat{p}(\mathbf{X})}^{-1}(1-\alpha) \right). \end{aligned}$$

This corresponds to formula (5.2) in Tasche [17].

For the Gini index in ML we need to calculate the denominator of (3.1). However, this formula only applies for a continuous distribution F_Y of Y . In the case of a discrete distribution of Y we need to modify (3.1). Starting from the right-hand side of (3.6), we define the empirical function in the discrete case by

$$\alpha \in (0, 1) \mapsto \widehat{\text{CAP}}_{Y,Y}^-(\alpha) = \frac{1}{\frac{1}{n} \sum_{i=1}^n Y_i} \frac{1}{n} \sum_{i=1}^{\lceil \alpha n \rceil} Y_{(i)},$$

for i.i.d. data $(Y_i, \mathbf{X}_i)$, $1 \leq i \leq n$. In the Bernoulli case, this function is identically equal to zero up to $\alpha \leq 1 - \sum_{i=1}^n Y_i/n$, these describes the number of zeros among the observations $(Y_i)_{1 \leq i \leq n}$, and afterwards it increases to 1. Since this increase is only described on the discrete grid with span $1/n$, we linearly interpolate between these points. This provides a straight line between $1 - \sum_{i=1}^n Y_i/n$ and 1 with slope $n / \sum_{i=1}^n Y_i$. Under this linear interpolation, we get the area (integral)

$$\int_0^1 \widehat{\text{CAP}}_{Y,Y}^-(\alpha) d\alpha = \frac{1}{2n} \sum_{i=1}^n Y_i.$$

By the law of large numbers, the latter converges to $p^\dagger/2 = \mathbb{E}[p^\dagger(\mathbf{X})]/2 = \mathbb{E}[Y]/2$, a.s., as $n \rightarrow \infty$. This motivates in the (discrete) binary classification case the following definition of the Gini index in ML

$$G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}} = \frac{1/2 - \int_0^1 F_{\hat{p}(\mathbf{X})|Y=1} \left(F_{\hat{p}(\mathbf{X})}^{-1}(1-\alpha) \right) d\alpha}{(1 - p^\dagger)/2}. \quad (3.8)$$

In the binary classification case, the CAP can be related to the receiver operating characteristics (ROC) curve. The area under the curve (AUC) of the ROC curve has a one-to-one relationship to the Gini index in ML (3.8) in the Bernoulli case, we refer to Section 5 in Tasche [17]. We mention this because the ML community more frequently uses the AUC than the Gini index for model selection.

In general, in the discrete case we replace the integral in the denominator in (3.1) by the term

$$\frac{1}{4\mathbb{E}[Y]} \mathbb{E} \left[\left| Y - \tilde{Y} \right| \right], \quad (3.9)$$

where $\tilde{Y}$ is an independent copy of Y . This latter quantity (3.9) can be calculated for any distribution F_Y of Y , and in the continuous case we precisely receive the denominator in (3.1). The binary classification case (3.9) provides us with $(1 - p^\dagger)/2$ which gives (3.8). ■

4 Auto-calibration and consistency of the Gini index

Let $(Y, \mathbf{X}) \sim F$. A regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$ is auto-calibrated for Y if, a.s.,

$$\hat{\mu}(\mathbf{X}) = \mathbb{E}[Y | \hat{\mu}(\mathbf{X})] .$$

Auto-calibration is an important property in insurance pricing, as it implies that every cohort of insurance policies paying the same price $\hat{\mu}(\mathbf{X})$ is in average self-financing, because the price $\hat{\mu}(\mathbf{X})$ exactly covers the expected claim Y of that cohort. I.e., we do not have any systematic cross-financing between the price cohorts. This is the core of risk classification in insurance. It also implies unbiasedness on the portfolio level

$$\mathbb{E}[\hat{\mu}(\mathbf{X})] = \mathbb{E}[Y] , \quad (4.1)$$

which is a minimal requirement in insurance pricing. Typically, there are many auto-calibrated regression functions $\hat{\mu}(\mathbf{X})$ for Y , i.e., there are many systems of self-financing pricing cohorts.

Lemma 4.1 *The true regression function $\mathbf{X} \mapsto \mu^\dagger(\mathbf{X}) = \mathbb{E}[Y | \mathbf{X}]$ is auto-calibrated for Y , and it strictly dominates in convex order any other auto-calibrated regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$ for Y .*

Proof. To prove auto-calibration of $\mu^\dagger$ we apply the tower property to the σ -algebras $\sigma(\mu^\dagger(\mathbf{X})) \subset \sigma(\mathbf{X})$ which gives, a.s.,

$$\mathbb{E}[Y | \mu^\dagger(\mathbf{X})] = \mathbb{E}[\mathbb{E}[Y | \mathbf{X}] | \mu^\dagger(\mathbf{X})] = \mathbb{E}[\mu^\dagger(\mathbf{X}) | \mu^\dagger(\mathbf{X})] = \mu^\dagger(\mathbf{X}).$$

For any convex function ψ , auto-calibration, the tower property for $\sigma(\hat{\mu}(\mathbf{X})) \subset \sigma(\mathbf{X})$ and Jensen's inequality give

$$\begin{aligned} \mathbb{E}[\psi(\hat{\mu}(\mathbf{X}))] &= \mathbb{E}[\psi(\mathbb{E}[Y | \hat{\mu}(\mathbf{X})])] = \mathbb{E}[\psi(\mathbb{E}[\mathbb{E}[Y | \mathbf{X}] | \hat{\mu}(\mathbf{X})])] = \mathbb{E}[\psi(\mathbb{E}[\mu^\dagger(\mathbf{X}) | \hat{\mu}(\mathbf{X})])] \\ &\leq \mathbb{E}[\mathbb{E}[\psi(\mu^\dagger(\mathbf{X})) | \hat{\mu}(\mathbf{X})]] = \mathbb{E}[\psi(\mu^\dagger(\mathbf{X}))], \end{aligned}$$

whenever these exist. This proves that $\mu^\dagger$ dominates in convex order any other auto-calibrated regression function $\hat{\mu}$ for Y . Assume that there exists an auto-calibrated regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$ for Y such that for any convex function ψ we have an equality in the previous calculation, whenever these exist. This implies that $\mu^\dagger(\mathbf{X})$ is $\sigma(\hat{\mu}(\mathbf{X}))$ -measurable. Auto-calibration and the tower property for $\sigma(\hat{\mu}(\mathbf{X})) \subset \sigma(\mathbf{X})$ then provide, a.s.,

$$\hat{\mu}(\mathbf{X}) = \mathbb{E}[Y | \hat{\mu}(\mathbf{X})] = \mathbb{E}[\mathbb{E}[Y | \mathbf{X}] | \hat{\mu}(\mathbf{X})] = \mathbb{E}[\mu^\dagger(\mathbf{X}) | \hat{\mu}(\mathbf{X})] = \mu^\dagger(\mathbf{X}).$$

This proves the statement of strict convex order. $\square$

The next proposition is a consequence of Lemma 4.1 and of Theorem 3.1 in Krüger–Ziegel [12].

Proposition 4.2 *The true regression function $\mathbf{X} \mapsto \mu^\dagger(\mathbf{X})$ forecast-dominates any auto-calibrated regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$ for Y meaning that*

$$\mathbb{E}[-D_\psi(Y, \mu^\dagger(\mathbf{X}))] \geq \mathbb{E}[-D_\psi(Y, \hat{\mu}(\mathbf{X}))],$$

for any convex function ψ where the above exists, and with Bregman divergence given by

$$D_\psi(y, m) = \psi(y) - \psi(m) - \psi'(m)(y - m) \geq 0,$$

for $y, m \in \mathbb{R}$ and ψ' is a (sub-)gradient of the convex function ψ .

Proposition 4.2 says that every negative Bregman divergence provides a consistent scoring rule (2.2) for the conditional mean regression functional T under auto-calibration for Y . This statement motivates the common practice in model selection of minimizing (out-of-sample) deviance losses, as deviance losses are special cases of Bregman divergences; see Chapters 2 and 4 in Wüthrich–Merz [18]. For more information on this topic we refer to Krüger–Ziegel [12], Theorem 7 in Gneiting [10] and Savage [15], the latter two references state that Bregman divergences provide the only strictly consistent scoring functions for mean estimation.

The definition of the Gini index [9] in economics slightly differs from the ML version (3.1). Assume $F_{\hat{\mu}(\mathbf{X})}$ is a continuous distribution. It is then based on the *Lorenz curve* [13] given by

$$\alpha \in (0, 1) \mapsto L_{\hat{\mu}(\mathbf{X})}(F_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha)) = \frac{1}{\mathbb{E}[\hat{\mu}(\mathbf{X})]} \mathbb{E} \left[\hat{\mu}(\mathbf{X}) \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) \leq F_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha)\}} \right] \in [0, 1].$$

Note that we have the property $\text{CAP}_{\hat{\mu}(\mathbf{X}), \hat{\mu}(\mathbf{X})}^- = L_{\hat{\mu}(\mathbf{X})}(F_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha))$, see (3.5).

The *Gini index in economics* has many (equivalent)¹ definitions, we use the following two

$$G_{\hat{\mu}(\mathbf{X})}^{\text{eco}} = 1 - 2 \int_0^1 L_{\hat{\mu}(\mathbf{X})}(F_{\hat{\mu}(\mathbf{X})}^{-1}(\alpha)) d\alpha = \frac{1}{2\mathbb{E}[\hat{\mu}(\mathbf{X})]} \mathbb{E} \left[\left| \hat{\mu}(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \right| \right], \quad (4.2)$$

where $\hat{\mu}(\mathbf{Z})$ is an independent copy of $\hat{\mu}(\mathbf{X})$. The first definition in (4.2) is based on a continuous distribution $F_{\hat{\mu}(\mathbf{X})}$, whereas the second one can be used for any distribution $F_{\hat{\mu}(\mathbf{X})}$, we also refer to (3.9).

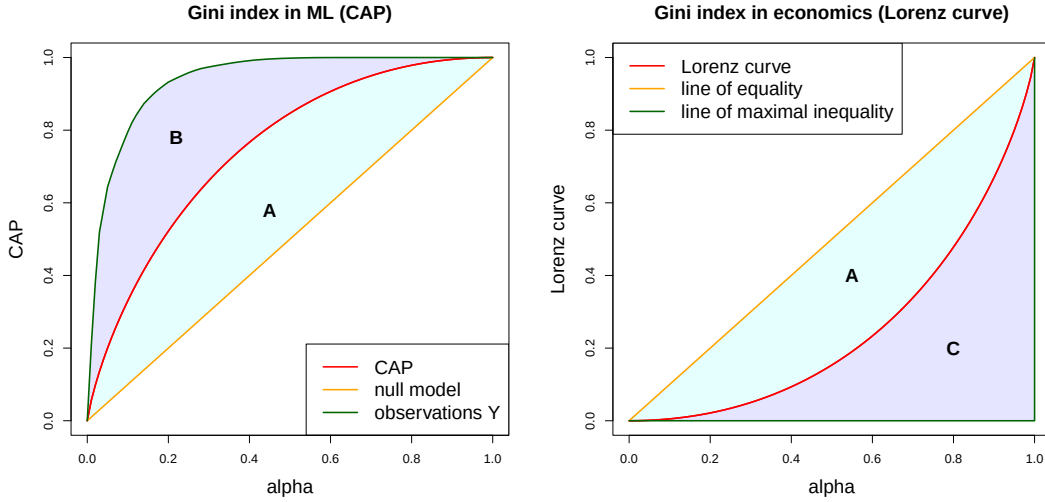


Figure 1: (lhs) cumulative accuracy profile (CAP) and (rhs) Lorenz curve.

There are three differences between the Gini index in ML and the one in economics, see Figure 1: (i) $G_{\hat{\mu}(\mathbf{X})}^{\text{eco}}$ considers a mirrored version of the curves compared to $G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}}$; (ii) $G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}}$ depends on Y and $\hat{\mu}(\mathbf{X})$, $G_{\hat{\mu}(\mathbf{X})}^{\text{eco}}$ only depends on $\hat{\mu}(\mathbf{X})$; (iii) scalings are different leading to areas B and C, respectively, in Figure 1. The two Gini indices are geometrically obtained by, see Figure 1,

$$G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}} = \frac{\text{area(A)}}{\text{area(A + B)}} \quad \text{and} \quad G_{\hat{\mu}(\mathbf{X})}^{\text{eco}} = \frac{\text{area(A)}}{\text{area(A + C)}} = 2 \text{area(A)} = 1 - 2 \text{area(C)}. \quad (4.3)$$

¹For an equivalence in (4.2) we need that $F_{\hat{\mu}(\mathbf{X})}$ is continuous, otherwise one should choose the term on the right-hand side as the definition of the Gini index in economics.

Property 3.1 of Denuit–Trufin [4] gives the following nice result.

Proposition 4.3 *Under auto-calibration of the regression function $\mathbf{X} \rightarrow \hat{\mu}(\mathbf{X})$ for Y we have the identity $\text{CAP}_{Y, \hat{\mu}(\mathbf{X})}(\alpha) = 1 - L_{\hat{\mu}(\mathbf{X})}(F_{\hat{\mu}(\mathbf{X})}^{-1}(1 - \alpha))$ for all $\alpha \in (0, 1)$.*

Proof. Using the tower property, auto-calibration of $\hat{\mu}$ for Y and unbiasedness (4.1) give us

$$\begin{aligned} \text{CAP}_{Y, \hat{\mu}(\mathbf{X})}(\alpha) &= \frac{1}{\mathbb{E}[Y]} \mathbb{E} \left[Y \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) > F_{\hat{\mu}(\mathbf{X})}^{-1}(1 - \alpha)\}} \right] = \frac{1}{\mathbb{E}[Y]} \mathbb{E} \left[\mathbb{E}[Y | \hat{\mu}(\mathbf{X})] \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) > F_{\hat{\mu}(\mathbf{X})}^{-1}(1 - \alpha)\}} \right] \\ &= \frac{1}{\mathbb{E}[\hat{\mu}(\mathbf{X})]} \mathbb{E} \left[\hat{\mu}(\mathbf{X}) \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) > F_{\hat{\mu}(\mathbf{X})}^{-1}(1 - \alpha)\}} \right] = 1 - \frac{1}{\mathbb{E}[\hat{\mu}(\mathbf{X})]} \mathbb{E} \left[\hat{\mu}(\mathbf{X}) \mathbb{1}_{\{\hat{\mu}(\mathbf{X}) \leq F_{\hat{\mu}(\mathbf{X})}^{-1}(1 - \alpha)\}} \right]. \end{aligned}$$

This proves the claim. $\square$

Thus, under auto-calibration for Y , the CAP and the Lorenz curve coincide (up to mirroring/sign switching). This gives us the following corollary.

Corollary 4.4 *Under auto-calibration of the regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$ for Y we have for the Gini indices*

$$G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}} = \frac{G_{\hat{\mu}(\mathbf{X})}^{\text{eco}}}{2 \int_0^1 \text{CAP}_{Y, Y}(\alpha) d\alpha - 1}. \quad (4.4)$$

Proof. Proposition 4.3 gives us for the Gini index in ML

$$\begin{aligned} G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}} &= \frac{\int_0^1 \text{CAP}_{Y, \hat{\mu}(\mathbf{X})}(\alpha) d\alpha - 1/2}{\int_0^1 \text{CAP}_{Y, Y}(\alpha) d\alpha - 1/2} = \frac{1/2 - \int_0^1 L_{\hat{\mu}(\mathbf{X})}(F_{\hat{\mu}(\mathbf{X})}^{-1}(1 - \alpha)) d\alpha}{\int_0^1 \text{CAP}_{Y, Y}(\alpha) d\alpha - 1/2} \\ &= \frac{1 - 2 \int_0^1 L_{\hat{\mu}(\mathbf{X})}(F_{\hat{\mu}(\mathbf{X})}^{-1}(1 - \alpha)) d\alpha}{2 \int_0^1 \text{CAP}_{Y, Y}(\alpha) d\alpha - 1} = \frac{1 - 2 \int_0^1 L_{\hat{\mu}(\mathbf{X})}(F_{\hat{\mu}(\mathbf{X})}^{-1}(\beta)) d\beta}{2 \int_0^1 \text{CAP}_{Y, Y}(\alpha) d\alpha - 1}, \end{aligned}$$

where the last step uses the change of variable $\alpha \mapsto \beta = 1 - \alpha$. This proves the claim. $\square$

This says that under auto-calibration for the response both Gini indices (the ML score and the version in economics) provide the same scoring rule because the (positive) denominator² in (4.4) does not depend on the specific choice of the regression function $\hat{\mu}(\cdot)$. Moreover, the same arguments apply to the Gini indices in non-continuous cases, e.g., in the binary classification (Bernoulli) case (3.8).

Theorem 4.5 *The true regression function $\mathbf{X} \mapsto \mu^\dagger(\mathbf{X})$ maximizes the Gini index (in ML) among all auto-calibrated regression functions $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$ for Y , i.e., $G_{Y, \mu^\dagger(\mathbf{X})}^{\text{ML}} > G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}}$ unless $\hat{\mu}(\mathbf{X}) = \mu^\dagger(\mathbf{X})$, a.s.*

Proof. Conditionally, given $\hat{\mu}(\mathbf{Z})$, $m \mapsto |m - \hat{\mu}(\mathbf{Z})|$ is a convex function in $m \in \mathbb{R}$. Using formula (4.2), independence between $\hat{\mu}(\mathbf{X})$ and $\hat{\mu}(\mathbf{Z})$ in (4.2) and Lemma 4.1 we obtain inequality, a.s.,

$$\mathbb{E} [|\hat{\mu}(\mathbf{X}) - \hat{\mu}(\mathbf{Z})| | \hat{\mu}(\mathbf{Z})] \leq \mathbb{E} \left[\left| \mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \right| \middle| \hat{\mu}(\mathbf{Z}) \right],$$

where $\mu^\dagger(\mathbf{X})$ is independent of $\hat{\mu}(\mathbf{Z})$. Using the tower property, applying the same argument to the exchanged role of $\mu^\dagger(\mathbf{X})$ and $\hat{\mu}(\mathbf{Z})$, using unbiasedness (4.1) and using Corollary 4.4 provides $G_{Y, \mu^\dagger(\mathbf{X})}^{\text{ML}} \geq G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}}$.

Assume there exists an auto-calibrated regression function $\hat{\mu}$ for Y such that $G_{Y, \mu^\dagger(\mathbf{X})}^{\text{ML}} = G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}}$. Using auto-calibration of $\hat{\mu}$ for Y and the tower property, we receive for $\mathbb{P}$ -a.e. $\omega \in \Omega$

$$\hat{\mu}(\mathbf{X})(\omega) = \mathbb{E}[Y | \hat{\mu}(\mathbf{X})](\omega) = \mathbb{E}[\mathbb{E}[Y | \mathbf{X}] | \hat{\mu}(\mathbf{X})](\omega) = \mathbb{E} \left[\mu^\dagger(\mathbf{X}) \middle| \hat{\mu}(\mathbf{X}) \right](\omega). \quad (4.5)$$

²Note that the denominator in (4.4) is positive for every non-deterministic Y . This follows from the fact that the denominator is equal to twice (3.9) which is positive unless Y is deterministic.

Denote by $\Omega_1 \subset \Omega$ a set of full measure 1 on which (4.5) holds. On Ω_1 , the predictor $\hat{\mu}(\mathbf{X})$ is between the conditional essential infimum and supremum of $\mu^\dagger(\mathbf{X})$, given $\hat{\mu}(\mathbf{X})$, because it corresponds to the conditional expectation of $\mu^\dagger(\mathbf{X})$, given $\hat{\mu}(\mathbf{X})$. Consider the case of sample points $\omega \in \Omega_1$ where the conditional essential infimum and supremum of $\mu^\dagger(\mathbf{X})$, given $\hat{\mu}(\mathbf{X})$, do not coincide, and denote the corresponding set of sample points by $\Omega_2 \subset \Omega_1$. On Ω_2 , the predictor $\hat{\mu}(\mathbf{X})$ is strictly between the conditional essential infimum and supremum of $\mu^\dagger(\mathbf{X})$, given $\hat{\mu}(\mathbf{X})$, due to the conditional expectation property (4.5). We have using (4.5) and independence between $\hat{\mu}(\mathbf{X})$ and $\hat{\mu}(\mathbf{Z})$

$$\begin{aligned} \mathbb{E} [|\hat{\mu}(\mathbf{X}) - \hat{\mu}(\mathbf{Z})|] &= \mathbb{E} \left[\left| \mathbb{E} \left[\mu^\dagger(\mathbf{X}) \middle| \hat{\mu}(\mathbf{X}) \right] - \hat{\mu}(\mathbf{Z}) \right| \right] = \mathbb{E} \left[\left| \mathbb{E} \left[\mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \middle| \hat{\mu}(\mathbf{X}), \hat{\mu}(\mathbf{Z}) \right] \right| \right] \\ &= \mathbb{E} \left[\left(\mathbb{1}_{\Omega_2} + \mathbb{1}_{\Omega_2^c} \right) \left| \mathbb{E} \left[\mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \middle| \hat{\mu}(\mathbf{X}), \hat{\mu}(\mathbf{Z}) \right] \right| \right]. \end{aligned} \quad (4.6)$$

We calculate the first term on the right-hand side of (4.6)

$$\mathbb{E} \left[\mathbb{1}_{\Omega_2} \left| \mathbb{E} \left[\mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \middle| \hat{\mu}(\mathbf{X}), \hat{\mu}(\mathbf{Z}) \right] \right| \right] = \int_{\Omega_2} \left(\int_{\Omega} \left| \mathbb{E} \left[\mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \middle| \hat{\mu}(\mathbf{X}), \hat{\mu}(\mathbf{Z}) \right] \right| (\omega, \tilde{\omega}) d\mathbb{P}(\tilde{\omega}) \right) d\mathbb{P}(\omega).$$

We study the inner integral for fixed sample point $\omega \in \Omega_2$. Jensen's inequality gives us

$$\int_{\Omega} \left| \mathbb{E} \left[\mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \middle| \hat{\mu}(\mathbf{X}), \hat{\mu}(\mathbf{Z}) \right] \right| (\omega, \tilde{\omega}) d\mathbb{P}(\tilde{\omega}) < \int_{\Omega} \mathbb{E} \left[\left| \mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \right| \middle| \hat{\mu}(\mathbf{X}), \hat{\mu}(\mathbf{Z}) \right] (\omega, \tilde{\omega}) d\mathbb{P}(\tilde{\omega}), \quad (4.7)$$

where we receive a strict inequality for $\omega \in \Omega_2$ because of the following items: (1) on Ω_2 , $\mu^\dagger(\mathbf{X})$ is non-deterministic, conditionally given $\hat{\mu}(\mathbf{X})$, (2) $m \mapsto |m - \hat{\mu}(\mathbf{Z})|$ is a convex function, (3) $\hat{\mu}(\mathbf{Z})$ has the same distribution (and support) as $\hat{\mu}(\mathbf{X})$, and (4) $\hat{\mu}(\mathbf{Z})$ and $(\mu^\dagger(\mathbf{X}), \hat{\mu}(\mathbf{X}))$ are independent. Items (1)-(4) imply that on a set of positive $\mathbb{P}(\tilde{\omega})$ -measure we receive a strict Jensen's inequality, because on this set, $\hat{\mu}(\mathbf{Z})$ is strictly within the conditional essential infimum and supremum of (the non-deterministic) $\mu^\dagger(\mathbf{X})$, given $\hat{\mu}(\mathbf{X})$.

Assume $\mathbb{P}[\Omega_2] > 0$, i.e., strict inequality (4.7) occurs on a set of positive measure. Applying Jensen's inequality also to the other term in (4.6) we receive strict inequality

$$\mathbb{E} [|\hat{\mu}(\mathbf{X}) - \hat{\mu}(\mathbf{Z})|] < \mathbb{E} \left[\left| \mathbb{E} \left[\mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \right] \middle| \hat{\mu}(\mathbf{X}), \hat{\mu}(\mathbf{Z}) \right| \right] = \mathbb{E} \left[\left| \mu^\dagger(\mathbf{X}) - \hat{\mu}(\mathbf{Z}) \right| \right].$$

This strict inequality contradicts our assumption $G_{Y, \mu^\dagger(\mathbf{X})}^{\text{ML}} = G_{Y, \hat{\mu}(\mathbf{X})}^{\text{ML}}$. Therefore, $\mathbb{P}[\Omega_2] = 0$, which implies

$$\mathbb{P}[\Omega_2^c \cap \Omega_1] = \mathbb{P}[\Omega_2^c] = 1.$$

On the set $\Omega_2^c \cap \Omega_1$, we have $\mu^\dagger(\mathbf{X}) = \hat{\mu}(\mathbf{X})$, which proves the claim. $\square$

Theorem 4.5 proves that the Gini index gives a strictly consistent scoring rule on the class of auto-calibrated regression functions that are $\mathbf{X}$ -measurable, because the true regression function $\mathbf{X} \mapsto \mu^\dagger(\mathbf{X})$ maximizes this Gini index. A bigger Gini index can only be achieved by a larger information set than the σ -algebra generated by $\mathbf{X}$.

The following proposition generalizes Property 5.1 of Denuit et al. [2], which gives a method of restoring auto-calibration for a general regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$.

Proposition 4.6 *Consider a regression function $\mathbf{X} \mapsto \hat{\mu}(\mathbf{X})$. The following regression function is auto-calibrated for Y*

$$\mathbf{X} \mapsto \hat{\mu}^{(\text{auto})}(\mathbf{X}) = \mathbb{E}[Y | \hat{\mu}(\mathbf{X})].$$

Proof. Note that $\hat{\mu}^{(\text{auto})}(\mathbf{X})$ is $\sigma(\hat{\mu}(\mathbf{X}))$ -measurable. This implies $\sigma(\hat{\mu}^{(\text{auto})}(\mathbf{X})) \subset \sigma(\hat{\mu}(\mathbf{X}))$. Henceforth, using the tower property, a.s.,

$$\mathbb{E}[Y | \hat{\mu}^{(\text{auto})}(\mathbf{X})] = \mathbb{E}[\mathbb{E}[Y | \hat{\mu}(\mathbf{X})] | \hat{\mu}^{(\text{auto})}(\mathbf{X})] = \mathbb{E}[\hat{\mu}^{(\text{auto})}(\mathbf{X}) | \hat{\mu}^{(\text{auto})}(\mathbf{X})] = \hat{\mu}^{(\text{auto})}(\mathbf{X}).$$

This completes the proof. $\square$

5 Conclusions

In general, one should not use the Gini index for model selection because it does not give a strictly consistent scoring rule and, thus, may lead to wrong decisions. We have shown in Theorem 4.5 that if we restrict Gini index scoring to the class of auto-calibrated regression functions for the given response, the Gini index allows for strictly consistent scoring. This also translates to the binary classification case where the (machine learning version of the) Gini index has an equivalent formulation in terms of the area under the curve (AUC) of the receiver operating characteristics (ROC) curve, we refer to Tasche [17]. We only need to ensure that the binary classification model is auto-calibrated for the Bernoulli response to receive a strictly consistent scoring rule from the AUC.

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A Comparison of the Most Commonly Used Measures of Association for Doubly Ordered Square Contingency Tables via Simulation

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Abstract

Spearman and Pearson correlation coefficient, Gamma coefficient, Kendall's tau-b, Kendall's tau-c, and Somers' d are the most commonly used measures of association for doubly ordered contingency tables. So far there has been no study expressing a priority on those measures of association. The aim of this study is to compare those measures of association for several types and different sample sizes of generated squared doubly ordered contingency tables and determine which measures of association are more efficient. It is found that both the sample sizes and the dimension of the doubly ordered contingency tables play a significant role on the effect of those measures of association.

1 Introduction

When categorical measures have a natural order (ex., strongly agree to strongly disagree; high, medium, low), additional information may be presented beside nominal variables. When there are two categorical variables that are both naturally ordered, a variety of effect size measures have been proposed for such ordinal data, including Gamma coefficient, Kendall's tau-b, Kendall's tau-c, and Somers' d (Garson, 2008).

An ordinal variable is also a type of a categorical variable. The only difference between the two is that there is a clear ordering of the ordinal variables, whereas there is no such ordering for ordinary categorical variables. For example, suppose you have a variable, patient's status, with three categories (worse, no difference and much better). In addition to being able to classify patients into these three categories, you can order the categories as worse, no difference and much better. Now think of a variable like educational background (with levels

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such as elementary school graduate, high school graduate, some college and university graduate). These also can be ordered as elementary school, high school, some college, and university graduate.

Even though the levels are ordered from lowest to highest, the distance between the levels need not to be the same across the levels of the variables. Suppose we assign scores for the levels of educational experience as 1, 2, 3 and 4 respectively and we compare the difference in education between levels one and two with the difference in educational experience between levels two and three, or the difference between levels three and four. The difference between levels one and two (elementary and high school) is perhaps much larger than the difference between categories two and three (high school and some college). In this example, we can order the people in level of educational experience but the size of the difference between levels is inconsistent (because the distance between levels one and two is larger than levels two and three) i.e the level of measuring is ordinal not interval (Ucla, 2007).

A doubly ordered categorical data or doubly ordered contingency tables are data with two variables that are both naturally ordered and cross tabulated. The most commonly and widely used measures of association for doubly ordered categorical data are measures of differences between probabilities of concordant and discordant pairs. Examples of these are Kendall's tau-b, Stuart's tau-c, Goodman-Kruskal's gamma, and Somers'd (Svensson, 2000). The difference among these measures lies in the power of overcoming of ties. One of the most well known non-parametric measures of association is called the Spearman rank-correlation ρ_s . Another famous measure of association is Kendall's tau which may be formulated as a Pearson product-moment correlation between signed indicators of X's and Y's, and Spearman's rank-correlation is the special case of the Pearson product-moment using the ranks instead of the actual variates the correlation with (Kruskal, 1958 and Hoeffding, 1948).

Kendall's tau which does not need to specify the ranking scores for both row and column and Somers' d coefficients are alternatives to Pearson's product-moment correlation coefficient and Spearman's rank-order correlation coefficient for ordinal data (Cyrus and Nitin, 1995).

2 The most commonly used measures of association

Spearman's rank-order correlation coefficient and Pearson's product-moment correlation coefficient, Goodman-Kruskal's gamma coefficient, Kendall's tau-b, Kendall's tau-c, and Somers' d are the most commonly used measures of association for doubly ordered contingency tables. This study was performed for the square doubly ordered contingency tables. What square term actually means is that the number of row categories equals to the number of column categories.

Notation

The following notations are used throughout this study:

X_i	Row variable arranged in ascending order: $X_1 < X_2 < \dots < X_R$
Y_j	Column variable arranged in ascending order: $Y_1 < Y_2 < \dots < Y_C$
f_{ij}	Frequency in row category i and column category j
c_j	$\sum_{i=1}^R f_{ij}$ - the subtotal of j -th column
r_i	$\sum_{j=1}^C f_{ij}$ - the subtotal of i -th row
W	$\sum_{j=1}^C c_j = \sum_{i=1}^R r_i$ - the general total of the sample size

2.1 Pearson correlation coefficient

Pearson Product Moment Correlation is the most widely and common used measure of correlation also called Pearson's correlation for short. The Pearson Product Moment correlation is represented by the Greek letter ρ (rho) when calculated from a population, whereas it is represented by the letter "r" if it is computed from a sample that is sometimes called "Pearson's r". Pearson's correlation reflects the degree of linear relationship between two variables. It varies from -1 to +1. A positive correlation means that as X and Y increases in the same direction. A correlation of +1 means that there is a perfect positive linear relationship between variables that is the degree of increment in X is proportional to the degree of increment of Y (Lane, 1997). A reverse explanation may be given for a -1 correlation.

Some assumptions are required and given below for the calculation of Pearson's product moment correlation r:

- Significant linear relationship between X and Y variables
- X and Y are continuous random variables
- Both variables must be normally distributed

There is a relationship between simple linear regression and Pearson's correlation coefficient. The main difference is that the variables used for the calculations are treated as response and explanatory for simple linear regression whereas there is no such discrimination for the Pearson's correlation. The square of r is called the goodness of fit or coefficient of determination and denotes the portion of total variance explained by the simple linear regression model.

The formula of Pearson's product moment correlation r may be given as in the definition (2.1),

$$r = \frac{\mathbf{cov}(\mathbf{X}, \mathbf{Y})}{\sqrt{\mathbf{S}(\mathbf{X})\mathbf{S}(\mathbf{Y})}} \equiv \frac{\mathbf{S}}{\mathbf{T}} \quad (2.1)$$

where $\mathbf{cov}(\mathbf{X}, \mathbf{Y})$ which is given below in equation (2.2) is also called the covariance of X and Y

$$\mathbf{cov}(\mathbf{X}, \mathbf{Y}) = \sum_{i,j} \mathbf{X}_i \mathbf{Y}_j \mathbf{f}_{ij} - \left(\sum_{i=1}^R \mathbf{X}_i \mathbf{r}_i \right) \left(\sum_{j=1}^C \mathbf{Y}_j \mathbf{c}_j \right) / \mathbf{W} \quad (2.2)$$

$\mathbf{S}(\mathbf{X})$ which is presented in equation (2.3) is also called the variance of X

$$\mathbf{S}(\mathbf{X}) = \sum_{i=1}^R \mathbf{X}_i^2 \mathbf{r}_i - \left(\sum_{i=1}^R \mathbf{X}_i \mathbf{r}_i \right)^2 / \mathbf{W} \quad (2.3)$$

and $\mathbf{S}(\mathbf{Y})$ in (2.4) is the variance of Y

$$\mathbf{S}(\mathbf{Y}) = \sum_{j=1}^C \mathbf{Y}_j^2 \mathbf{c}_j - \left(\sum_{j=1}^C \mathbf{Y}_j \mathbf{c}_j \right)^2 / \mathbf{W} \quad (2.4)$$

The variance of r is

$$\mathbf{var}_1 = \frac{1}{\mathbf{T}^4} \sum_{i,j} \mathbf{f}_{ij} \left\{ \mathbf{T}(\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{Y}_j - \bar{\mathbf{Y}}) - \frac{\mathbf{S}}{2\mathbf{T}} [(\mathbf{X}_i - \bar{\mathbf{X}})^2 \mathbf{S}(\mathbf{Y}) + (\mathbf{Y}_j - \bar{\mathbf{Y}})^2 \mathbf{S}(\mathbf{X})] \right\}^2 \quad (2.5)$$

If the null hypothesis which is “ $\mathbf{H}_0 : \rho = 0$ ” against the alternative hypothesis which is either “ $\mathbf{H}_1 : \rho \neq 0$ ” or “ $\mathbf{H}_1 : \rho > 0$ ” or “ $\mathbf{H}_1 : \rho < 0$ ” is true, the variance of r may be presented as in (2.6),

$$\mathbf{var}_0 = \frac{\sum_{i,j} \mathbf{f}_{ij} \mathbf{X}_i^2 \mathbf{Y}_j^2 - \left(\sum_{i,j} \mathbf{f}_{ij} \mathbf{X}_i \mathbf{Y}_j \right)^2 / \mathbf{W}}{\left(\sum_i \mathbf{r}_i \mathbf{X}_i^2 \right) \left(\sum_j \mathbf{c}_j \mathbf{Y}_j^2 \right)} \quad (2.6)$$

where $\bar{\mathbf{X}} = \sum_{i=1}^R \mathbf{X}_i \mathbf{r}_i / \mathbf{W}$ and $\bar{\mathbf{Y}} = \sum_{j=1}^C \mathbf{Y}_j \mathbf{c}_j / \mathbf{W}$ are the mean of X and the mean of Y respectively. Under the null hypothesis that there is no correlation,

$$t_{\text{calculated}} = \frac{r\sqrt{W-2}}{\sqrt{1-r^2}} \quad (2.7)$$

statistics has a t distribution with $W - 2$ degrees of freedom.

2.2 Spearman rank correlation coefficient

Calculating the Pearson's correlation coefficient needs the assumption that the two samples are normally distributed. If the assumption of normality is violated, Pearson's correlation coefficient will produce unreliable results. Hence a very best alternative for Pearson's correlation coefficient may be the use of Spearman's rank correlation r_s which can be calculated under the first assumption of Pearson's product moment correlation (Lohninger, 1999). There is no need of satisfaction of the second and third assumptions of the Pearson's product moment correlations for the use of Spearman rank correlation. Dependency of the ordinal variables is denoted as a rank correlation and their intensity is expressed by correlation coefficients. One of the most used ordinal coefficients is Spearman's correlation coefficient (Rezankova, 2009). The Spearman's rank correlation coefficient r_s is computed by using rank scores R_i for X_i and rank scores C_j for Y_j . These rank scores are defined as follows:

$$R_i = \sum_{k < i} r_k + (r_i + 1)/2 \quad \text{for } i = 1, 2, \dots, R \quad (2.8)$$

$$C_j = \sum_{h < j} c_h + (c_j + 1)/2 \quad \text{for } j = 1, 2, \dots, C \quad (2.9)$$

The formulas for r_s can be obtained from the Pearson formula given in (2.1) by substituting R_i and C_j for X_i and Y_j , respectively. And its asymptotic variance of the Spearman correlation can be obtained under the null hypothesis of no correlation from the formula presented in (2.6) by substituting R_i and C_j for X_i and Y_j , respectively.

$$r_s = \frac{\text{cov}(R, C)}{\sqrt{S(R)S(C)}} \equiv \frac{S}{T} \quad (2.10)$$

If there are no ties, another simple formula for obtaining Spearman's rank correlation is given in (2.11) as follows:

$$r_s = 1 - \frac{6 \sum d_i^2}{W(W^2 - 1)} \quad (2.11)$$

Where d_i in Spearman's rank correlation coefficient represents the difference in the ranks assigned to the values of the variable for each item of the certain data. When W is fairly small, the computation of the formula is very straightforward. In case of numerically equal observations an arithmetic average of the rank numbers associated with the ties are assigned to the values of the variables. This formula of Spearman's rank correlation coefficient is applied in cases when there are no tied ranks. When there are tied ranks the formula in (2.11) is not algebraically equivalent to the formula in (2.10). However, when there are a reasonable number of ties in the pairs of values of the variables, this approximation of Spearman's rank correlation coefficient is often used as fairly good approximations.

The Spearman's rank correlation coefficient may be used to test for association between both ordinals and continues variables. The underlying relationship between variables must be monotonic. In other words, generally speaking, the variables should either increase in values together, or when one gets increased, and then the other should get decreased.

Some difficulties of calculating Spearman's rank correlation coefficient arise, when the sample is large. For large data it can be hard to rank the data for both variables and consequently it is time consuming to perform Spearman's rank correlation coefficient test.

Since Spearman's rank correlation coefficient is a non parametric test, it does not depend upon the assumptions given for the Pearson's product moment correlation coefficient. Hence it is distribution free. It can be used to test whether there is a statistically significant association between variables. The null hypothesis we are testing is that there is no association between the variables under study. Thus, the main purpose of Spearman's rank correlation coefficient is to investigate the existence of any association in the underlying variables. To this end, the null hypothesis is constructed as having no rank correlation between the variables while using Spearman's rank correlation coefficient. Under the null hypothesis that there is no correlation,

$$t_{\text{calculated}} = \frac{r_s \sqrt{W-2}}{\sqrt{1-r_s^2}} \quad (2.12)$$

statistics has a t distribution with $W - 2$ degrees of freedom (Kendall and Stuart, 1973).

2.3 Goodman and Kruskal gamma (γ or G)

The Gamma (γ) statistics is proposed in a series of papers from 1954 to 1972 by Leo Goodman and William Kruskal. It is now mostly described just as Gamma that is used to investigate an association in a given doubly ordered contingency table.

The estimator of gamma uses only the number of concordant and discordant pairs of observations. It ignores tied pairs. In other words, pairs of observations that have equal values of X and equal values of Y are called tied pairs. Gamma can be calculated for only when both variables lie on an ordinal scale. It has the range $-1 \leq \gamma \leq 1$ just as Spearman's rank correlation coefficient. If there is no association between the two variables, then the estimator of gamma should be close to zero. The estimation of Gamma (γ) may be given as follows:

$$\gamma = \frac{P - Q}{P + Q} \quad (2.13)$$

where P has the form as $P = \sum_{i,j} f_{ij} C_{ij}$ and it is the probability that a randomly selected pair of observations will place in the same order and Q has the form as $Q = \sum_{i,j} f_{ij} D_{ij}$ and it is the probability that a randomly selected pair of observations will place in the opposite order, where f_{ij} is the frequency of i -th row and j -th column of the doubly order contingency table, C_{ij} is $\sum_{k>i} \sum_{l>j} f_{kl} + \sum_{k<i} \sum_{l<j} f_{kl}$ and D_{ij} is $\sum_{k>i} \sum_{l<j} f_{kl} + \sum_{k<i} \sum_{l>j} f_{kl}$. Its general standard error may be given as follows:

$$ASE_1 = \frac{4}{(P + Q)^2} \sqrt{\sum_{i,j} f_{ij} (QC_{ij} - PD_{ij})^2} \quad (2.14)$$

Under the null hypothesis of independence or no association, its standard error becomes as follows:

$$ASE_0 = \frac{2}{(P + Q)} \sqrt{\sum_{i,j} f_{ij} (C_{ij} - D_{ij})^2 - \frac{1}{W} (P - Q)^2} \quad (2.15)$$

For 2×2 tables, gamma is equivalent to Yule's Q which may be presented as follows (Goodman and Kruskal, 1979; Agresti, 2010; Brown and Benedetti, 1977b);

$$Q = \frac{f_{11}f_{22} - f_{12}f_{21}}{f_{11}f_{22} + f_{12}f_{21}} \quad (2.16)$$

Gamma coefficient can also be calculated for even small or perhaps for zero frequency of a 2×2 table.

Suppose that we have a value of gamma to be .582. It can be inferred that knowing the independent variable reduces our errors in predicting the rank (not

value) of the dependent variable by 58.2%. Under statistical independence, gamma will be zero, but there are some other times in which gamma coefficient may be zero whenever the number of concordant equal to the number of discordant. Meanwhile, using gamma coefficient a perfect association is present whenever the number of discordant pairs is zero. Under the null hypothesis that there is no correlation,

$$Z_{\text{calculated}} = \frac{\hat{\gamma}}{ASE_0} \quad (2.17)$$

statistics has standard normal distribution.

2.4 Kendall's Tau-b

Kendall's tau-b (τ_b) is similar to gamma except that tau-b uses a correction for ties. The rule of both variables lie on an ordinal scale for calculation Tau-b is just the same as gamma coefficient. Tau-b has also the range $-1 \leq \tau_b \leq 1$ as both gamma and Spearman's rank correlation. It is estimated by,

$$\tau_b = \frac{P - Q}{\sqrt{D_r D_c}} \quad (2.18)$$

where D_r stands for $W^2 - \sum_{j=1}^c r_j^2$ and r_j is the total count or the total frequency of row i in the doubly ordered cross table, D_c stands for $W^2 - \sum_{j=1}^c c_j^2$ and c_j is the total count or the total frequency of column j in the doubly ordered cross table. Its general standard error may be obtained as follows:

$$ASE_1 = \frac{1}{(D_r D_c)} \sqrt{\sum_{i,j} f_{ij} (2\sqrt{D_r D_c} (C_{ij} - D_{ij}) + \tau_b v_{ij})^2 - W^3 \tau_b^2 (D_r + D_c)^2} \quad (2.19)$$

where v_{ij} is defined as $r_i D_r + c_j D_c$. Under the null hypothesis of independence or no association, the standard error takes its form as follows:

$$ASE_0 = 2 \sqrt{\frac{\sum_{i,j} f_{ij} (C_{ij} - D_{ij})^2 - \frac{1}{W} (P - Q)^2}{D_r D_c}} \quad (2.20)$$

and under the null hypothesis of independence the asymptotic test statistics has standard normal distribution which is given as,

$$Z_{\text{calculated}} = \frac{\tau_b}{\text{ASE}_0} \quad (2.21)$$

The test statistics given in (2.21) is used to test whether the degree of association of the cross tabulations when both variables are measured in ordinal scale is significant (Kendall, 1955; Brown and Benedetti, 1977a; SAS, 2010).

It adjusts the ties and is most appropriate for square tables what means that the number of row categories equals to the number of column categories. Value of -1 is 100% negative association or perfect inversion whereas value of $+1$ is 100% positive association, or perfect agreement. A value of zero indicates no association.

If $\tau_b = \pm 1$ then there is no ties and subjects from different cells form strict concordant and discordant pairs in these two extreme cases. When both $\tau_b = \pm 1$ and $\gamma = \pm 1$, it is generally concluded that τ_b is stronger than γ . If $\tau_b = 1$, then the table is diagonal and if $\tau_b = -1$, the table is skewed diagonal (Tu, 2007).

2.5 Kendall's Tau-c

Stuart's tau-c (τ_c) makes an adjustment for table size as well as a correction for ties. Tau-c is also appropriate only when both variables lie on an ordinal scale. Tau-c has the range $-1 \leq \tau_c \leq 1$ as well as Spearman's rank correlation, Gamma and Tau-b. It is estimated by

$$\tau_c = \frac{q(P-Q)}{W^2(q-1)} \quad (2.22)$$

where q is defined as $\min(R, C)$. Its general standard error may be written as follows:

$$\text{ASE}_1 = \frac{2q}{(q-1)W^2} \sqrt{\sum_{i,j} f_{ij}(C_{ij} - D_{ij})^2 - \frac{1}{W}(P-Q)^2} \quad (2.23)$$

Under the null hypothesis of no association ASE_1 is identical to ASE_0 . Therefore the test statistics which may be used to investigate the degree of association for two ordinal variables under the null hypothesis of no association can be expressed as

$$Z_{\text{calculated}} = \frac{\tau_c}{\text{ASE}_0} \quad (2.24)$$

where $Z_{\text{calculated}}$ statistics has standard normal distribution. Besides making adjustments for ties it is most suitable for rectangular tables. Value of -1 is 100% negative association or perfect inversion whereas value of $+1$ is 100% positive association, or perfect agreement. A value of zero indicates no association (Brown and Benedetti, 1977a; SAS, 2010).

Kendall's tau-c, also called Stuart's tau-c or Kendall-Stuart tau-c, is a special case of tau-b for larger tables. It also makes adjustments for the size of the cross table (Lohninger, 1999).

2.6 Somers' d

Somers' $d(C|R)$ and Somers' $d(R|C)$ are asymmetric modifications of tau-b. $C|R$ represents that the row variable X is treated as an independent variable, whereas the column variable Y is treated as dependent. Similarly, $R|C$ represents the reverse interpretation. Somers' d differ from tau-b in that it only makes a correction for tied pairs on the independent variable. Somers' d can be calculated only when both variables are ordered. It varies in the range $-1 \leq d \leq 1$. Formulas for Somers' d is obtained according to the position of independent variable. For instance, if the row variable X is treated to be independent then Somers' d can be calculated as

$$d_{Y/X} = \frac{P - Q}{D_r} \quad (2.25)$$

and its general standard error is defined as below:

$$\text{ASE}_1 = \frac{2}{D_r^2} \sqrt{\sum_{i,j} f_{ij} \{D_r (C_{ij} - D_{ij}) - (P - Q)(W - R_i)\}^2} \quad (2.26)$$

or, under the null hypothesis of independence its standard error may be written as:

$$\text{ASE}_0 = \frac{2}{D_r} \sqrt{\sum_{i,j} f_{ij} (C_{ij} - D_{ij})^2 - \frac{1}{W} (P - Q)^2} \quad (2.27)$$

by interchanging the roles of X and Y , the formulas for Somers' d with X as the dependent variable can be obtained with only a minor change in the denominator by replacing D_r with D_c .

If both variables are ignored to be either independent or dependent, symmetric version of Somers' d is appropriate and it is calculated as follows:

$$d_{\text{symetric}} = \frac{(P - Q)}{\frac{1}{2}(D_c + D_r)} \quad (2.28)$$

and its standard error is simplified as follows:

$$ASE_1 = \frac{2\sigma_{\tau_b}^2}{(D_r + D_c)} \sqrt{D_r D_c} \quad (2.29)$$

where $\sigma_{\tau_b}^2$ is the variance of Kendall's τ_b . Under the null hypothesis of no association its standard error may be obtained as follows:

$$ASE_0 = \frac{4}{(D_c + D_r)} \sqrt{\sum_{i,j} f_{ij} (C_{ij} - D_{ij})^2 - \frac{1}{W} (P - Q)^2} \quad (2.30)$$

Somers' d value of -1 is 100% negative association or perfect inversion whereas value of $+1$ is 100% positive association, or perfect agreement (Somers, 1962; Goodman and Kruskal, 1963; Liebetrau, 1983; SAS, 2010).

A value of zero indicates no association. Under the null hypothesis of independence, the following statistics asymptotically has standard normal distribution

$$Z_{\text{calculated}} = \frac{d_{\text{symetric}}}{ASE_0} \quad (2.31)$$

3 Generation of doubly ordered contingency table

In order to generate a doubly ordered contingency table, there are lots of techniques in the literature of Statistical simulation. For instance, a doubly ordered contingency table may be generated from the uniform association model (Agresti, 2010). In our study we present a new way of generating a doubly ordered contingency table using bivariate standard normal distribution. In the first step we generate two identically independently distributed random variables, as $X_1 \sim N(0,1)$ and $X_2 \sim N(0,1)$. To generate two random variables (X and Y) from the bivariate normal distribution with certain correlation (ρ) for a specific sample size, we apply the followings:

$$\mathbf{X} = \mathbf{aX}_1 + \mathbf{bX}_2 \quad (3.1)$$

$$\mathbf{Y} = \mathbf{bX}_1 + \mathbf{aX}_2 \quad (3.2)$$

where $\mathbf{a}^2 + \mathbf{b}^2 = 1$ and $2\mathbf{ab} = \rho$, and hence $\mathbf{a}$ and $\mathbf{b}$ are obtained as $\mathbf{a} = \frac{\sqrt{1+\rho} + \sqrt{1-\rho}}{2}$ and $\mathbf{b} = \frac{\sqrt{1+\rho} - \sqrt{1-\rho}}{2}$.

To generate two random variables for certain correlation from the bivariate normal distribution, $\mathbf{a}$ and $\mathbf{b}$ are calculated and presented in Table 1.

Table 1: For specific correlations the values of $\mathbf{a}$ and $\mathbf{b}$.

ρ	$\mathbf{a}$	$\mathbf{b}$
0	1	0
0.5	0.9659258263	0.25881190451
0.9	0.8473163206	0.5310885546

If we would like to generate a doubly ordered contingency table for a certain number of rows $\mathbf{R}$ and certain number of column $\mathbf{C}$, say $\mathbf{R} \times \mathbf{C}$ table, we split the range of generated data for $\mathbf{X}$ variable into $\mathbf{R}$ sub equal intervals and for $\mathbf{Y}$ variable into $\mathbf{C}$ sub equal intervals. And then we recode the variables into new variables according to the sub equal intervals. How we do that is quite simple. The recoding is performed for instance if a datum falls into the first interval then its recode value is 1, for general if it falls into i -th interval then its recode value is i and so on. An example of generating 4×4 doubly ordered contingency table for 100 sample size when there is no correlation has been given step by step below.

Table 2 presents both the generated data from the uncorrelated bivariate standard normal distribution for the sample size 100 and the recoded new variables according to the subintervals presented in Table 3.

Table 3 presents the subintervals of each variable and their code values. For instance the range of the generated data from $\mathbf{X}$ is -2.69058 for lower bound and 2.97257 for upper bound. This range has been split into four equal subintervals as (-2.69058;-1.27479) for the first subinterval and (-1.27479;0.14100) for the second subinterval and so on.

Table 4 presents the generated doubly ordered contingency table that is obtained by cross tabulating the $\mathbf{X}$ coded and $\mathbf{Y}$ coded variable presented in Table 3.

Table 2: Uncorrelated X and Y from the bivariate normal distribution with their codes.

NO	X	X coded	Y	Y coded	NO	X	X coded	Y	Y coded
1	-0.2536	2	-0.7163	2	51	1.0623	3	0.0464	3
2	-1.0397	2	-0.2695	2	52	1.5019	3	-0.3173	2
3	-1.0620	2	-0.7119	2	53	1.0088	3	0.4799	3
4	-0.1434	2	-0.2043	2	54	-0.6655	2	-1.5755	1
5	0.5554	3	-0.6441	2	55	0.4410	3	-0.5301	2
6	-0.9337	2	0.5972	3	56	-0.2411	2	-0.2260	2
7	-0.6532	2	-1.3105	1	57	-0.6315	2	-1.6726	1
8	-0.0312	2	-0.5532	2	58	1.2348	3	-0.0832	2
9	0.1456	3	-0.7766	2	59	2.9726	4	-1.1554	2
10	-0.1454	2	-1.7343	1	60	-0.2394	2	-0.0542	2
11	-0.5058	2	0.6856	3	61	-0.1062	2	-0.1400	2
12	-1.2511	2	1.1864	3	62	0.9309	3	0.0107	3
13	-0.5854	2	-0.4971	2	63	0.2709	3	-0.2638	2
14	0.9921	3	0.1856	3	64	-0.5009	2	0.5375	3
15	0.4573	3	1.0416	3	65	0.7381	3	-1.2461	1
16	-0.8605	2	-1.3636	1	66	0.0861	2	-0.1343	2
17	0.3545	3	1.0972	3	67	-0.2575	2	-1.3048	1
18	0.4592	3	-0.7049	2	68	-0.1921	2	-0.0969	2
19	0.0779	2	0.1284	3	69	-0.9413	2	1.6775	4
20	-1.2302	2	0.1972	3	70	0.8649	3	1.5616	4
21	-1.6566	1	-0.6006	2	71	-0.3182	2	0.1286	3
22	-0.3763	2	2.3653	4	72	2.1642	4	-1.5743	1
23	0.1518	3	0.4474	3	73	1.4203	3	-1.3141	1
24	2.8291	4	0.1628	3	74	-0.8289	2	-2.4796	1
25	-0.2974	2	0.3574	3	75	-1.7606	1	0.8185	3
26	-1.7357	1	0.2520	3	76	0.7911	3	-0.6351	2
27	-1.3409	1	-1.2586	1	77	-0.9899	2	0.7008	3
28	0.0192	2	1.2798	4	78	-0.3139	2	-0.7316	2
29	-2.6906	1	0.5039	3	79	1.5227	3	0.1013	3
30	1.0547	3	1.3173	4	80	-1.5821	1	1.2279	3
31	-0.5241	2	-1.1634	2	81	-1.8928	1	0.2019	3
32	-0.1484	2	0.0275	3	82	-0.9889	2	0.4336	3
33	-0.3196	2	0.1960	3	83	0.5171	3	-0.3009	2
34	1.7460	4	0.1461	3	84	1.3806	3	0.4284	3
35	1.2623	3	0.5115	3	85	0.7285	3	-0.6359	2
36	0.0753	2	-1.8280	1	86	0.4379	3	-1.4755	1
37	1.0775	3	-0.9509	2	87	0.2492	3	-0.6649	2
38	-0.5648	2	0.5449	3	88	0.1890	3	-0.7737	2
39	-0.8080	2	1.0685	3	89	1.7847	4	0.4966	3
40	-1.4783	1	0.1946	3	90	-0.3535	2	-0.6291	2
41	-0.0221	2	0.4106	3	91	0.1400	2	-1.4845	1
42	-1.2005	2	-0.5804	2	92	-1.8057	1	2.4985	3
43	-0.1499	2	-0.0788	2	93	-1.0551	2	-0.4196	2
44	0.0909	2	-0.2291	2	94	-1.6230	1	0.0321	3
45	0.8470	3	0.4373	3	95	0.4015	3	-1.4419	1
46	-1.3933	1	0.2019	3	96	0.0687	2	-0.8363	2
47	-0.7364	2	-1.4237	1	97	-0.3731	2	-1.6131	1
48	0.2612	3	1.2662	4	98	0.2875	3	-2.0094	1
49	-1.2693	2	-0.4225	2	99	-0.3149	2	0.7019	3
50	0.7355	3	0.1085	3	100	0.6373	3	-0.8100	2

Table 3: The sub intervals of each variable and their code values.

X				Y			
	Lover Bound	Upper Bound	Code		Lover Bound	Upper Bound	Code
Interval 1	-2.69058	-1.27479	1	Interval 1	-2.47961	-1.23508	1
Interval 2	-1.27479	0.14100	2	Interval 2	-1.23508	0.00946	2
Interval 3	0.14100	1.55679	3	Interval 3	0.00946	1.25399	3
Interval 4	1.55679	2.97257	4	Interval 4	1.25399	2.49853	4

Table 4: The generated 4x4 doubly ordered contingency table.

		Y				
		1	2	3	4	Total
X	1	1	1	9	0	11
	2	11	20	16	3	50
	3	5	14	12	3	34
	4	1	1	3	0	5
Total		18	36	40	6	100

4 Simulation study

The simulation work has been designed in terms of sample size, table dimension and degree of association. For a fair degree of ordinal association, the correlation between variables generated has been declared as 0.5 and for a strong degree of ordinal association it is declared to be 0.9. Seven different square table dimensions which are 3x3, 4x4, 5x5, 6x6, 7x7, 8x8, 9x9 have been generated for each correlation. Also for each correlation and table dimension, eight different sample sizes which are 50, 100, 150, 200, 250, 500, 750, 1000 are studied. For each correlation, table dimension and sample size, the process has been repeated 10000 times. The comparisons have been made according to the mean of 10000 replications of the degree of ordinal measure of associations.

Since it is not easy to judge the results recorded in tables, it is decided to perform line plots. Therefore the results obtained are presented in line plots to clarify the effect of both the sample size and the table dimension. Actually the results are presented in two types of line plots. The first type has been performed to investigate the effect of table dimension, whereas the second has been performed to investigate the effect of sample size. For instance the line plots are presented in Figure 1 ($\rho=0.5$) and Figure 2 ($\rho=0.9$) give an idea of how the table dimension affects the result of each of the expected mean of ordinal measure of association. Figure 3 ($\rho=0.5$) and Figure 4 ($\rho=0.9$) give an idea of how the sample size affects the result of each of the expected mean of ordinal measure of association.

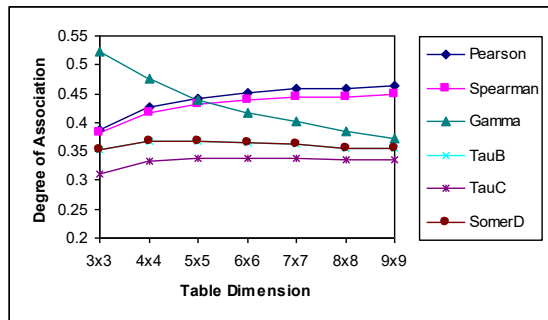


Figure 1a: Table dimension against degree of the ordinal measure of associations for $\rho=0.5$ and $n=50$.

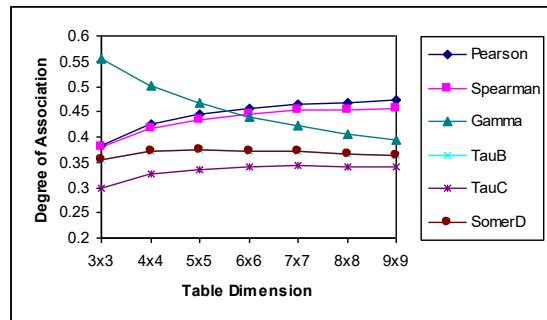


Figure 1b: Table dimension against degree of the ordinal measure of associations for $\rho=0.5$ and $n=100$.

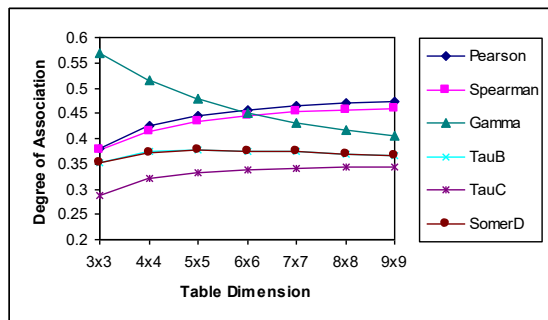


Figure 1c: Table dimension against degree of the ordinal measure of associations for $\rho=0.5$ and $n=150$.

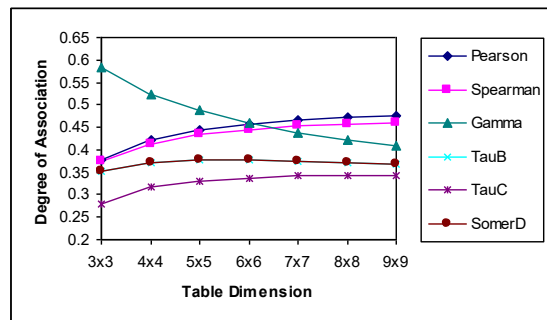


Figure 1d: Table dimension against degree of the ordinal measure of associations for $\rho=0.5$ and $n=200$.

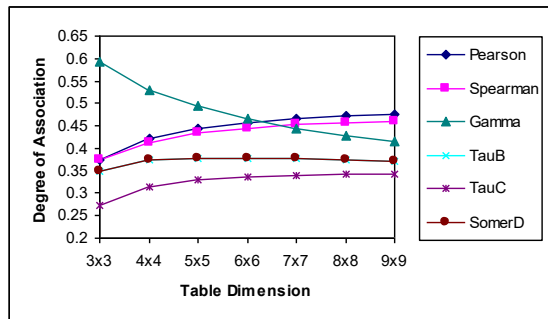


Figure 1e: Table dimension against degree of the ordinal measure of associations for $\rho=0.5$ and $n=250$.

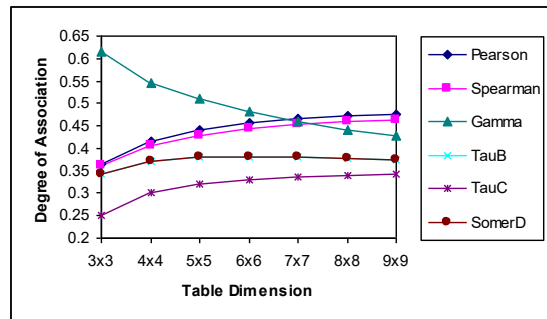


Figure 1f: Table dimension against degree of the ordinal measure of associations for $\rho=0.5$ and $n=500$.

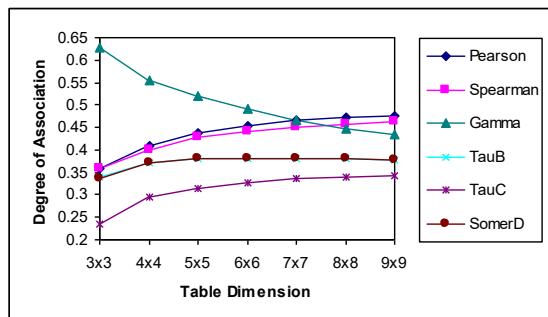


Figure 1g: Table dimension against degree of the ordinal measure of associations for $\rho=0.5$ and $n=750$.

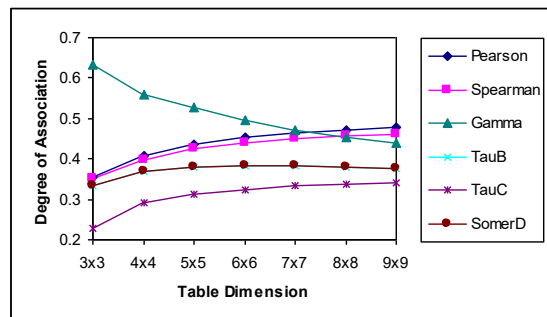


Figure 1h: Table dimension against degree of the ordinal measure of associations for $\rho=0.5$ and $n=1000$.

Figure 1: Table dimension against the mean of the ordinal measure of associations $\rho=0.5$ and sample size $n=50, 100, 150, 200, 250, 500, 750, 1000$.

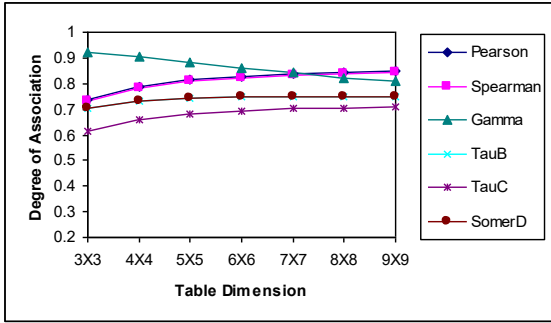


Figure 2a: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=50$.

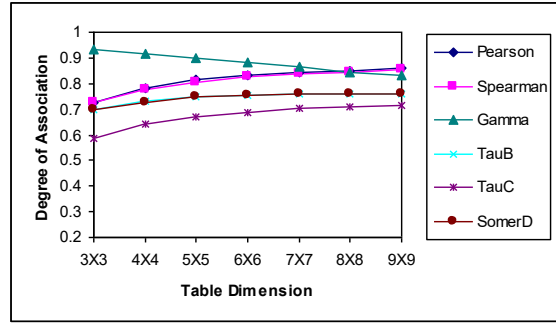


Figure 2b: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=100$.

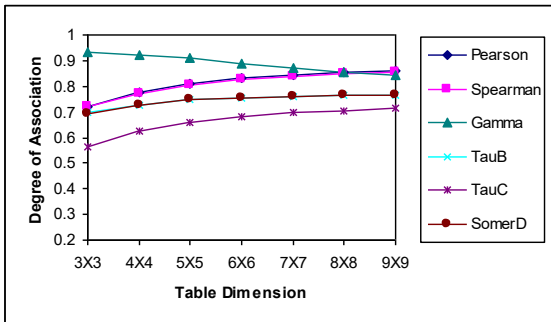


Figure 2c: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=150$.

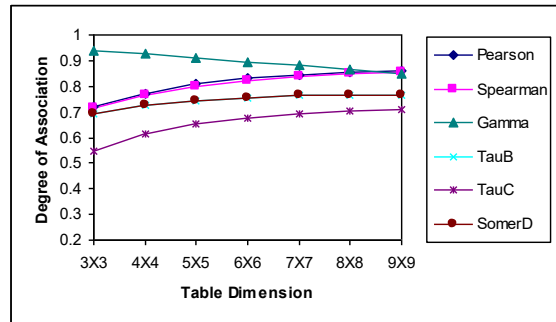


Figure 2d: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=200$.

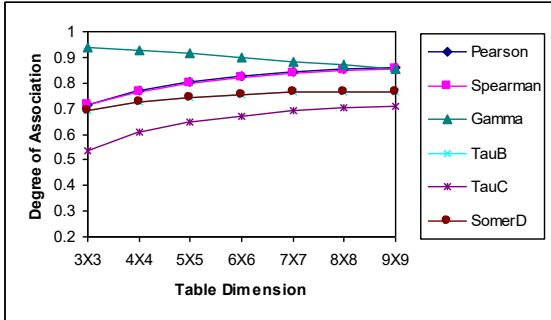


Figure 2e: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=250$.

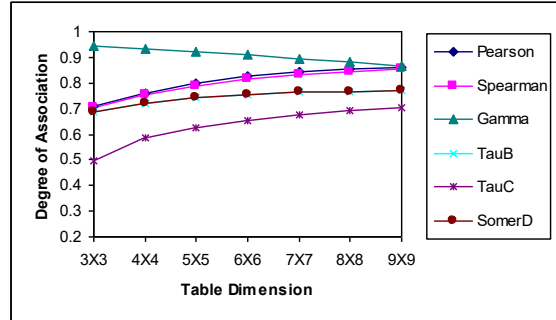


Figure 2f: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=500$.

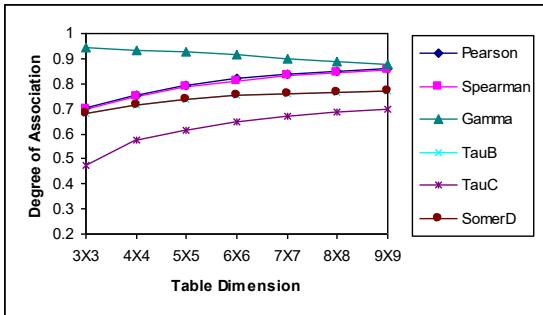


Figure 2g: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=750$.

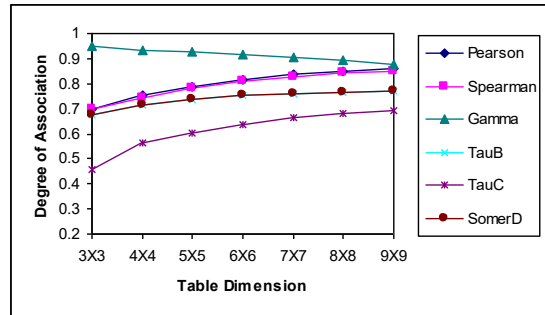


Figure 2h: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=1000$.

Figure 2: Table dimension against degree of the ordinal measure of associations for $\rho=0.9$ and $n=50, 100, 150, 200, 250, 500, 750, 1000$.

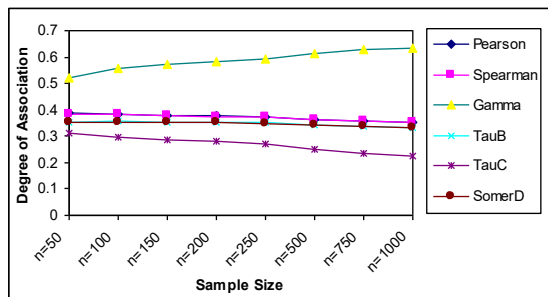


Figure 3a: Sample size against degree of the ordinal measure of associations for $\rho = 0.5$ and 3x3 Figure.

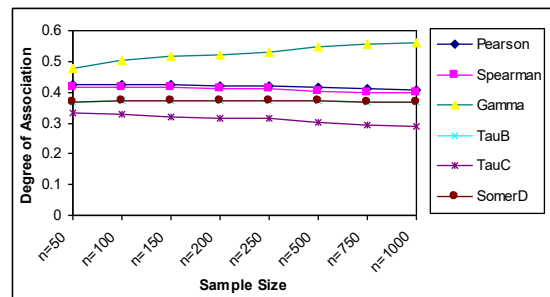


Figure 3b: Sample size against degree of the ordinal measure of associations for $\rho = 0.5$ and 4x4 Figure.

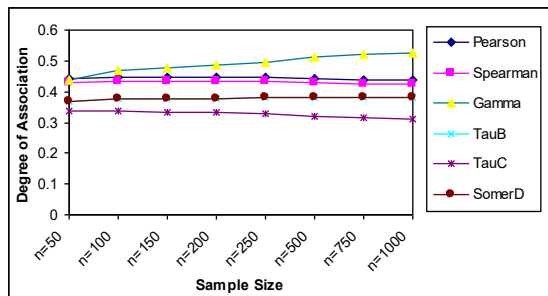


Figure 3c: Sample size against degree of the ordinal measure of associations for $\rho = 0.5$ and 5x5 Figure.

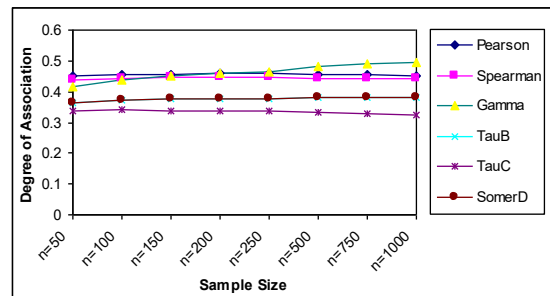


Figure 3d: Sample size against degree of the ordinal measure of associations for $\rho = 0.5$ and 6x6 Figure.

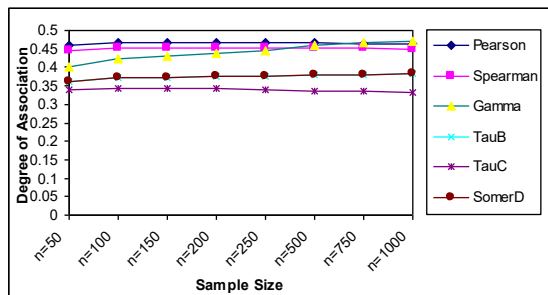


Figure 3e: Sample size against degree of the ordinal measure of associations for $\rho = 0.5$ and 7x7 Figure.

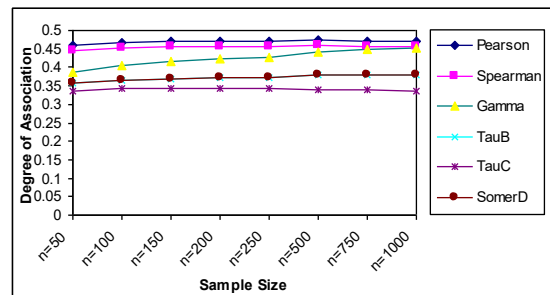


Figure 3f: Sample size against degree of the ordinal measure of associations for $\rho = 0.5$ and 8x8 Figure.

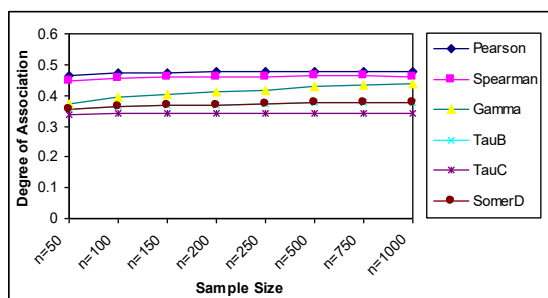


Figure 3g: Sample size against degree of the ordinal measure of associations for $\rho = 0.5$ and 9x9 Figure.

Figure 3: Sample size against degree of the ordinal measure of associations for $\rho = 0.5$ and 3x3, 4x4, 5x5, 6x6, 7x7, 8x8, 9x9.

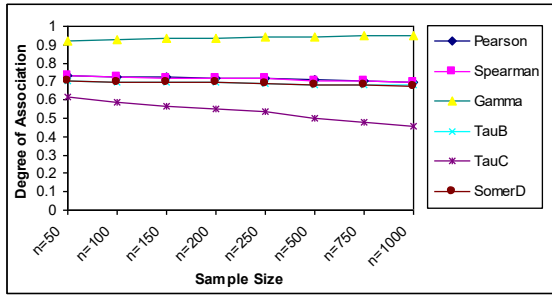


Figure 4a: Sample size against degree of the ordinal measure of associations for $\rho=0.9$ and 3x3 Figure.

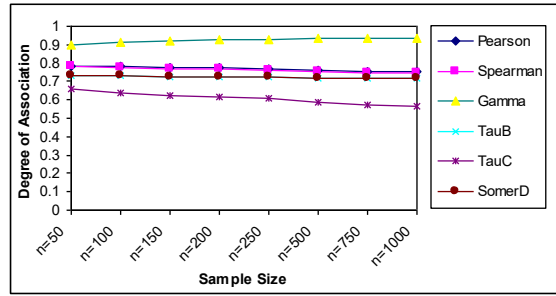


Figure 4b: Sample size against degree of the ordinal measure of associations for $\rho=0.9$ and 4x4 Figure.

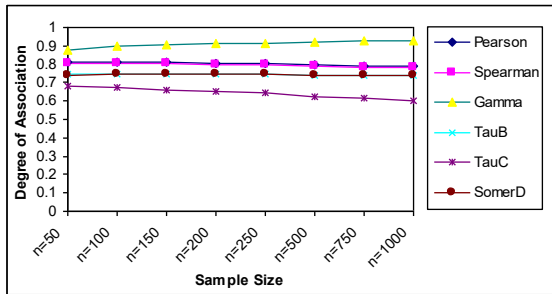


Figure 4c: Sample size against degree of the ordinal measure of associations for $\rho=0.9$ and 5x5 Figure.

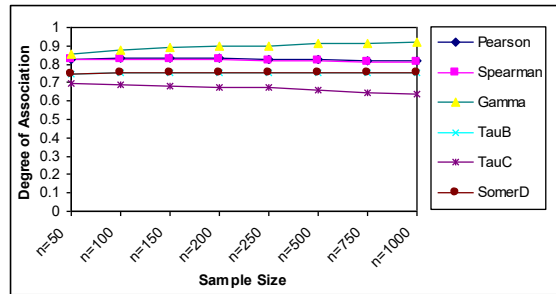


Figure 4d: Sample size against degree of the ordinal measure of associations for $\rho=0.9$ and 6x6 Figure.

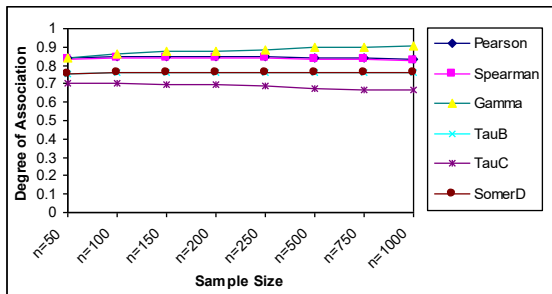


Figure 4e: Sample size against degree of the ordinal measure of associations for $\rho=0.9$ and 7x7 Figure.

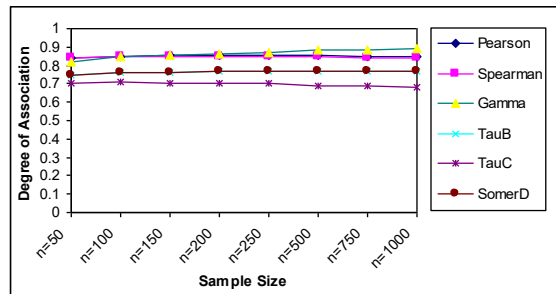


Figure 4f: Sample size against degree of the ordinal measure of associations for $\rho=0.9$ and 8x8 Figure.

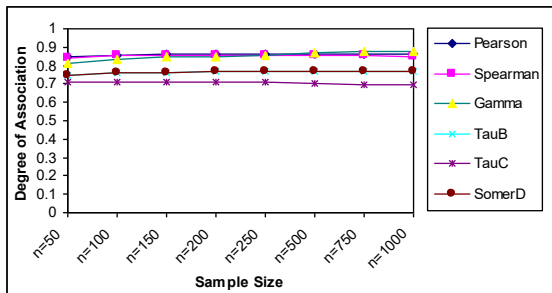


Figure 4g: f Sample size against degree of the ordinal measure of associations or $\rho=0.9$ and 9x9 Figure.

Figure 4: Sample size against degree of the ordinal measure of associations for $\rho=0.9$ and 3x3, 4x4, 5x5, 6x6, 7x7, 8x8, 9x9.

5 Results and remarks

As can be seen from Figure 1a for Table dimension 3x3 and 4x4 Gamma ordinal measure of association presents closer estimate of the degree of association which is expected to be 0.5 in comparison with the other ordinal measures. However it can be concluded from the same figure that Pearson's correlation and Spearman's rank correlation presents a good estimate of the actual degree of association as table dimension increases. In the meantime, Kendall's Tau-b, Kendall's Tau-c and Somers' d for such a small sample present a poor estimate of the actual degree of ordinal association in average for any table dimension. As table dimension increases Gamma coefficient decreases too.

When the sample size increases from 50 to 100 sizes, the results are presented in Figure 1b. A similar line plot has clearly been observed. There is not much difference at all. In fact, when the sample size increases to a fair size, for small table dimensions Gamma overestimate the actual the degree of ordinal association. Pearson's correlation and Spearman's rank correlation always tend to increase to a much better estimation as table dimension increases. Although as table dimension increases, Kendall's Tau-b, Kendall's Tau-c and Somers' d for such a fair and large sample present a slight different estimate of the actual degree of association. They are all underestimating the actual degree of ordinal measure of association. Meanwhile for large samples Kendall's Tau-c is the worst no matter how large the table dimension is.

Increasing the actual degree of association from 0.5 to 0.9 does not make any difference in the results obtained. The picture in Figure 2 shows that the lines representing the ordinal measure of association are shifted up by the amount of the increment of the actual degree of association.

Figure 3 has been prepared from the same results of Figure 1 except that the horizontal axis represents the sample size. The aim of drawing the picture in Figure 3 is to investigate the effect of sample size. It can be seen from Figure 3a that is drawn for the table dimension 3x3 as the sample size increases the Gamma ordinal measure of association not only increases but also overestimates the actual degree of association. However as the sample size increases Pearson's correlation, Spearman's rank correlation, Kendall's Tau-b, Kendall's Tau-c and Somers' d slightly decrease and underestimate the actual degree of association for the 3x3 contingency table in average. The picture in Figure 3b that is drawn for the table dimension 4x4 shows no significant difference in comparison with the 3x3 contingency table. When the results obtained for the rest of the contingency table dimensions are analyzed, it is found that there is no much difference at all except a little shift up. To this end, when the table dimension is large enough, Pearson's correlation and Spearman's rank correlation tend to present a better estimate of the actual degree of ordinal measure of association no matter what the sample size is, but the other ordinal measure of associations except Gamma always underestimate

the actual degree of association. Gamma tends to increase as the sample size increases. Figure 4 presents similar results of the results presented in Figure 3 apart from that the actual degree of association has changed from 0.5 to 0.9. There cannot be observed any different results in comparison with Figure 3.

6 Conclusions and discussion

As the number of table dimension increases, no matter what the sample size of the contingency table is Pearson's correlation and Spearman rank correlation coefficients of the generated tables from the bivariate standard normal distribution increase in average, but they still slightly underestimate the actual degree of ordinal measure of association.

Although Kendall's Tau-b, Tau-c and Somers' d increase as the table dimension increases, those measures of association always underestimate the actual degree of association of the generated tables.

Pearson's correlation is slightly larger than Spearman correlation. Those two measures are good at fairly large dimensional doubly ordered tables.

Gamma coefficient is good when the table dimension is small for relatively small sample sizes. It increases and overestimates as the sample size increases for any certain type of table dimension. In overall, for square tables Gamma presents the best estimation of the actual degree of the association in average.

There is another measure of association called polychoric correlation which has not been included in this study. Since that measure of association uses iterative methods which are hard to estimate the actual degree of association, it has been excluded from the study. However, a new study may be performed with this correlation to make a comparison with the ordinal measure of associations that are presented in our study.

A future study may be performed to develop a new measure of ordinal association that is free of both table dimension and sample size.

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Credit Scores: Performance and Equity*

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Abstract

Credit scores are critical for allocating consumer debt in the United States, yet little evidence is available on their performance. We benchmark a widely used credit score against a machine learning model of consumer default and find significant misclassification of borrowers, especially those with low scores. Our model improves predictive accuracy for young, low-income, and minority groups due to its superior performance with low quality data, resulting in a gain in standing for these populations. Our findings suggest that improving credit scoring performance could lead to more equitable access to credit.

JEL Codes: C45; D14; E27; E44; G21; G24.

Keywords: Consumer default; credit scores; machine learning; equity; fairness.

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1 Introduction

Credit scores are key in the allocation of consumer credit in the United States. Lenders report the status of consumer loans to three credit bureaus— private corporations that collect administrative data on most consumer loans.¹ Using proprietary models, these credit scores rank borrowers based on their likelihood of future default, providing a single, standardized measure of individual risk. The Fair Isaac Corporation introduced the first credit score in 1958 and launched the widely used FICO score in 1989. The FICO score and the VantageScore, introduced in 2006, are the most commonly used.² When consumers apply for a new loan, lenders check their credit score. While the credit score may not be the only input in the underwriting decision, the amount of credit and the interest rate mostly depend on the consumer’s credit score, even for secured debts, such as mortgages and vehicle loans (Edelberg (2006)). The emergence of algorithmic underwriting of consumer loans based on credit scores is seen as a key factor in the growth of consumer debt in the past three decades (Avery et al. (2003)).

Despite their ubiquitous use in the financial industry, there is very little information on credit scores with no performance metrics publicly available. Emerging evidence suggests that currently used credit scores have severe limitations. The Consumer Financial Protection Bureau estimates that 11% of consumers are unscored, and therefore excluded from conventional credit markets, with these borrowers concentrated among young, low-income, and minority populations (Brevoort, Grimm, and Kambara (2016)). Additionally, Albanesi, DeGiorgi, and Nosal (2022) show that during the 2007-2009 housing crisis, there was a marked rise in mortgage delinquencies among high credit score borrowers, suggesting that credit scoring models at the time did not accurately reflect the probability of default for

¹Credit reporting started in a very basic form in the 1880s and developed to a comprehensive nationwide system in the 1930s (Lauer (2017)). The Fair Credit Reporting Act of 1970 systematized the information that can be collected in credit reports and determined which financial entities are subject to mandatory reporting Hunt (2005).

²There are also several other scores that are product specific, for example, for vehicle loans or credit cards (see CFPB (2012)). Credit scoring models are also updated regularly and change over time. More information on credit scores is reported in Section 4 and Appendix B.

these borrowers.

We assess the performance of a widely used credit score by developing an alternative scoring model based on machine learning with substantially better predictive performance. We find that credit scores misclassify 41% of consumers by placing them in a risk category that does not align with their actual default probability. The misclassification is more severe for consumers with low credit scores. For example, 47% of Subprime and 70% of Near Prime borrowers are misclassified, while only 26% of Superprime borrowers are. We show that credit score performance is worse for young, low-income, and minority borrowers and our model improves performance most for these borrowers. Our model also improves the ranking of these marginalized borrowers, which could contribute to a fairer and more equitable distribution of consumer credit.

The machine learning model of consumer default that we deploy to benchmark credit score performance uses the same information as standard credit scoring models and is designed for environments with high-dimensional data and complex non-linear patterns of interaction among variables.³ Our model targets the same default outcome as conventional credit scoring models: a 90+ days delinquency within the subsequent eight quarters, and strictly uses only the information in credit reports that is permitted under current legislation. Since credit scores provide only an ordinal ranking of consumers based on their default risk, we use only the ordinal ranking of consumers with respect to their predicted probability of default based on our model.

Our model performs significantly better than conventional credit scores. The average AUC score for the credit score is about 85%, but it drops notably during the 2007-2009 crisis, while the average AUC score for our model is approximately 91% and stable over time. Most importantly, the credit score generates significant disparities between the implied predicted probability of default and the realized default rate for large groups of borrowers, particularly at the low end of the credit score distribution. We show that, among borrowers with a

³For excellent reviews of how machine learning can be applied in economics, see Mullainathan and Spiess (2017) and Athey and Imbens (2019).

Subprime credit score, who comprise 21% of the population, 22% display default behavior consistent with Near Prime borrowers, and 15% display default behavior consistent with Deep Subprime. The realized default rates for Deep Subprime, Subprime, and Near Prime borrowers are 68%, 44%, and 22%, respectively. This suggests that the credit score is unable to differentiate between borrowers with substantially different default risks. By contrast, the discrepancy between predicted and realized default rates for our model is never more than five percentage points within any risk category. Our analysis points to severe limitations of conventional credit scores in their ability to differentiate consumers by default risk.

We use interpretability techniques to identify which factors are associated with variation in consumer default ranking for our model, and compare these to the credit score. By law, credit scoring companies have to reveal the four most important factors driving credit score variation, which are reported to be amounts owed, credit mix, incidence of new credit, and length of the credit history. Our model places more weight on amounts owed, which explains 49% of the variation in our model implied rankings, and only 30% of the variation in credit scores. By contrast, credit mix and the incidence of new credit, which are viewed as indicators of credit demand, each only account for 5% of the variation in rankings for our model, while they each account for 10% of the variation in credit scores. Additionally, the length of the credit history only accounts for 8% of variation in rankings for our model, but 15% of the variation in credit scores.

One standing concern with the adoption of machine learning-based scoring models is that a more sophisticated statistical technology might exacerbate disparities in access to credit for disadvantaged consumers such as young, low-income, or minority borrowers (see Fuster et al. (2018)). We show that, on the contrary, our model provides a more favorable risk assessment to young, low-income, and in most cases minority borrowers, particularly for those who do not default. This result likely stems from the property that credit demand factors and length of the credit history have a sizable negative impact on the conventional credit score, while they play a lesser role in default predictions made by our model. Young,

minority, and low-income borrowers, who often have short credit histories and high demand for credit, are more at risk of being placed lower in the ranking by the credit score. Additionally, we show that our machine learning model improves performance relative to the credit score more for young, low-income, and minority consumers compared to the rest of the population, and that this result is a function of its better ability to deliver accurate predictions when confronted with low quality data, such as thin files, low credit mix and a history of default, attributes that are prevalent among these traditionally marginalized populations. Contrary to Blattner and Nelson (2021), these findings suggest that a credit scoring algorithm based on machine learning can improve performance in the presence of data bias, benefitting vulnerable consumers.

Our analysis contributes to the literature on credit scores and disparities in credit access. Credit scores are viewed as a key tool for improving consumer credit market efficiency (Chatterjee, Corbae, and Rios-Rull (2011)), yet this role is based on the premise that they are an unbiased if imperfect signal of the variation in default risk amongst consumers. However, we show that widely used credit scores place too much weight on factors that are not strongly associated with default and misclassify a large fraction of consumers. Additionally, the statistical performance of the credit score is worse for populations historically marginalized on credit markets and these consumers would gain in standing if ranked by a better performing model. This implies that our default predictions could help improve credit allocation in a way that benefits both lenders, in the form of lower losses from default, and borrowers, in the form of more access to credit for vulnerable populations. Our results also speak to the perils associated with using conventional credit scores outside the consumer credit sphere. As it is well known, credit scores are used to screen job applicants, in insurance applications, and a variety of additional settings. Economic theory would suggest that this is helpful as long as the credit score provides information correlated with hidden characteristics of interest to the party using the score (Corbae and Glover (2018)). However, as we show, conventional credit scores misclassify borrowers by a substantial degree based on their default risk and

are biased against young, low-income, and, in most cases, minority borrowers, which implies that they may not be accurate and may not include appropriate information or use adequate methodologies. The expanding use of credit scores could amplify the economic disparities resulting from these limitations.

The remainder of this paper is structured as follows. Section 2 describes our data. Section 3 describes our prediction problem and our model. Section 4 compares our model to conventional credit scores. Section 5 investigates the trade-off between performance and equity.

2 Data

We use anonymized credit file data from the Experian credit bureau. The data is quarterly. It starts in 2004Q1 and ends in 2015Q4. The data comprises over 200 variables for a nationally representative panel of 1 million households, constructed with a random draw from the universe of borrowers with an Experian credit report. The data covers credit cards and other revolving credit, auto loans, installment loans, business loans, first and second mortgages, home equity lines of credit, student loans, and collections. There is information on the number of trades for each type of loan, the outstanding balance and available credit, the monthly payment, hard inquiries, and whether any of the accounts are in a state of delinquency. All balances are adjusted for joint accounts to avoid double counting. We also have each borrower’s credit score for each quarter in the sample. Because this is data drawn from credit reports, we do not know the gender, marital status, or any other demographic characteristic, though we know a borrower’s address at the zip code level. We also do not have any information on asset holdings. The data also includes an estimate of individual and household labor income based on IRS data.

Table 1 reports basic demographic information on our sample, including age, household income, credit score, and incidence of default, defined as the fraction of households who

Table 1: Descriptive Statistics

Feature	Mean	Std. Dev.	Min	25%	50%	75%	Max
Age	45.530	16.751	18.0	31.0	45.0	58.0	84.0
Household Income (Imputed)	78.331	55.092	15.0	43.0	64.0	91.0	316.0
Credit Score	679.317	108.475	300.0	595.0	691.0	776.0	839.0
Credit History (Months)	197.672	129.669	0.0	95.0	175.0	274.0	988.0
$\mathbb{I}_{\{90+ \text{ DPD Debt}\}}$	0.078	0.269	0.0	0.0	0.0	0.0	1.0
$\mathbb{I}_{\{90+ \text{ DPD Debt, next 8 Quarters}\}}$	0.184	0.387	0.0	0.0	0.0	0.0	1.0

Notes: Household income is in USD thousands, winsorized at the 99th percentile. Source: Authors' calculations based on Experian Data.

report 90 or more days past due delinquency on any trade, excluding collections. This will be our baseline definition of default, as it is the outcome targeted by credit scoring models.

3 Model

Credit scores are an ordinal measure that ranks consumers by the probability that they will default over a certain future horizon based on information in a borrowers' credit report. The definition of default used in this context is 90 days or more past due on any debt, and the horizon is 8 quarters. Given that credit scoring models are proprietary, little is known about how the models are constructed. The Fair Credit Reporting Act of 1970 and the Equal Opportunity in Credit Access Act of 1974, and their subsequent additions, regulate credit scores and in particular determine which information can be included and must be excluded in credit scoring models. Such models can incorporate only information in a borrower's credit report, except for age and location. These restrictions are intended to prevent discrimination by age and factors related to location, such as race.⁴ The law also mandates that entities that provide credit scores make public the four most important factors affecting scores. In marketing information, these are reported to be payment history, which is stated to explain about 35% of variation in credit scores, followed by amounts owed, length of credit history,

⁴Credit scoring models are also restricted by law from using information on race, color, gender, religion, marital status, salary, occupation, title, employer, employment history, nationality.

new credit and credit mix, explaining 30%, 15%, 10% and 10% of the variation in credit scores respectively.⁵ There is no mandated requirement to reveal any performance information on credit scores.⁶

To evaluate the performance of credit scores, we develop a model to predict consumer default based on machine learning. We then use the ordinal ranking of consumers based on their predicted probability of default as a credit score alternative and compare the performance of conventional credit scores to our model.

Predicting consumer default maps well into a supervised learning framework, one of the most widely used techniques in machine learning. In supervised learning, a learner takes in pairs of input/output data. The input data, typically a vector, represent preidentified attributes, also known as features, used to determine the output value. The supervised learning problem is referred to as a “regression problem” when the output is continuous and a “classification problem” when the output is discrete. Once the learner is presented with input/output data, its task is to find a function that maps the input vectors to the output values. The goal of supervised learning is to find a function that generalizes beyond the training set to forecast out-of-sample outcomes correctly. Adopting this machine-learning methodology, we build a model that predicts defaults for individual consumers.

We now formalize our prediction problem. We define default as a 90+ days delinquency on any debt in the subsequent 8 quarters, the outcome targeted by conventional credit scoring models. We adopt a discrete-time formulation for periods $0, 1, \dots, T$, each corresponding to a quarter. We let the variable D_t^i prescribe the state at time t for individual i with $D \subset \mathbb{N}$ denoting the set of states. We define $D_1^i = 1$ if a consumer is 90+ days past due on any trade and $D_1^i = 0$ otherwise. Consumers will transition between these two states quarter to

⁵See for example <https://www.myfico.com/credit-education/whats-in-your-credit-score>.

⁶The FACT Act of 2003 required the Federal Reserve Board to study some basic performance properties of credit cores. The findings are summarized in FRB (2007).

quarter. Our target outcome corresponds to:

$$Y_t^i = \begin{cases} 0 & \text{if } \sum_{n=t}^{t+7} D_n^i = 0 \\ 1 & \text{otherwise} \end{cases} \quad (1)$$

We posit that the dynamics of this process are influenced by a vector of explanatory variables $X_{t-1}^i \in \mathbb{R}^{d_x}$, which includes the previous realization of the state D_{t-1}^i . In our empirical implementation, X_{t-1}^i represents the set of features, which we will discuss below. We fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and an information filtration $(\mathcal{I}_t)_{(t=0,1,\dots,T)}$. Then, we specify a probability transition function $f_\theta : \mathbb{R}^{d_x} \rightarrow [0, 1]$ satisfying

$$\mathbb{P}[Y_t^i = y | \mathcal{I}_{t-1}] = f_\theta(X_{t-1}^i), y \in D \quad (2)$$

where θ is a vector of parameters to be estimated. Equation 2 gives the marginal conditional probability for the transition of individual i from state D_{t-1}^i at time $t-1$ to state y at time t given the explanatory variables X_{t-1}^i .⁷ The vector output of the function g is a probability distribution on D . Our model outputs a continuous variable between 0 and 1 that can be interpreted as an estimate of the probability of default for a particular borrower at a given time, given input variables from their credit reports.

Equation 2 defines a theoretical transition matrix for the default outcome we consider. Its empirical counterpart is reported in Table 2. The fraction of consumers in our sample in default according to this definition is 18.4%. The default and no-default state are very persistent, with a 85% of consumers in default and 87% of consumers not in default remaining in the same state in the subsequent quarter.

We adopt a deep learning model, as models in this class display high performance in settings with high volume data and multidimensional non-linear interactions.⁸ Specifically,

⁷The state y encompasses realizations of the state between time t and $t+7$.

⁸Sirignano, Sadhwani, and Giesecke (2018) and Fuster et al. (2022) apply these models to mortgage default risk.

Table 2: Y_t^i Frequency and Transitions

Frequency		Transition probability	No default	Default
Default	0.184	No default	0.8729	0.1271
No default	0.816	Default	0.1460	0.8540

Notes: Average default rates and average quarterly transition frequency 2006Q1-2016Q2. Source: Authors' calculations based on Experian Data.

we model the transition function f_θ with a hybrid deep neural network/gradient boosting model, which combines the predictions of a deep neural network and an extreme gradient boosting model. This hybrid approach is particularly well-suited to our prediction problem. The deep neural network excels at learning intricate patterns, while the gradient boosting model improves its predictive accuracy by correcting errors from previous iterations, making this combination the best-performing model in this class for our specific prediction problem and data (Albanesi and Vamossy (2019)). Appendix A explains each of the component models and their properties and the rationale for combining them.⁹

We implement our approach as follows. Given the 8-quarter horizon for the prediction problem, we allow for fully out-of-sample performance evaluation by separating our training/validation and testing data by 8 quarters. Additionally, we only use one quarter of data to train and validate our model. That is, we train and validate our model on quarter t data and then test it on quarter $t + 8$ data.¹⁰ For example, default predictions for quarter 2008Q2 will be generated with a model trained on 2006Q2 data. This allows us to provide an out of sample default prediction for every consumer with non-empty credit record in a given quarter for 2006Q1-2016Q2. We select a broad set of 79 features, listed in Table 13, in order to maximize the number of consumers with a prediction and cover the factors that are listed by conventional credit scoring models, such as credit mix, length of credit history, utilization,

⁹Albanesi and Vamossy (2019) extensively discusses alternative machine learning models and their performance properties for this prediction problem. Additionally, it explains the rationale for adopting this class of models.

¹⁰This type of temporal cross-validation approach within the credit scoring context was also adopted by Khandani, Kim, and Lo (2010).

Data	Training	Validation	Testing
Purpose	fitting	hyperparameter tuning	out of sample evaluation
Time	t	t random 80/20 split	$t + 8$

and new credit. Appendix A presents detailed performance metrics for our model.

4 Comparison with Credit Score

We now compare the performance of our model to a widely used conventional credit score. Our goal is to provide a conceptual assessment of credit score performance based on its economic role in consumer credit markets. We are not able to provide a comprehensive statistical comparison for two reasons. First, we do not know the precise attributes or the particular time frame used to build credit scoring models, as the law only requires disclosing basic features of those models (see Section 4.3). Second, there might be additional constraints that credit scoring companies impose on their models, driven by monotonicity and palatability requirements.¹¹ These requirements are imposed on credit scoring models to enhance transparency and facilitate equitable treatment of consumers. We apply explainability techniques to make our model transparent, and as we will show in Section 5, ranking consumers based on our model advances the standing of traditionally marginalized consumers. So, even while not explicitly imposing these constraints, our model satisfies them indirectly.

Since the credit score is an ordinal ranking of individuals with respect to default risk, we only use the ordinal ranking of consumers with respect to the probability of default implied by our model for our comparison with the credit score. Specifically, when comparing the ranking of consumers implied by our model and by the credit score, we proceed as follows. We utilize a credit score that is a whole number, ranging from 300 to 850, with higher values corresponding to lower default risk, which we segment into percentiles. Since our model’s

¹¹See <https://www.fico.com/blogs/trusted-ai-challenge-monotonicity-and-palatability> for a discussion.

output, $\hat{Y}(X)$, is a probability of default, we rank consumers by: $1 - \hat{Y}(X)$, segmenting the corresponding distribution into 100 equal-sized bins (percentiles).

Our analysis proceeds in three steps. We first evaluate the statistical performance of the credit score and our model using standard performance metrics. We then evaluate the ability of the credit score to place consumers in the standard risk profiles used in the industry, such as Subprime and Prime, and show that the credit score misclassifies a substantial fraction of borrowers, especially at the bottom of the credit score distribution. We also examine the factors that lead to different default predictions for the credit score and our model. Finally, we study the impact on vulnerable populations by age, income and race.

4.1 Statistical Performance

A common way to measure performance for conventional credit scoring models is the Gini coefficient, which measures the dispersion of the credit score distribution and, therefore, its ability to separate borrowers by their default risk. The Gini coefficient is related to a key performance metric for machine learning algorithms, the AUC score, with $Gini = 2 * AUC - 1$ so that we can compare the credit score performance to our model along this dimension. Figure 1 plots the Gini coefficient for the credit score and our model prediction over time. The Gini coefficient for our model is about 0.8 between 2006Q1 and 2008Q3 and then rises to 0.84 for the rest of the sample. For the credit score, the Gini coefficient is above 0.7 until 2011Q1, when it drops to approximately 0.69, then slowly recovers to 0.72, suggesting a drop in credit score performance in the aftermath of the Great Recession. Appendix B presents similar results for the Spearman and Kendall rank correlations. This analysis highlights the superior separating power of our model compared to the conventional credit score, particularly during periods of economic instability.

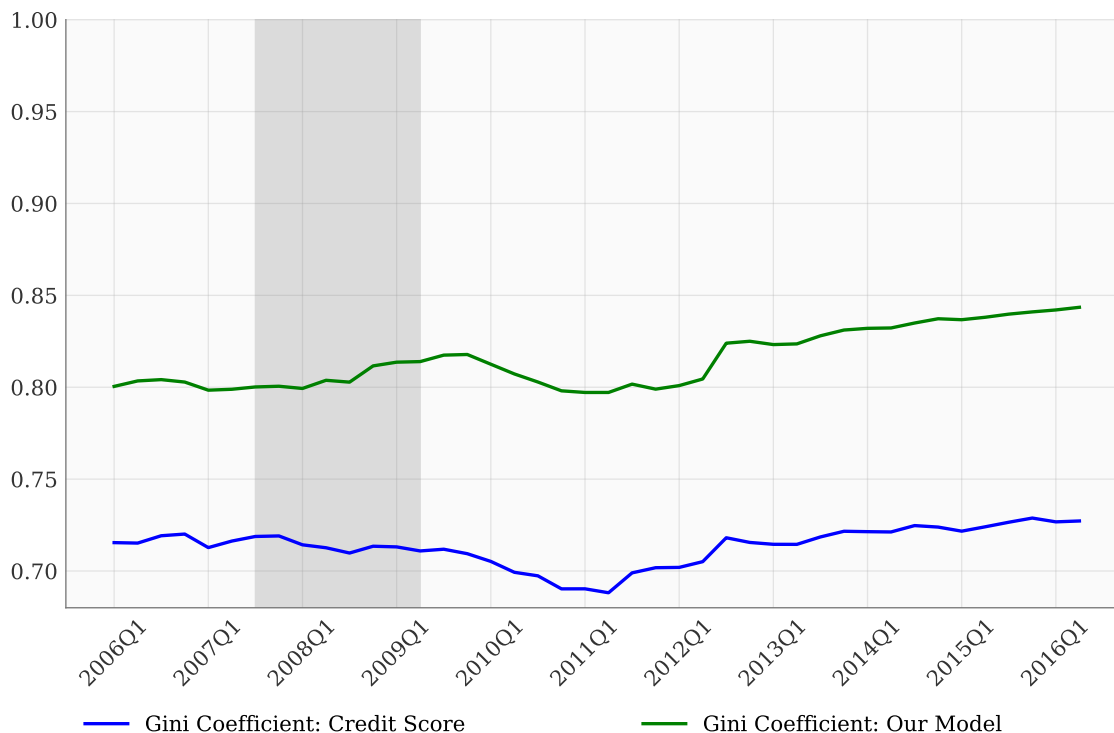


Figure 1: Gini Coefficient, Credit Score and Prediction Model

Notes: Gini coefficients for the credit score and our model's by quarter. Source: Authors' calculations based on Experian data.

4.2 Performance by Risk Profile

Credit scores are used to place borrowers into five industry defined risk profiles– Deep Subprime, Subprime, Near Prime, Prime, and Super Prime– that are critical for the determination of credit limits and interest rates on consumer loans. We now examine the performance of the credit score in relation to these risk profiles.

The credit score threshold levels that correspond to the risk profiles in descending order of default risk for the credit score are reported in Table 3. As shown in the table, these categories account for respectively, 6%, 20.1%, 14.0%, 36.3% and 23.6% of all borrowers. The thresholds that define these categories can be restated in percentiles of the credit score distribution, which are reported in the second row of the table. Based on this classification, we can create five corresponding risk profiles for the predicted probability of default based

on our model, with the lowest category containing the 6% of borrowers with the highest predicted default risk, corresponding to Deep Subprime, all the way to the highest category containing the 23.6% of all borrowers with the lowest predicted default risk, corresponding to Super Prime. We then calculate where borrowers in each credit score risk profile are placed by our model.

Table 3: Credit Score Risk Profiles

	Deep Subprime	Subprime	Near Prime	Prime	Super Prime
Credit score ranges	300-499	500-600	601-660	661-780	781-850
Credit score percentiles	0-6.0	6.01-27.10	27.11-41.10	41.11-77.40	77.41-100
Percent of borrowers	6.0	20.1	14.0	36.3	23.6

Notes: Industry-defined credit score risk profiles. Definition by credit score range, credit score percentiles. All rates, fractions, and shares in percentage. Total # of observations: 26,240,879. Time period 2006Q1-2016Q2. Source: Authors' calculations based on Experian Data.

The results are displayed in Table 4, with items on the diagonal showing the fraction of borrowers that our model would place in the same risk profile as the credit score, and off-diagonal items showing the fraction of borrowers in each credit score risk profile who are misclassified according to our model. The last column shows the overall percentage of borrowers who are misclassified by the credit score for each credit score risk profile.

The findings suggest that credit scores misclassify a significant portion of borrowers, especially those with lower credit scores. According to our model, only 45% of Deep Subprime borrowers would remain in that category, while 44% would be reclassified as Subprime, 9% as Near Prime, and 2% as Prime. For Subprime borrowers, our model places 53% in the Subprime category, with 15% reclassified as Deep Subprime and 22% as Near Prime. Among Near Prime borrowers, our model categorizes 38% as Subprime, 28% as Prime, and only 30% as Near Prime. In the Prime category, 67% of borrowers remain in that profile, while 13% are reclassified as Near Prime and 16% as Super Prime. Finally, for Super Prime borrowers, our model places 74% in that category, with 26% reclassified as Prime. The credit score classifies between 26% and 70% of borrowers into the wrong risk category, with higher misclassification rates for those with lower credit scores.

Table 4: Credit Score and Model Risk Profile Comparison

Credit Score	Model Based Risk Profile					Disagreement
	Deep Subprime	Subprime	Near Prime	Prime	Super Prime	
Deep Subprime	45.06	43.57	8.89	2.44	0.03	54.94
Subprime	14.67	52.71	22.46	10.07	0.08	47.29
Near Prime	1.66	38.05	30.35	28.08	1.86	69.65
Prime	0.17	4.43	12.83	66.50	16.07	33.50
Super Prime	0.02	0.05	0.15	25.71	74.08	25.92

Notes: Rows correspond to industry-defined credit score risk profiles, defined in Table 3. Columns are fractions in each corresponding model-based risk profile. Column “Disagreement” reports the fraction of borrowers in each credit score risk profile that would be placed in a different risk profile by our model. All values in percentage. Total # of observations: 26,240,879. Time period 2006Q1-2016Q2. Source: Authors’ calculations based on Experian Data.

The differences in assigned risk profiles between the credit score and our model are significant, reflecting substantial variations in default risk, as illustrated in Table 5. For example, Subprime borrowers have an average default rate of 44%. However, borrowers who are Subprime based on the credit score but Near Prime based on our model have a 17% realized default rate, whereas those who are Subprime according to our model have a 48% default rate. Similarly, borrowers with Prime credit score have a default rate of 6% overall. However, among them, those who have a Near Prime profile based on our model have a 13% realized default rate, whereas those who are Prime according to our model have a 4% realized default rate. This suggests that the credit score is unable to distinguish borrowers with very different risk profiles, whose default rates often vary by more than 10 percentage points. This is a severe limitation, as the primary goal of the credit score is to stratify borrowers based on their default risk. As we will document in Section 4.4, being placed in an incorrect risk category has severe implications for consumers in terms of access to credit.

Table 5 also reports our model-based prediction for the default rate of borrowers in each credit score/model risk profile. Our model-predicted default rate is always less than 5 percentage points away from the realized default rate for each risk profile. This confirms that our model’s statistical performance is high at every point of the default risk distribution. Our model outperforms the conventional credit score, particularly in accurately categorizing

borrowers with lower credit scores, who are traditionally more prone to misclassification. This superior performance underscores the potential for our model to improve credit risk assessment and reduce misclassification, particularly for vulnerable populations.

Table 5: Default Risk Variation by Credit Score Profile

Risk Profile		Default Rate		Risk Profile		Default Rate	
Credit Score		Realized	Predicted	Model Based		Realized	Predicted
Deep Subprime	68.19	66.88		Deep Subprime		94.75	93.86
				Subprime		54.40	52.48
				Near Prime		18.18	17.75
				Prime		6.94	5.99
				Super Prime		5.82	0.94
Subprime	43.68	43.10		Deep Subprime		92.48	92.46
				Subprime		48.25	47.59
				Near Prime		17.09	16.61
				Prime		8.27	7.20
				Super Prime		4.00	1.00
Near Prime	22.38	22.56		Deep Subprime		88.18	90.85
				Subprime		34.89	37.76
				Near Prime		18.31	16.46
				Prime		7.36	5.97
				Super Prime		1.17	1.00
Prime	6.05	6.62		Deep Subprime		90.87	91.53
				Subprime		25.88	32.72
				Near Prime		13.17	15.17
				Prime		4.38	4.40
				Super Prime		0.92	0.91
Super Prime	1.02	1.24		Deep Subprime		92.43	91.48
				Subprime		42.23	44.90
				Near Prime		10.15	14.44
				Prime		2.04	2.39
				Super Prime		0.61	0.76

Notes: Realized and model predicted default rates by credit score and model-based risk profile. Industry-defined credit score risk profiles, defined in Table 3. All values in percentage. Total # of observations: 26,240,879. Time period 2006Q1-2016Q2. Source: Authors' calculations based on Experian Data.

4.3 Feature Attribution

We now examine the possible factors leading to differences in predictions of default risk ranking for our model and the credit score. We base our analysis on public information from the FICO and Vantage Score consumer websites, that cite, following the legislative requirements, the most important factors driving the credit score distribution. Credit scoring companies

are required to list at least four. We group the features we include in our model into categories to correspond to these same factors and aggregated the absolute SHAP values¹² for each instance across each category for our testing dataset across all periods.¹³

Credit scoring companies cite payment history, amounts owed, length of credit history, credit mix, and new credit as the most important factors driving credit score variation. Their stated contribution towards credit scores is reported in Table 6. Payment history explains 35% of the credit score variation and amounts owed 30%. Length of credit history explains 15% of the credit score variation, while credit mix and new credit each explain 10% of the variation.¹⁴

We report the feature attribution statistics for our model in Table 6. The most striking difference in feature attribution for our model in comparison to the credit score is that Amounts Owed explains 50% of the variation in default risk rankings, compared to only 30% for the credit score. Also, our model gives lower weight to Length of Credit History, which only explains 9% of the variation in rankings compared to 15% for the credit score, and New Credit, which accounts for only 3% of variation in the model’s rankings compared to 10% for the credit score. The high weight placed by credit scores on length of credit history and new credit may explain why credit scores tend to misclassify younger and low-income borrowers to a greater degree, as we will document in the next section.

¹²For more details on SHAP, see Lundberg and Lee (2017).

¹³Credit scoring companies do not need to reveal which specific features enter the factors they cite as driving credit score variation. For our model, we made the groupings based on the nature of each of the features.

¹⁴See <https://www.myfico.com/credit-education/whats-in-your-credit-score>. Payment history measures how a borrower has paid their accounts over the length of their credit history. Amounts owed includes five factors: amounts all on all accounts and on different types of accounts, accounts with balances, credit utilization ratio on existing accounts and remaining payments on installment debt. Length of credit history measures how long credit accounts have been open, the age of the newest account, and an average age of all accounts, as well as how long specific credit accounts have been open and how long it has been since accounts have been used. Credit mix refers to the number of different products, such as credit cards or mortgages, with a wider mix reported to marginally improve the credit score. New credit measures the number of new accounts and the number of inquiries in the last 12 months.

Table 6: Feature Attribution Differences

Feature Group	# of Features	<u>Model</u>	
		Our Model	Credit Score
Payment History	21	0.33	0.35
Amounts Owed	43	0.49	0.3
Length of Credit History	6	0.08	0.15
Credit Mix	5	0.05	0.1
New Credit	4	0.05	0.1

Notes: SHAP values for five feature groups. For each prediction window, we compute the SHAP value for 100,000 randomly sampled observations and for each feature. We then calculate the sum of the absolute value for each feature, aggregate it across the feature groups, and report the results for the group. We normalized the results so that for each model the five groups sum up to 1. Feature groups are defined in Table 13 in Appendix A. Credit score variation information from <https://www.myfico.com/credit-education/whats-in-your-credit-score>. Source: Authors' calculations based on Experian data.

4.4 Costs of Risk Profile Misclassification

We have shown that the credit score's lower statistical performance leads to risk profile misclassification for a large fraction of consumers concentrated primarily at the low end of the credit score distribution. What are the costs of such risk profile misclassification? We provide an initial answer to this question by analyzing how access to credit varies as a function of credit scores for credit cards and mortgages, the most common types of consumer loans.¹⁵

Table 14 in Appendix C reports summary statistics on indicators of credit demand and supply by credit score risk profile for the period 2006Q1-2016Q2. We report both the dollar values and the ratio to household income on average for each credit score risk profile. Credit limits on credit cards vary from \$7,032 on average for Deep Subprime borrowers to \$44,711 for Superprime borrowers. Mortgage balances average to \$ 172,753 for Subprime borrowers, rising to \$ 208,992 for Prime borrowers. Additionally, though demand for credit, as approximated by inquiries, is decreasing in the credit score, success at obtaining credit increases with the credit score. For example, only 6.1% of Subprime borrowers who place a

¹⁵Since we use data from credit reports, we do not have access to interest rate information.

credit inquiry originate a new credit card account, while 13.6% of Prime borrowers do. The corresponding values for mortgages are 3.9% and 10.6%.

These values are population averages affected by variation in characteristics, such as age and place of residence, within a given credit score profile. To control for these factors, we estimate a regression of various credit demand and credit supply outcomes, where the main explanatory variable is the credit score risk profile, controlling for age, zip code fixed effects and quarter time effects interacted with an indicator for the credit score risk profile. The details of the regression specification and the full estimation results are reported in Appendix C. Table 7 reports the average age-adjusted credit score risk profile effects for the entire sample period, estimated from these regression. Only 38% of Subprime borrowers have a credit card, whereas 80% of Prime borrowers do. This despite the fact that 11% of Subprime borrowers show a credit card inquiry in the previous quarter, whereas 8% of Prime borrowers do. Only 17% of Subprime borrowers who placed a credit card inquiry at quarter t or $t - 1$ display a credit card origination at t , while 34% of Prime borrowers do. Additionally, the average credit card balances for Subprime borrowers are \$7,730, while those for Prime borrowers are \$26,300. These values vary monotonically across risk profiles, and suggest that though demand for credit card borrowing is higher for lower credit score borrowers, their access to credit card lending is severely restricted compared to borrowers in more favorable risk profiles.

The pattern is similar for mortgages. Only 14% of Subprime borrowers have a mortgage, while 33% of Prime borrowers do. Additionally, only 5% of Subprime borrowers submit a mortgage inquiry in a quarter, and only 3% of them show a mortgage origination in the same or subsequent quarter. By contrast, 6% of Prime borrowers submit a mortgage inquiry in a quarter, and 9% of them show an origination in the same or subsequent quarter. Consequently, average mortgage balances for Prime borrowers are \$66,420, while they are \$28,330 for Subprime borrowers. The variation in these statistics is monotone in the risk profile.

Figure 5 in Appendix C displays the time effects by risk profile. The time variation is interesting, as after the Great Recession, there was a tightening of standards for consumer loans, particularly mortgages. The figure shows that the gap in access to credit card and mortgage credit between borrowers at the bottom and at the top of the credit score distribution widened substantially from 2009 to 2016.

This analysis suggests that borrowers in non-Prime credit score risk profiles, who should have a more favorable ranking based on our model, have faced substantial restrictions in accessing credit card and mortgage loans. On the other hand, borrowers with Prime and Superprime scores, whose default probabilities are consistent with a worse risk rating based on our model, borrowed excessively, based on current industry standards, potentially exposing their lenders to unexpected losses.

Table 7: Access to Credit by Risk Profile

	(1)	(2)	(3)	(4)
	Fraction with	Credit Limit / Balances	Inquiries	Originations if Inquired
<u>Panel A: Credit Cards</u>				
Superprime	96.13	39.43	5.90	35.15
Prime	79.58	26.30	8.04	33.88
Near Prime	66.88	19.76	12.37	29.87
Subprime	38.15	7.73	11.40	16.70
Deep Subprime	25.00	5.99	10.93	5.81
<u>Panel B: Mortgages</u>				
Superprime	37.86	61.18	4.58	12.24
Prime	32.93	66.42	5.99	9.08
Near Prime	25.92	50.48	6.83	6.10
Subprime	13.64	28.33	4.82	3.06
Deep Subprime	11.14	25.55	4.15	1.77

Notes: This table reports age-adjusted estimated coefficients for the average credit risk profile effects for credit cards (Panel A) and mortgages (Panel B). Deep Subprime, Subprime, Near Prime, Prime and Superprime are indicator variables for the consumer belonging to that category at quarter $t - 1$ while outcomes are measured at quarter t . Column (2) is credit limits for credit cards and first mortgage balances for mortgages measured in thousands of USD. All other values in percentage. Column (4) restricts the sample to individuals who inquired in quarters $t - 1, t$ and measures originations at quarter t . Full specification and regression results are reported in Appendix C.

This exercise quantifies how, given current lending conditions, access to credit would change for consumers whose credit score risk profile differs from the one predicted by our model. Interest rates also vary significantly with credit scores, (Edelberg (2006)), but since we do not have individual interest rate data, we are unable to document this variation in our sample. Changing the reference credit score and its performance properties would presumably lead to equilibrium changes in lending conditions. While we abstract from this possible effect, our analysis provides an assessment of how much being placed in the wrong risk profile can affect borrowers within the current system.

5 Relation between Performance and Equity

We have shown that ranking consumers by predicted default risk based on our model delivers significant performance improvements, particularly for borrowers at the bottom of the credit score distribution. However, there is growing concern about the potential negative impact of improved scoring algorithms for marginalized populations, such as young, minority, and low-income consumers. Fuster et al. (2018) examine the racial implications of mortgage approvals based on a random forest model of default risk compared to a lower performing model based on logistic regression. They find that though each group gains from improved predictive accuracy, Black and White Hispanic borrowers lose standing relative to White and Asian borrowers with the more statistically advanced approach. They argue that the increased flexibility of the higher-performing model leads to an unequal distribution of the gains associated with improved performance. Additionally, Blattner and Nelson (2021) argue that sophisticated scoring models based on machine learning are not helpful at improving performance for marginalized borrowers because the main factor leading to worse performance for those borrowers is compositional differences in the attribute distribution, referred to as data bias.

We examine how using our model to rank individuals by default risk affects the standing

of different demographic groups traditionally marginalized in consumer credit markets in comparison to the credit score, and how the performance gains vary across these groups. We consider three criteria to identify marginalized borrowers: income, age and minority status. Our credit report data contains limited demographic information, but does include date of birth and estimated annual income. This information allows us to construct indicator variables for being under 30 years old (Young) and for being in the bottom quintile of the income distribution (Income_{p20}). Our data does not include details on race due to its codification as a protected category in U.S. law. To address this, we construct a proxy for race derived from the 2010 Census data, where we combine Black and Hispanic populations into a single Minority indicator variable.¹⁶ We also match our Experian data with the Home Mortgage Disclosure Act (HMDA) that includes direct information on the race of mortgage borrowers and applicants. This restricts the sample to consumers with a first mortgage, but allows us to observe race directly. The details of our matching procedure are described in the Online Appendix.

5.1 Variation in Standing

To compare the standing of consumers in our model to the credit score, we rely on the percentile ranking introduced in Section 4 and calculate the ranking differences between our model and the credit scores. A positive difference suggests that our model considers a borrower less risky than the credit score would indicate. This allows us to understand whether using our model’s ranking generates disparities among different demographic groups, providing insights into the equity implications of improved performance.

Our analysis is based on regressions of the form:

$$Y_{ist} = \alpha + \beta_1 x_{ist} + \beta_2 z_{ist} + \beta_3 z_{ist} \times x_{ist} + \lambda_t + \delta_s + \delta_{st} + \epsilon_{ist} \quad (3)$$

¹⁶We construct our minority indicator following the approach proposed by the Consumer Financial Protection Bureau and described in <https://www.consumerfinance.gov/data-research/research-reports/using-publicly-available-information-to-proxy-for-unidentified-race-and-ethnicity/>.

where i corresponds to an individual, s corresponds to the state, and t corresponds to time in quarters. The main regressor is x_{ist} , an indicator for marginalized group membership. We include state-level controls as various regulations affecting the implication of default vary at the state level (Albanesi and Nosal (2018)). In selected specifications, we also control for realized default by including an indicator variable for whether an individual i is ever 90 days or more past due in the subsequent 8 quarters at quarter t , the variable z_{ist} . Adding the default indicator as an explanatory variable allows us to assess how the differences in standing of consumers between our model and the credit score vary across consumers who default and those who do not. In most specifications, we also include an interaction term between x_{ist} and z_{ist} , which allows us to evaluate whether rankings across the credit score and our model vary systematically for defaulting and non-defaulting consumers by demographic group. The variable λ controls for any common time-varying factors, while δ controls for any time-invariant differences and any state-level trends, such as variations in economic conditions that might impact default probabilities.

We conduct our analysis on the overall sample, known as the “Experian Sample,” using the race proxy and then replicate it with the matched data from both Experian and HMDA, termed the “HMDA Match Sample.” Descriptive statistics for these two samples by the demographic group are reported in Table 17. In each demographic group, borrowers in the HMDA Match sample have lower default rates, higher credit scores and higher household income. This suggests that they are positively selected relative to the overall Experian sample, as we would have expected, since appearing in the HMDA data requires that consumers have at least applied for a mortgage. The descriptive statistics also reveal that younger, low income and minority borrowers tend to have higher default rates, lower total balances and overdue debt balances, and lower household incomes. They are also ranked lower both by the credit score and by our model. These differences appear both the Experian and in the HMDA Match samples.

Table 8 reports our baseline regression results for both the Experian (Panel A) and

HMDA Match (Panel B) samples. Column (1), where the only regressor in addition to the quarter $\times$ state effects is a constant, confirms that for the overall sample there are no differences in consumer rankings on average, which is a consequence of our construction of the percentile rankings. Column (2) adds the future default indicator, which displays a negative and significant coefficient, with a value equivalent to approximately 29 credit score points.¹⁷ The negative sign of this coefficient suggests that our model systematically ranks consumer who default lower than the credit score, consistent with its higher performance. Column (3) adds the low income indicator that displays a positive and significant coefficient. Its value implies that consumers in the lowest quantile of the income distribution would gain on average about 5 percentiles in the ranking with our model, equivalent to approximately 25 credit score points. Column (4) adds an interaction term between the low income and the future default indicators. The estimates for this specification suggest that low income consumers who do not default would gain 6 percentiles when ranked by our model, equivalent to approximately 33 credit score points, while those who default would lose 7 percentiles in the ranking, equivalent to approximately 39 credit score points. The estimates suggest that the improvement in standing of low income borrowers when ranked by our model is obtained by the model’s better ability to separate those with high default risk from those with low default risk within that group.

The findings are similar for young borrowers. The estimates in column (5) suggest our model would place young borrowers 2 percentiles higher in the ranking compared to the credit score, equivalent to approximately 11 credit score points. Column (6) adds the interaction with the default indicator, implying that young borrowers who do not default would gain 2 percentiles in rankings with our model, equivalent to 11 credit score points, but those who do default would lose 4.5 percentiles or 25 credit score points. Finally, column (7) deploys the Minority indicator. Overall, minority consumers would gain 1 percentile in ranking with our model. Column (8) introduces the interaction between minority and future default.

¹⁷One percentile in ranking is equivalent to approximately 5.5 credit score points.

Table 8: Vulnerable Populations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DV: Our Model Ranking - Credit Score Ranking								
Panel A: Experian Sample								
Default		-5.164*** (0.032)		-6.392*** (0.038)		-5.585*** (0.036)		-5.111*** (0.028)
Income _{p20}			4.485*** (0.069)	6.067*** (0.077)				
Income _{p20} × Default				-0.438*** (0.048)				
Young					2.296*** (0.070)	2.387*** (0.074)		
Young × Default						0.983*** (0.045)		
Minority							0.998*** (0.035)	1.861*** (0.041)
Minority × Default								-0.902*** (0.030)
Constant	0.001*** (0.000)	0.951*** (0.006)	-0.857*** (0.013)	0.047*** (0.013)	-0.498*** (0.015)	0.461*** (0.016)	-0.184*** (0.007)	0.643*** (0.008)
Observations	26,240,879	26,240,879	26,240,879	26,240,879	26,240,879	26,240,879	26,240,879	26,240,879
R ²	0.0019	0.0181	0.0145	0.0391	0.0055	0.0228	0.0024	0.0196
Panel B: HMDA Match Sample								
Default		-3.387*** (0.052)		-3.790*** (0.057)		-3.229*** (0.059)		-3.532*** (0.061)
Income _{p20}			2.979*** (0.096)	3.454*** (0.105)				
Income _{p20} × Default				-0.131 (0.112)				
Young					2.430*** (0.084)	2.540*** (0.088)		
Young × Default						-0.709*** (0.099)		
Minority							-1.113*** (0.044)	-0.973*** (0.048)
Minority × Default								0.907*** (0.090)
Constant	-5.375*** (0.000)	-4.986*** (0.006)	-5.746*** (0.012)	-5.366*** (0.014)	-5.979*** (0.021)	-5.614*** (0.024)	-5.183*** (0.008)	-4.831*** (0.010)
Observations	756,179	756,179	756,179	756,179	756,179	756,179	756,179	756,179
R ²	0.0416	0.0487	0.0475	0.0564	0.0482	0.0556	0.0426	0.0494

Notes: This table reports regressions of the form specified by Equation (3). The dependent variable is the percentile ranking difference between our model and the credit score. The independent variable, "Default", is a binary indicator showing whether the individual defaulted within the next two years. The variable "Young" is a binary indicator for individuals under 30 years old. The variable "Income_{p20}" indicates whether the individual is in the lowest quintile of the estimated income distribution. In Panel A, the Minority indicator is based on ZIP code level Black and Hispanic population being over 50%. In Panel B, the Minority indicator is 1 if the individual is Black or Hispanic. Panel A presents the full Experian sample, while Panel B provides the HMDA Match sample results. We control for State x Quarter fixed effects. Standard errors are clustered by state and quarter. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Minority borrowers who do not default would gain 2 percentiles in the rankings with our model, equivalent to 11 credit score points, whereas those who default would lose 6 percentiles in the ranking, equivalent to 33 credit score points.

Taken together, the results for the Experian sample suggest that marginalized consumers, because of low income, youth or minority status, would on average gain in standing based on the ranking implied by our model compared to the credit score. The gain in ranking is driven by our model’s superior ability to separate consumers by default risk in each demographic group. Findings are similar for the HMDA matched sample, as shown in Panel B. Consumers in this sample are positively selected by credit score and, consistent with our analysis in Section 4, our model tends to give them a lower ranking, as shown by the negative value of the constant term. However, low income and young borrowers in this sample gain on average in ranking, and these gains are concentrated among those who do not default, whereas those who default decline in ranking. The magnitude of the estimated coefficients is similar to the Experian sample. Minorities lose approximately 1 percentile in ranking in the HMDA sample, approximately 5 credit score points, even those who do not default, whereas those who default lose 3 percentiles in ranking equivalent to approximately 16 credit score points.

Appendix D reports the estimates for Prime and non-Prime borrowers.¹⁸ Tables 18 shows that non-Prime consumers on average are ranked higher by our model by approximately 3 percentiles or 17 credit score points, with estimates by defaults status and demographic group very similar to the overall population. Prime borrowers, as shown in Table 19, are on average ranked 2 percentiles lower by our model, and low income, young and minority borrowers in this group do not gain in ranking compared to the credit score on average. However, both in the aggregate and for each demographic group, the estimates confirm that are model drops the ranking of those who will default, relative to the credit score, by approximately 7 percentiles.

The results for non-Prime and Prime borrowers shed light on the discrepancy of the

¹⁸Non-Prime consumers include those with a Deep Subprime, Subprime and Near Prime credit score, while Prime consumers include those with a Prime and Super Prime credit score.

change in ranking for minorities in the Experian and HMDA Match sample. The latter, including only consumers with a mortgage, is a positively selected sample for conventional credit scoring models, as having a mortgage boosts the credit mix and typically improves the credit score. Credit mix is a less important factor for our model, as discussed in Section 4, which then tends to rank these consumers lower than the credit score. The positive selection associated with having a mortgage is likely stronger for minority consumers due to the history of redlining and housing discrimination in the United States.

5.2 Variation in Performance

We now discuss performance by demographic group. Table 9 reports AUC scores for the credit score and our model in the aggregate and for each of the marginalized groups we consider. For both the credit score and our model, statistical performance is lower for the marginalized groups compared to the rest of the population, though the gap in AUC scores between marginalized and non-marginalized groups is much larger for the credit score than for our model. For the overall sample, AUC is 5 percentage points higher for our model in comparison with the credit score, while the increase is much larger for the marginalized groups. For consumers in the bottom quintile of the income distribution, the AUC score rises to 0.86 for our model from 0.74 for the credit score, for consumers younger than 30 it rises to 0.88 for our model from 0.82 for the credit score, and for minority consumers it rises to 0.88 for our model from 0.80 for the credit score. The other consumers also experience an improvement in AUC score with our model compared to the credit score, though this is smaller, typically 4-5 percentage points, compared to 6-12 percentage points for the marginalized groups.

We now examine what drives this difference in performance. Table 10 reports feature attribution statistics for each demographic group, based on SHAP values, showing virtually no difference across groups. This suggests that the conditional expectation function for default is similar across groups. Any differences in predictions are likely due to the distribution of

Table 9: Performance by Demographic group

AUC score	All	Income		Age		Minority	
		$Income_{p20}$	$Income_{\geq p20}$	Age < 30	Age $\geq$ 30	Yes	No
Credit Score	0.856	0.735	0.867	0.824	0.865	0.802	0.864
Our model	0.906	0.855	0.908	0.881	0.913	0.881	0.91

Notes: AUC score for our model and the credit score for each demographic group, averages for 2006Q1-2016Q2. Source: Authors’ calculations based on Experian data.

attributes, consistent with Blattner and Nelson (2021).

Table 10: Feature Attribution by Demographic group

Feature Group	Income		Age		Minority	
	$Income_{p20}$	$Income_{\geq p20}$	Age < 30	Age $\geq$ 30	Yes	No
Payment History	0.33	0.34	0.34	0.34	0.34	0.34
Amounts Owed	0.49	0.49	0.49	0.49	0.49	0.49
Length of Credit History	0.08	0.08	0.08	0.08	0.08	0.08
Credit Mix	0.05	0.05	0.05	0.05	0.05	0.05
New Credit	0.05	0.05	0.05	0.05	0.05	0.05

Notes: SHAP values for five feature groups. For each prediction window, we compute the SHAP value for 100,000 randomly sampled observations and for each feature. We then calculate the sum of the absolute value for each feature, aggregate it across the feature groups, and report the results for the group. We normalized the results so that the five groups sum up to 1. Feature groups are defined in Table 13 in Appendix A. Time period 2006Q1-2016Q2. Source: Authors’ calculations based on Experian data.

To examine the performance impact of the distribution of attributes across demographic groups, we calculate model and credit score AUC scores stratifying by attribute composition. The analysis is based on the notion that credit score performance is worse for consumers with delinquencies on their record, for those with thin files and for those with a limited credit mix, following Blattner and Nelson (2021). To capture this, we calculate AUC scores for our model and the credit score separately for consumers who are current on all accounts and those who have any delinquencies, for those with thick files and those with thin files¹⁹ and for those with a mortgage and those without a mortgage, where we interpret having a mortgage as another indicator of a broader credit mix, for each demographic group. Figure 2 reports

¹⁹Thin file consumers are those with a credit history of less than 10 years or less than 3 types of products on the credit report, among credit cards, bank cards, auto loans, HELOCs, first mortgages and installment loans.

the difference in AUC scores across each feature composition category: Current-Delinquent, Thick files-Thin files, Mortgage - No mortgage. A positive gap suggests that performance is better for the favored feature composition category, such as current vs delinquent. The figure clearly illustrates that, for the credit score, performance as measured by the AUC score is significantly better for the favored attribute composition, while the differences are minimal for our model.

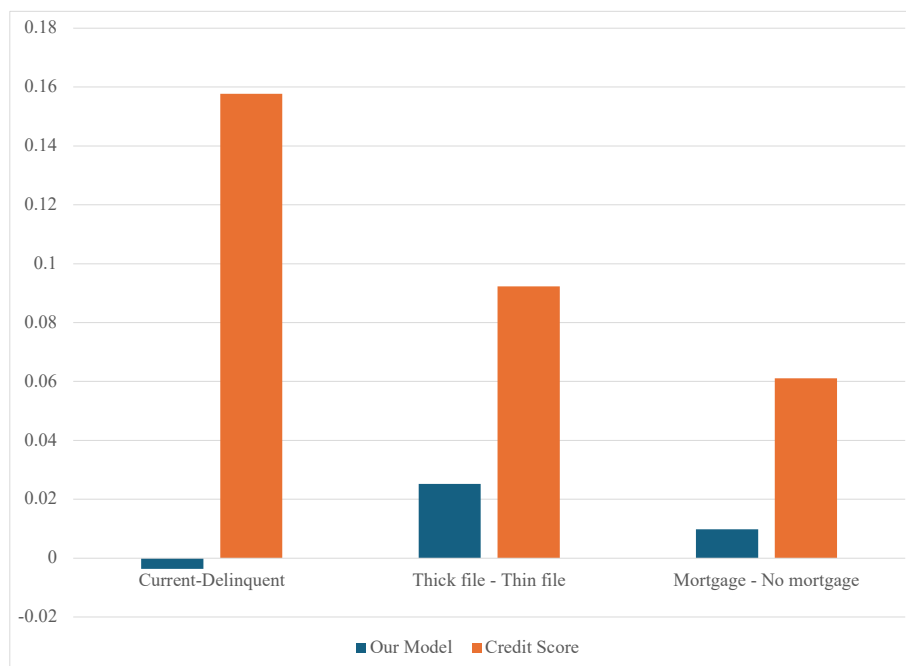


Figure 2: Performance by Feature Composition

Notes: AUC scores for the credit score and our model's by feature composition. Averages 2006Q1-2016Q2. Source: Authors' calculations based on Experian data.

Table 11 reports the share of consumers in each feature composition category by demographic group. Low income, young and minority consumers are over-represented in the Delinquent, Thin File, No Mortgage categories, compared to older, higher income and non-minority consumers, with sizable differences in the share in each unfavorable feature composition group. The variation in AUC scores for our model and the credit score across the feature composition groups for each demographic group are reported in Table 16 in Appendix D. The gap in AUC scores across the favored and unfavorable feature composition categories

are similar across demographic groups for our model, and substantially smaller than for the credit score.

Table 11: Feature Composition by Demographic Group

Feature Category	Status	Demographic Group					
		Income		Age		Minority	
		$Income_{p20}$	$Income_{\geq p20}$	Age < 30	Age $\geq$ 30	Yes	No
Default History	Current	0.64	0.87	0.80	0.83	0.73	0.84
	Delinquent	0.36	0.13	0.20	0.17	0.27	0.16
Credit History	Thin file	0.93	0.39	0.89	0.39	0.64	0.47
	Thick file	0.07	0.61	0.11	0.61	0.36	0.53
Mortgage Status	No Mortgage	0.95	0.61	0.91	0.62	0.78	0.66
	Mortgage	0.05	0.39	0.09	0.38	0.22	0.34

Notes: Share of borrowers in each feature composition category by demographic group. Feature composition categories defined by Default History, Credit History, Mortgage Status and Credit History interacted with Mortgage Status, with shares within each category adding up to 1. Time period 2006Q1-2016Q2. Source: Authors' calculations based on Experian data.

To evaluate how feature composition affects model performance for marginalized consumer groups, we conduct a straightforward counterfactual analysis. In this analysis, we calculate the AUC score for each marginalized group by assigning to it the same feature composition as the non-marginalized group. We then compare the difference between the actual and counterfactual AUC scores for both our model and the credit score. If the counterfactual AUC score is higher than the actual, it indicates that feature composition plays a role in reducing statistical performance. The results are presented in Table 12 and show that the gap between the actual and counterfactual credit score are substantially smaller for our model compared to the credit score, in most cases by an order of magnitude. This suggests that for the credit score, feature composition is a significant factor lowering performance for marginalized groups whereas this is not the case for our model.

This analysis establishes that a credit scoring algorithm with superior statistical performance based on machine learning can yield greater performance improvements for groups traditionally marginalized on consumer credit markets, contrary to the findings in Blattner and Nelson (2021). The performance improvements for marginalized group are driven by

Table 12: Contribution of Feature Composition to Model Performance

Group		Default	Credit History	Mortgage Status
Young	Our Model	-0.001	0.014	0.004
	Credit Score	0.004	0.040	0.018
I_{p20}	Our Model	-0.017	-0.005	0.006
	Credit Score	0.012	0.040	0.035
Minority	Our Model	-0.005	0.004	0.001
	Credit Score	0.013	0.019	0.009

Notes: Gap between counterfactual and actual AUC score for marginalized consumers by group for our model and the credit score by feature component. Counterfactual AUC score applies feature distribution of non-marginalized group to marginalized group. Time period 2006Q1-2016Q2. Source: Authors’ calculations based on Experian data.

the model’s improved predictive ability with low quality data, stemming from default status, thin files and low credit mix. The model’s ability to minimize data bias raises performance for marginalized groups more than it does for the rest of the population and results in substantial improvements in ranking for these groups compared to the credit score. While these findings are promising, it is important to note that incorporating features more informative for young, low-income, or minority borrowers could potentially improve the AUC score for these marginalized groups even further, as suggested by Blattner and Nelson (2021).

6 Conclusion

Our analysis demonstrates that widely used credit scores exhibit a significant degree of misclassification, wherein a substantial proportion of consumers are assigned to risk categories that do not accurately reflect their true default risk. This phenomenon is particularly pronounced among individuals with lower credit scores. Furthermore, our research indicates that the utilization of a machine learning-based scoring algorithm, characterized by superior predictive performance even with low data quality, would result in a more favorable credit risk assessment for low-income, young, and minority consumers. This suggests that the implementation of such an algorithm has the potential to ameliorate existing disparities in credit access, promoting a more equitable allocation of consumer credit.

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Appendix

A Model Details

We use a Hybrid Deep Neural Network-Gradient Boosted Trees model for our analysis. This is an ensemble model, comprised of two components. The first is based on deep learning, in the class used by Sirignano, Sadhwani, and Giesecke (2018). We restrict attention to feed-forward neural networks, composed of an input layer, which corresponds to the data, one or more interacting hidden layers that non-linearly transform the data, and an output layer that aggregates the hidden layers into a prediction. Layers of the networks consist of neurons, with each layer connected by synapses that transmit signals among neurons of subsequent layers. A neural network is, in essence, a sequence of nonlinear relationships. Each layer in the network takes the output from the previous layer and applies a linear transformation followed by an element-wise nonlinear transformation.

The second component of our model is Extreme Gradient Boosting, which builds on decision tree models. Tree-based models split the data several times based on certain cut-off values in the explanatory variables.²⁰ Gradient Boosted Trees (GBT) are an ensemble learning method designed to correct the tendency of tree-based models to overfit training data. This is achieved by recursively combining the predictions of multiple, simpler trees. Although each individual shallow tree is a “weak learner” with limited predictive power, the ensemble of these weak learners forms a strong model with improved stability over a single complex tree.

These two components — Deep Neural Network (DNN) and Gradient Boosted Trees (GBT) — are combined to improve predictive performance. Each model is trained independently, and their final predicted probabilities are then averaged. This approach, akin to the method proposed by Kvamme et al. (2018), which combined a convolutional neural network with a random forest by averaging, ensures that our methodology benefits from the strengths of both models. Specifically, we achieve this by following two key steps:

1. For each observation, run DNN and GBT separately and obtain predicted probabilities for each of the models;
2. Compute a weighted average of the predicted probabilities, with the weights determined based on model performance, as explored in Albanesi and Vamossy (2019).

Table 13 lists the features from the model we use as inputs from the credit report data. They include information on balances and credit limits for different types of consumer debt, severity and number of delinquencies, credit utilization by type of product, public record items such as bankruptcy filings, collection items, and length of the credit history. In order to be consistent with the restrictions of the Fair Credit Reporting Act of 1970 and the Equal Opportunity in Credit Access Act of 1984, we do not include information on age or zip code, and we do not include any income information, to be consistent with current credit scoring models. Table 13 lists the complete set of features used in our machine learning models.

²⁰Splitting means that different subsets of the dataset are created, where each observation belongs to one subset.

Table 13: Model Inputs

Amt. past due on bankcard revolving and charge trades presently 30 days delinquent	Monthly pmt. on open non-def. student trades
Amt. past due on credit card trades 90-180 DPD+	Monthly pmt. on second mortgage trades
Amt. past due on installment trades 90-180 DPD+	Mo. since the most recent 30-180 DPD+ on credit card trades
Amt. past due on joint mortgage type trades	Mo. since the most recent 30-180 DPD+ on trades
Amt. past due on revolving trades presently 30 days delinquent	Mo. since the most recent 90+ days delinquency
Amt. past due on revolving trades 90-180 DPD+	Mo. since the most recent foreclosure proceeding started on first mortgage trades
Amt. past due on trades presently 30 DPD+	Mo. since the most recently closed, transferred, or refinanced first mortgage trade
Amt. past due on trades presently 90-180 days delinquent	Mo. since the most recently opened credit card trade
Balance on bankcard revolving and charge trades 90-180 DPD+	Mo. since the most recently opened first mortgage trade
Balance on collections	Mo. since the most recently opened HELOC trade
Balance on collections, placed with the collector in the last 24m	Mo. since the oldest trade was opened
Balance on credit & bankcards	Mortgage to total debt
Balance on HELOC trades	Mortgage type inquiries made in the last 3 m
Balance on installment trades	No. of auto loan trades
Balance on installment trades presently 90-180 days delinquent	No. of collections
Balance on open auto loan trades	No. of credit & bankcards
Balance on revolving trades presently 90-180 days delinquent	No. of installment trades
Balance on second mortgage trades	No. of occurrences of 90 days delinquencies in the last 36m
Balance on trades presently 30 days delinquent	No. of occurrences of 90 days delinquencies in the last 6m
Balance on trades presently 60 days delinquent	No. of open mortgage type trades
Balance on trades presently 90+ days delinquent or derogatory	Open HELOC trades
Bankcard revolving and charge inquiries made in the last 3m	Public record bankruptcies
Credit amt. on HELOC	Public record discharged bankruptcies
Credit amt. on open credit card trades	Public record dismissed bankruptcies
Credit amt. on open def. student trades	Public records filed in the last 24m
Credit amt. on open non-def. student trades	Ratio of inquiries (no deduplication) to trades opened in the last 6m
Credit amt. on open trades	Total debt balances
Credit amt. on revolving trades	Trades legally paid in full for less than the full balance
Credit amt. paid down on open first mortgage trades	Unsatisfied collections
Credit card utilization	Utilization ratio
Fraction of 30 days delinquent debt	Worst ever status on a credit card trade in the last 24m
Fraction of 60 days delinquent debt	Worst ever status on a mortgage type trade in the last 24m
Fraction of 90+ days delinquent debt	Worst ever status on a trade in the last 24m
Heloc utilization	Worst present status on a credit card trade
Inquiries made in the last 12m (no deduplication)	Worst present status on a mortgage type trade
Installment utilization	Worst present status on a trade
Monthly pmt. on credit card trades	Worst present status on a trade (excl. collections)
Monthly pmt. on debt	Worst present status on an installment trade
Monthly pmt. on joint mortgage type trades	Worst present status on an open trade
Monthly pmt. on open first mortgage trades	

Notes: Amt=Amount, pmt=Payment, dpd=Days past due, HELOC=Home equity line of credit, def.=Deferred, revolving=Revolving credit, Mo.=Months, m=Months, No.=Number, excl.=Excluding.

B Comparison with Credit Scores

The credit score is a summary indicator used to predict the likelihood of a borrower defaulting, and it is widely used in the financial industry. For most unsecured debts, lenders typically check a prospective borrower’s credit score at the time of application. They may also review a recent short sample of their credit history. For larger unsecured debts, as well as secured debts like mortgages and auto loans, lenders usually require some form of income verification. Nevertheless, the credit score often plays a crucial role in determining key terms of the borrowing contract, such as the interest rate, down payment, or credit limit.

The most widely known credit score is the FICO score, a measure generated by the Fair Isaac Corporation, which has existed since 1989. Each of the three major credit reporting bureaus— Equifax, Experian, and TransUnion— also develops its proprietary credit scores. Credit scoring models are not public, though they are restricted by the law, mainly the Fair Credit Reporting Act of 1970 and the Consumer Credit Reporting Reform Act of 1996. These laws mandate that consumers are informed of the four main factors affecting their credit score. According to available materials from FICO and the credit bureaus, payment history and outstanding debt together account for more than 65% of the variation in credit scores. This is followed by the length of credit history, which explains 15% of the variation, and by new accounts and types of credit used (10%), along with new “hard” inquiries —credit report inquiries initiated by prospective lenders.

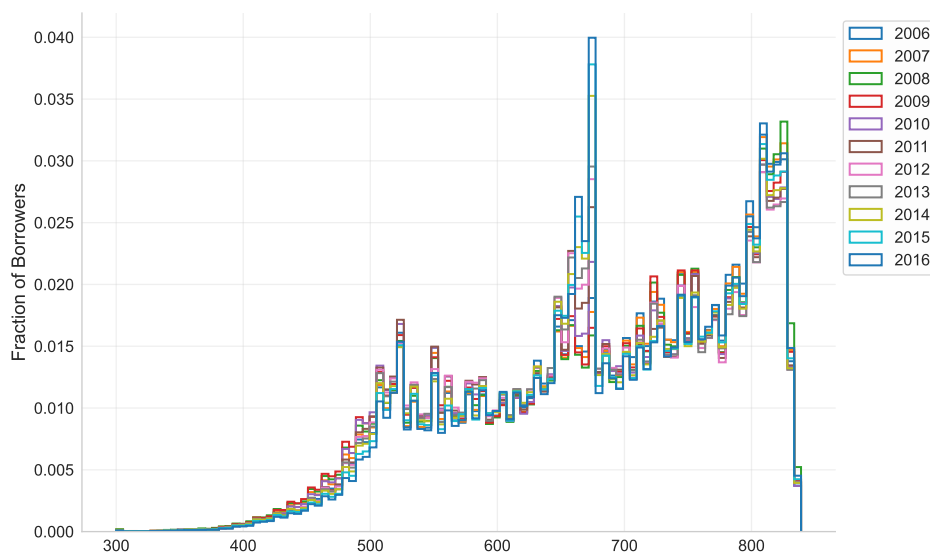


Figure 3: Credit Score Histogram by Years

Notes: Histogram of the credit score in our data by year for selected years. Source: Authors’ calculations based on Experian Data.

U.S. law prohibits credit scoring models from considering a borrower’s race, color, religion, national origin, sex and marital status, age, address, public assistance receipt, or any consumer right exercise under the Consumer Credit Protection Act. The credit score cannot be based on information not found in a borrower’s credit report, such as salary, occupation,

title, employer, date employed or employment history, or interest rates charged on particular accounts. Additionally, items related to child or family support obligations and “soft” inquiries—including “consumer-initiated” inquiries (e.g., requests to view one’s credit report), “promotional inquiries” (e.g., pre-approved credit offers), or “administrative inquiries” (e.g., lender reviews of open accounts) — are excluded from the scoring model. Requests marked as coming from employers are also not counted. In general, any information that is not proven to be predictive of future credit performance cannot be included in the credit score calculation.

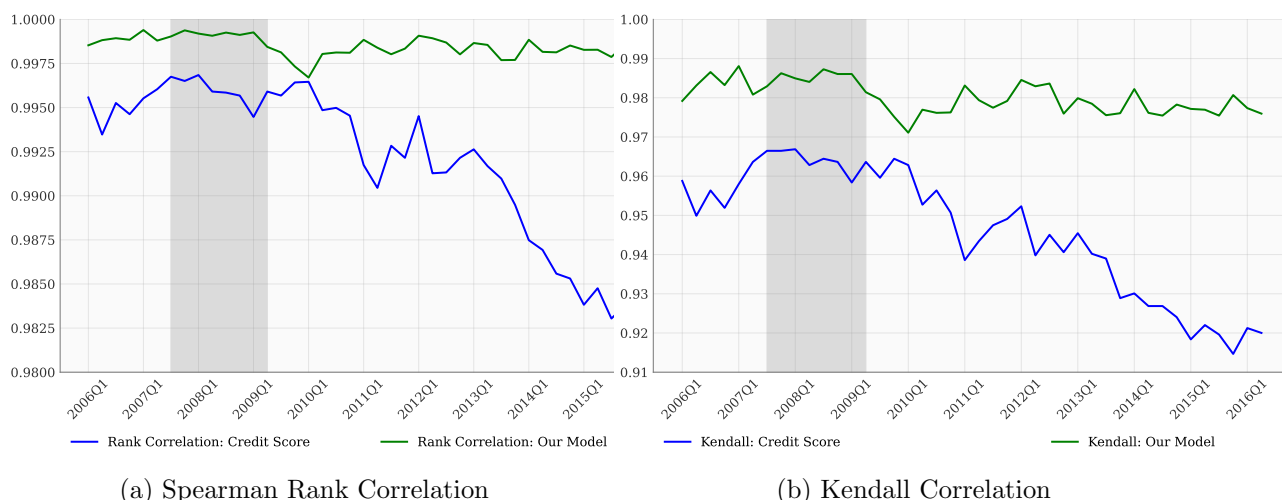


Figure 4: Gini Coefficient and Rank Correlation with Realized Default Rate

Notes: Panel a): Spearman’s rank correlation for the credit score and our model’s by quarter. Panel b): Kendall correlation for the credit score and our model’s by quarter. Source: Authors’ calculations based on Experian data.

C Costs of Misclassification

Table 14: Descriptive Statistics on Credit Access by Credit Score Risk Profile

	(1) Credit Limit / Balances	(2) % Inquired	(3) Origination	(4) Origination if Inquired	(5) % of Borrowers with	(6) Credit Limit or Balances / Income (%)
Panel A: Credit Cards						
Deep Subprime	7,031.5	11.4	6.2	2.4	1.36	12.0
Subprime	13,170.4	11.6	17.3	6.1	7.36	20.1
Near Prime	27,685.4	12.6	31.0	13.2	9.30	36.0
Prime	32,923.3	8.2	34.5	13.6	28.94	35.8
Superprime	44,710.7	5.0	34.9	14.0	23.01	39.1
Panel B: Mortgages						
Deep Subprime	171,036.42	4.7	0.5	2.3	0.56	268.02
Subprime	172,753.09	5.0	0.8	3.9	2.62	238.81
Near Prime	188,108.00	6.9	1.6	7.6	3.60	224.40
Prime	208,991.91	6.0	1.9	10.6	11.79	195.83
Superprime	170,236.19	4.1	2.1	13.4	9.30	123.35

Notes: The table presents descriptive statistics categorized by credit score risk profiles from 2006Q1 to 2016Q2. The credit score risk profiles are defined in Table 3. The sample consists of 26,147,712 borrowers. The credit risk profile is determined based on the previous quarter. % Originated represents the fraction of borrowers within each credit risk profile category who originated in the current quarter. % Inquired indicates the fraction of borrowers within each credit risk profile category who made an inquiry in the current quarter. % Originated (Borrowers with Inquiries) reflects the fraction of borrowers who had an inquiry in the current or previous quarter and subsequently originated in the current quarter. % of Borrowers with reports the percentage of borrowers in each credit risk category who have a first mortgage or credit limit. For example, 1.36% of borrowers with a non-zero credit limit are classified as deep subprime borrowers, even though they represent 6% of the population. Therefore, only 22.7% of deep subprime borrowers have a non-zero credit limit. The table reports credit limits, first mortgage balances, and their ratios to household income for those with non-zero credit card limits and first mortgage balances. Source: Authors' calculations based on Experian Data.

C.1 Access to Credit

The specification for the access to credit regressions in Section 4.4 is as follows:

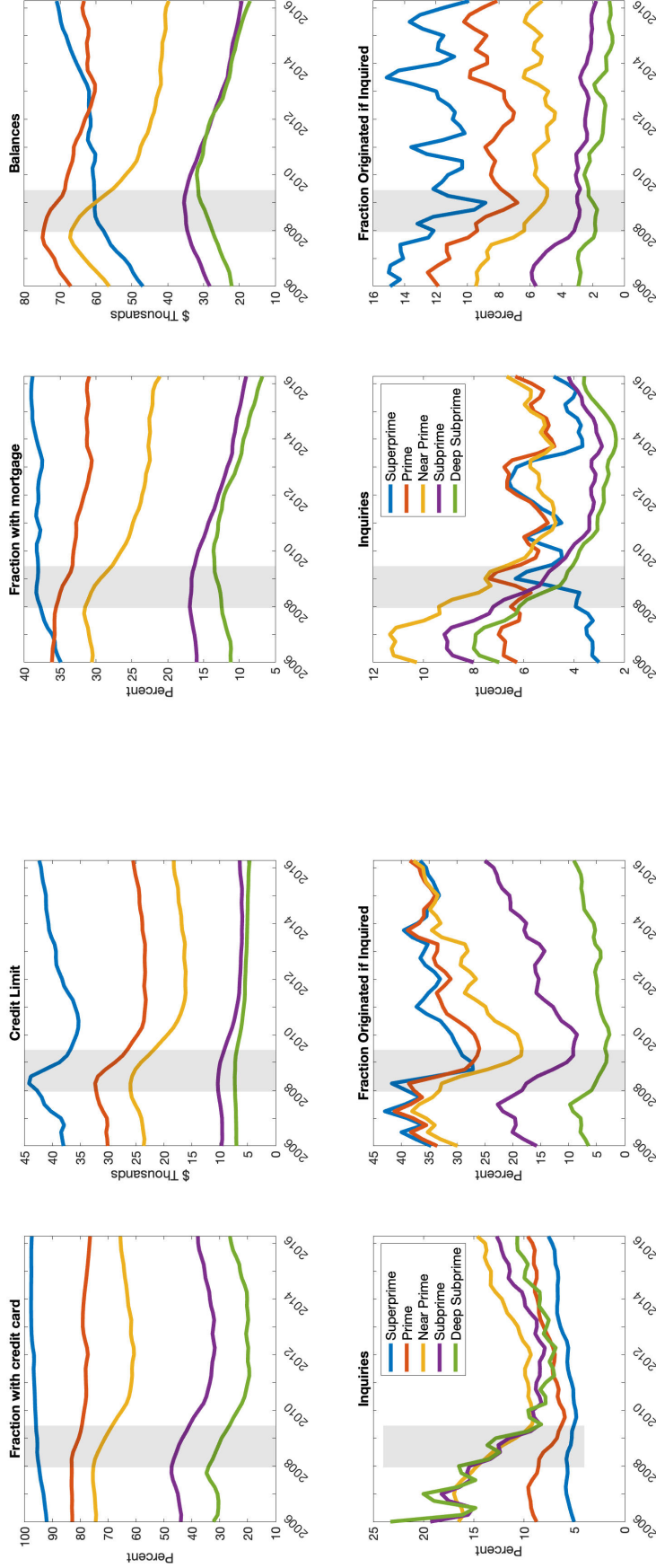
$$y_{it} = \alpha + \beta crp_{i,t-1} + \gamma a_{it} + \lambda_t \times crp_{i,t-1} + \epsilon_{it} \quad (4)$$

where i corresponds to an individual and t corresponds to time in quarters. The main regressor is $crp_{i,t-1}$, an indicator for credit score risk profile at $t - 1$, varying from Deep Subprime to Superprime. The credit profile is measured at $t - 1$ to avoid joint endogeneity in risk profile and credit outcomes (see Albanesi, DeGiorgi, and Nosal (2022) for a discussion). The variable a_{it} is an indicator for the consumer's age bin. We allow for five age bins, corresponding to the following age ranges in years: 18-29, 30-39, 40-49, 50-59, 60-69 and 70+. The variable λ controls time-varying factors affecting access to credit over the sample period, and is interacted with the credit risk profile, to detect any risk profile specific variations in access to credit over the sample period. Additionally, we include zip code fixed effect to account for geographical variation in home values and consumer prices and other factors that might affect the level of credit card and mortgage balances. We consider four possible outcomes: credit limits for credit cards and first mortgage balances for mortgages; inquiries and originations for credit cards and mortgages; and originations at quarter t among consumers with inquiries at quarter $t - 1, t$, to allow for a lag between the posting of an inquiry and the possible realization of an origination. The estimated are reported in Table 15. The age adjusted estimated values for the time effects interacted with the credit risk profile are displayed in Figure 5.

Table 15: Access to Credit by Risk Profile

	Credit Cards			Mortgages				
	(1) Credit Limit / Balances	(2) Inquiries	(3) Origination	(4) Originations if Inquired	(5) Credit Limit / Balances	(6) Inquiries	(7) Origination	(8) Originations if Inquired
Prime	-7917.347 (130.652)	0.037 (0.001)	0.011 (0.001)	-0.011 (0.005)	20126.692 (494.724)	0.033 (0.001)	0.007 (0.001)	-0.031 (0.005)
Near Prime	-14383.424 (173.008)	0.115 (0.001)	0.015 (0.002)	-0.047 (0.006)	9405.433 (646.995)	0.073 (0.001)	0.010 (0.001)	-0.055 (0.005)
Subprime	-28332.291 (121.642)	0.143 (0.001)	-0.071 (0.001)	-0.189 (0.005)	-18646.570 (420.904)	0.050 (0.001)	-0.003 (0.001)	-0.093 (0.005)
Deep Subprime	-31015.469 (127.105)	0.183 (0.002)	-0.109 (0.001)	-0.282 (0.005)	-24759.575 (505.414)	0.040 (0.001)	-0.010 (0.001)	-0.120 (0.005)
30-39	9685.490 (70.860)	0.008 (0.000)	0.011 (0.000)	-0.001 (0.001)	48881.809 (417.223)	0.037 (0.000)	0.009 (0.000)	-0.002 (0.001)
40-49	15675.728 (108.016)	0.002 (0.000)	0.013 (0.000)	-0.005 (0.001)	65633.694 (608.152)	0.032 (0.000)	0.008 (0.000)	-0.009 (0.001)
50-59	18744.131 (123.438)	-0.009 (0.000)	0.009 (0.000)	-0.012 (0.001)	53565.830 (601.340)	0.019 (0.000)	0.005 (0.000)	-0.013 (0.001)
60-69	17673.841 (134.698)	-0.021 (0.000)	-0.001 (0.000)	-0.013 (0.001)	32022.655 (541.962)	0.006 (0.000)	0.001 (0.000)	-0.018 (0.001)
70+	7636.702 (123.213)	-0.042 (0.000)	-0.031 (0.000)	-0.021 (0.001)	-30.810 (404.867)	-0.015 (0.000)	-0.006 (0.000)	-0.033 (0.001)
Constant	-11502.599 (314.503)	0.062 (0.001)	0.087 (0.001)	0.362 (0.003)	11335.228 (526.828)	0.014 (0.001)	0.013 (0.000)	0.159 (0.004)
Observations	26,147,712	26,147,712	26,147,712	3,761,219	26,147,712	26,147,712	26,147,712	2,235,063
R ²	0.2518	0.0305	0.0367	0.0705	0.1241	0.0208	0.0073	0.0298

Notes: This table reports estimated for equation 4 for credit cards (columns 1 to 4), and mortgages (columns 5 to 8). Column (1) is credit limits for credit cards, while Column (5) is first mortgage balances for mortgages. Column (4) and (8) restrict the sample to individuals who inquired in the current or previous quarter and estimates the coefficients for origination. Credit risk profiles based on previous quarter. Standard errors are clustered by ZIP code.



(a) Credit Cards

(b) Mortgages

Figure 5: Credit Access By Risk Profile Overtime

Age-adjusted quarter effects interacted with credit risk profile from estimation of equation (4) for credit cards and mortgages. Age adjustment obtained by assuming aggregate age average age distribution for each credit risk profile and using age bin effects reported in Table 15. See text for more detail.

D Trade-off Between Performance and Equity

Table 16: AUC by Feature Composition

AUC Score: Our Model							
Feature Category	Status	Income		Age		Minority	
		$Income_{p20}$	$Income_{\geq p20}$	Age < 30	Age $\geq$ 30	Yes	No
Default History	Current	0.81	0.86	0.78	0.85	0.81	0.85
	Delinquent	0.84	0.85	0.85	0.85	0.85	0.85
Credit History	Thin file	0.88	0.90	0.85	0.90	0.87	0.90
	Thick file	0.90	0.92	0.84	0.91	0.90	0.92
Mortgage Status	No Mortgage	0.88	0.91	0.85	0.91	0.88	0.91
	Mortgage	0.89	0.91	0.87	0.91	0.89	0.91

AUC Score: Credit Score							
Feature Category	Status	Income		Age		Minority	
		$Income_{p20}$	$Income_{\geq p20}$	Age < 30	Age $\geq$ 30	Yes	No
Default History	Current	0.76	0.81	0.66	0.80	0.74	0.81
	Delinquent	0.64	0.65	0.61	0.67	0.63	0.65
Credit History	Thin file	0.82	0.81	0.74	0.83	0.77	0.82
	Thick file	0.89	0.91	0.81	0.90	0.88	0.91
Mortgage Status	No Mortgage	0.82	0.85	0.73	0.85	0.79	0.85
	Mortgage	0.88	0.90	0.84	0.90	0.87	0.90

Notes: AUC score for marginalized consumers by group for our model and the credit score by feature component. Source: Authors' calculations based on Experian data.

Table 17: Descriptive Statistics (Vulnerable Populations)

Age: < 30		Age ≥ 30	Diff _{Age}	Income _{p20}	Income _{>p20}	Diff _{Income}	Minority	Non-Minority	Diff _{Minority}
Panel A: Experian Sample									
Default	0.23 (0.42)	0.17 (0.38)	-0.06***	0.37 (0.48)	0.14 (0.35)	-0.23***	0.28 (0.45)	0.16 (0.37)	-0.12***
Model Percentile	40.46 (24.13)	52.65 (29.87)	12.19***	26.33 (18.62)	55.60 (28.38)	29.27***	39.23 (27.52)	52.46 (28.96)	13.22***
Credit Score Percentile	38.68 (24.15)	53.14 (29.65)	14.47***	22.70 (18.20)	56.46 (27.48)	33.76***	38.54 (27.50)	52.61 (28.89)	14.08***
Total Balances	22.63 (65.33)	85.08 (183.69)	62.45***	10.15 (39.37)	86.02 (182.08)	75.87***	48.66 (122.40)	76.71 (175.56)	28.05***
90 DPD+ Balances	1.25 (14.03)	3.48 (36.10)	2.23***	1.99 (15.06)	3.23 (35.51)	1.24***	4.32 (35.51)	2.69 (31.91)	-1.63***
Household Income	46.95 (32.93)	87.05 (56.81)	40.10***	29.68 (6.54)	89.84 (55.23)	60.16***	62.54 (43.39)	81.93 (56.82)	19.40***
Age	23.81 (3.31)	51.56 (13.71)	27.76***	32.70 (13.17)	48.57 (16.06)	15.87***	43.72 (16.15)	45.94 (16.86)	2.22***
N	5,704,564	20,536,315	26,240,879	5,021,250	21,219,629	26,240,879	4,873,733	21,367,146	26,240,879
Panel B: HMDA Matched Sample									
Default	0.13 (0.33)	0.11 (0.31)	-0.01***	0.22 (0.42)	0.10 (0.30)	-0.12***	0.19 (0.39)	0.10 (0.30)	-0.09***
Model Percentile	44.22 (21.45)	52.94 (26.17)	8.73***	30.75 (18.30)	53.62 (24.94)	22.88***	40.22 (24.71)	52.98 (24.94)	12.70***
Credit Score Percentile	47.37 (21.45)	59.06 (25.08)	11.68***	33.13 (18.72)	59.43 (23.74)	26.30***	46.68 (24.62)	58.13 (24.31)	11.46***
Total Balances	44.52 (83.85)	158.61 (187.30)	114.09***	23.59 (59.03)	145.42 (180.39)	121.83***	104.03 (139.25)	135.73 (180.83)	31.70***
90 DPD+ Balances	0.97 (14.25)	2.39 (26.35)	1.42***	1.57 (15.30)	2.11 (24.91)	0.54***	3.38 (29.96)	1.76 (22.45)	-1.62***
Household Income	50.82 (30.85)	99.56 (57.67)	48.75***	30.30 (6.45)	95.57 (55.64)	65.27***	71.37 (44.36)	90.81 (58.03)	19.44***
Age	24.32 (3.21)	46.87 (11.51)	22.55***	28.04 (9.82)	43.15 (13.53)	15.10***	40.07 (13.39)	41.52 (14.16)	1.45***
N	188,005	568,174	756,179	94,176	662,003	756,179	130,869	625,310	756,179

Notes: *** denotes statistical significance at the 1% level. Differences in means between subgroups are tested via a t-test (results in "Diff" columns). Income and debt figures are in thousands. In Panel A, 'Minority status' is based on a ZIP code's Black and Hispanic population being over 50%. In Panel B, it is assigned if an individual is Black or Hispanic.

Table 18: Vulnerable Populations: Non-Prime Borrowers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DV: Our Model Ranking - Credit Score Ranking								
Panel A: Experian Sample								
Default		-10.616*** (0.040)		-11.317*** (0.045)		-10.699*** (0.047)		-10.612*** (0.037)
Income _{p20}			3.367*** (0.062)	3.444*** (0.056)				
Income _{p20} × Default				1.234*** (0.033)				
Young					2.974*** (0.080)	2.591*** (0.086)		
Young × Default						0.452*** (0.061)		
Minority							0.216*** (0.029)	0.942*** (0.032)
Minority × Default								-0.157*** (0.035)
Constant	3.555*** (0.000)	7.787*** (0.016)	2.252*** (0.024)	6.528*** (0.023)	2.707*** (0.023)	7.030*** (0.029)	3.498*** (0.008)	7.555*** (0.015)
Observations	10,508,324	10,508,324	10,508,324	10,508,324	10,508,324	10,508,324	10,508,324	10,508,324
R ²	0.0042	0.1199	0.0157	0.1360	0.0119	0.1266	0.0042	0.1204
Panel B: Experian-HMDA Sample								
Default		-7.197*** (0.066)		-7.091*** (0.071)		-5.958*** (0.064)		-7.629*** (0.074)
Income _{p20}			2.950*** (0.105)	3.028*** (0.126)				
Income _{p20} × Default				-0.341** (0.134)				
Young					4.348*** (0.094)	4.846*** (0.112)		
Young × Default						-3.262*** (0.128)		
Minority							-1.811*** (0.064)	-1.926*** (0.085)
Minority × Default								1.791*** (0.118)
Constant	-1.766*** (0.000)	0.496*** (0.021)	-2.588*** (0.029)	-0.352*** (0.038)	-3.173*** (0.031)	-1.170*** (0.039)	-1.291*** (0.017)	0.963*** (0.029)
Observations	217,516	217,516	217,516	217,516	217,516	217,516	217,516	217,516
R ²	0.0320	0.0938	0.0416	0.1032	0.0542	0.1140	0.0354	0.0962

Notes: This table reports regressions of the form specified by Equation (3). The dependent variable is the percentile ranking difference between our model and the credit score. The independent variable, "Default", is a binary indicator showing whether the individual defaulted within the next two years. The variable "Young" is a binary indicator for individuals under 30 years old. The variable "Income_{p20}" indicates whether the individual is in the lowest quintile of the estimated income distribution. In Panel A, the Minority indicator is based on ZIP code level Black and Hispanic population being over 50%. In Panel B, the Minority indicator is 1 if the individual is Black or Hispanic. Panel A presents the full Experian sample, while Panel B provides the matched Experian-HMDA sample results. We control for State x Quarter fixed effects. Standard errors are clustered by state and quarter. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 19: Vulnerable Populations: Prime Borrowers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DV: Our Model Ranking - Credit Score Ranking								
Panel A: Experian Sample								
Default		-11.031*** (0.099)		-11.249*** (0.110)		-11.823*** (0.120)		-10.914*** (0.090)
Income _{p20}			-2.550*** (0.112)	-2.256*** (0.115)				
Income _{p20} × Default				2.950*** (0.131)				
Young					-0.135** (0.064)	-0.138** (0.062)		
Young × Default						3.513*** (0.105)		
Minority							-0.731*** (0.036)	-0.370*** (0.037)
Minority × Default								-0.386*** (0.091)
Constant	-2.373*** (0.000)	-1.924*** (0.004)	-2.218*** (0.007)	-1.794*** (0.008)	-2.350*** (0.011)	-1.901*** (0.011)	-2.276*** (0.005)	-1.876*** (0.005)
Observations	15,732,555	15,732,555	15,732,555	15,732,555	15,732,555	15,732,555	15,732,555	15,732,555
R ²	0.0029	0.0227	0.0045	0.0238	0.0029	0.0230	0.0031	0.0227
Panel B: Experian-HMDA Sample								
Default		-7.080*** (0.111)		-7.146*** (0.119)		-7.369*** (0.128)		-6.886*** (0.126)
Income _{p20}			-1.335*** (0.103)	-1.227*** (0.106)				
Income _{p20} × Default				1.128*** (0.340)				
Young					0.773*** (0.094)	0.726*** (0.094)		
Young × Default						1.298*** (0.250)		
Minority							-2.085*** (0.050)	-1.918*** (0.050)
Minority × Default								-0.219 (0.235)
Constant	-6.833*** (0.000)	-6.589*** (0.004)	-6.750*** (0.006)	-6.514*** (0.007)	-7.001*** (0.021)	-6.747*** (0.020)	-6.547*** (0.007)	-6.331*** (0.008)
Observations	538,663	538,663	538,663	538,663	538,663	538,663	538,663	538,663
R ²	0.0523	0.0636	0.0530	0.0641	0.0530	0.0643	0.0556	0.0664

Notes: This table reports regressions of the form specified by Equation (3). The dependent variable is the percentile ranking difference between our model and the credit score. The independent variable, "Default", is a binary indicator showing whether the individual defaulted within the next two years. The variable "Young" is a binary indicator for individuals under 30 years old. The variable "Income_{p20}" indicates whether the individual is in the lowest quintile of the estimated income distribution. In Panel A, the Minority indicator is based on ZIP code level Black and Hispanic population being over 50%. In Panel B, the Minority indicator is 1 if the individual is Black or Hispanic. Panel A presents the full Experian sample, while Panel B provides the matched Experian-HMDA sample results. We control for State x Quarter fixed effects. Standard errors are clustered by state and quarter. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Online Appendix: Matched Experian-HMDA Panel

HMDA Data

The Home Mortgage Disclosure Act mandates that almost all mortgage lenders report comprehensive information on the applications they receive and whether they approve the loan.²¹ Mortgage lenders submit applicants' racial and ethnic background, personal attributes, and loan application details, such as the requested loan size, income, loan purpose (purchase, refinancing, improvement), co-applicants, loan priority (first or second lien), and the census tract location of the property, to the HMDA database. If a loan is granted, any loan sale is reported, along with an indicator for sale to any quasi-government entity. Our HMDA data ranges from 2010 to 2017.

Credit Bureau - HMDA Matching

Both the credit bureau and HMDA data are anonymized and lack a unique identifier to directly link the two datasets. However, the detailed information on originated mortgages in both datasets facilitates the matching of mortgages based on their characteristics, as outlined by Butler, Mayer, and Weston (2022). Following Butler, Mayer, and Weston (2022), we implement a series of filters on the mortgages extracted from our HMDA and credit bureau data. For HMDA, we ensure each mortgage is a first lien and located within a Metropolitan Statistical Area (MSA), where HMDA data is most comprehensive. The borrower must reside in one of the 50 states, including Washington D.C., following loan origination. We focus on home purchase and refinancing loans, excluding home improvement loans due to their less defined nature in both datasets. Additionally, we limit our analysis to owner-occupied properties to align the property location with the borrower's location in the credit bureau data and consider only mortgages with a single applicant or borrower, ensuring that the demographic data is directly relevant to the individual matched in the credit bureau data. We apply similar filters to our credit bureau data. We consider only first-lien, non-joint mortgages, where balances for both types of mortgages are available in the Experian data. The residential ZIP code is restricted to those covered in our final HMDA sample, ensuring it falls within an MSA in one of the 50 states or Washington D.C. Moreover, to enhance the accuracy of matching balances in refinancing cases and increase the likelihood of identifying owner-occupied properties, we restrict our sample to individuals with only a single first mortgage trade in the current quarter and at most one in the previous two quarters. This careful selection process aims to enhance the accuracy of our analysis. For itemized sample restrictions, see Table 20.

After applying the filters to both the HMDA and credit bureau data, the target population for the matched sample comprises borrowers obtaining a first-lien home purchase or refinance loan individually (no co-applicant), for their primary residence located within an MSA in the United States excluding its territories, between 2010 and 2017.

²¹Exceptions are granted only for very small or strictly rural lenders (Butler, Mayer, and Weston (2022)). Approximately 95% of all first-lien mortgages are reported to the HMDA database (Avery et al. (2017)), with the coverage rate likely higher for properties in MSAs.

Table 20: Itemized Sample Restrictions: Credit Bureau and HMDA Data

Credit Bureau Data [2010-2017]	
Removal Reason	# of Individuals
Initial Data	769,423
No Mortgage	310,542
First Lien Mortgage Balance = Joint Mortgage Balance	290,481
Non-HMDA zip codes	267,042
More than one first mortgage (past two quarters and current)	127,891
First mortgage balance is zero or it is not different from the past two quarters	79,226
Foreclosed balances	77,481
Refinance or Purchase Activity over 6 months ago	44,192
HMDA Data [2010-2017]	
Removal Reason	# of Loans
Initial Data	62,610,270
Missing MSA	55,412,483
Home improvement loans	52,565,523
Joint loans	25,909,081
Non-first lien loans	25,501,830
Missing geographic information or outside of the 50 States + D.C.	25,500,339
Non-owner occupied loans	22,565,758
Duplicate observations along the match variables at the census tract level, 20,250,348	
Tracts with no associated ZCTAs	19,334,339
Tract-ZCTA5 Expansion	35,446,686
Duplicates along match variables at the ZCTA5 level	27,413,377

Notes: This table documents the sample restrictions applied to the credit bureau and HMDA datasets. The final credit bureau data contains 50,166 observations of 44,192 individuals. The final HMDA dataset, after deduplication at the ZCTA level, contained 27,413,377 observations, which included 15,763,550 of the original census tract level loans.

Following the procedure outlined in Butler, Mayer, and Weston (2022) as a reference, we made a slight modification in our matching process by substituting the census tract with ZCTA5 as in LaVoice and Vamossy (2024). Specifically, we matched mortgages in the credit bureau data to HMDA data based on six characteristics: origination year, ZCTA5, loan amount, loan purpose (purchase or refinancing), mortgage type (conventional, FHA, or VA), and if/to which quasi-government entity the loan is sold (purchaser type).²² We implement an exact matching algorithm, as opposed to using nearest matches or propensity scores, to ensure the highest accuracy for our matched dataset. 89.73% of originated mortgages in the HMDA data are unique when using our six matching variables, with ZCTA5 being replaced by the census tract. We refine our HMDA dataset by removing duplicates based on these six matching variables, where we use the census tract instead of ZCTA5.²³

Because our Experian credit report data only reports an individual’s zip code, we first matched our zip code level credit bureau data to Census 5-Digit Zip Code Tabulation Areas (ZCTA5s) using the Missouri Census Data Center’s Census Tract - ZIP/ZCTA crosswalk files.²⁴ This process increased our HMDA sample from 19.33M to 35.44M observations as it assigns each individual to all potential ZCTA5s. We observe a one-to-one match in 51.6% of cases, a one-to-two match in 29.2%, a one-to-three match in 11.01%, a one-to-four match in 4.31%, and 3.88% of cases involve more than four ZCTAs. However, duplication issues arise, particularly when two (or more) loans differ only at the census tract level but are assigned to the same ZCTA5. Upon expanding our dataset to the ZCTA5 level, we engage in a second phase of duplicate removal, this time targeting the matching variable level. This results in a dataset comprising 27.4 million observations. Notably, this process impacts 32.76% of the original census tract level observations. By implementing this two-stage duplicate elimination approach — initially at the census tract level and subsequently at the ZCTA5 level post-expansion — we reduce the incidence of false positives, enhancing the reliability of our data.

Butler, Mayer, and Weston (2022) identifies two potential sources of error when matching HMDA with credit bureau data. First, a data error in one of the matching variables might lead to a mismatch. Second, HMDA-reporting and non-reporting lenders may originate identical, otherwise unique mortgages. They argue that such mismatches should not introduce bias in estimates beyond pure noise.

In our matching process, expanding HMDA data from the census tract to ZCTA5 level introduces two potential errors. The first concerns assigning individuals from a specific cen-

²²These six variables are directly observable in the HMDA data. We provide additional details and validation for our credit bureau data in the Appendix D

²³To address the issue of integer-based origination balances in the HMDA data, we adopt a range-based approach for origination balances in the credit bureau data. This method, which includes adjacent integer values, helps to avoid inaccuracies stemming from rounding differences. For all other matching variables, we ensure precise, exact matches.

²⁴USPS zip codes are not areal features used by the Census but a collection of mail delivery routes that identify the individual post office or metropolitan area delivery station associated with mailing addresses. ZIP Code Tabulation Areas (ZCTAs) are generalized areal representations of United States Postal Service (USPS) ZIP Code service areas. HMDA data reports 2010 census tracts starting in 2013. For loans originated from 2003 to 2012, HMDA data uses the 2000 census tracts. We successfully matched ZCTAs for 95.5% of Census Tracts observations in the HMDA data. Crosswalk URL: <https://mcdc.missouri.edu/applications/geocorr2014.html>.

sus tract to multiple ZIP codes, raising the chance of matching with an incorrect ZCTA5. The risk is partially offset by two factors: first, the high degree of uniqueness in the HMDA data at the census tract level, evidenced by 89.73% of mortgages being distinct. Second, the comprehensive coverage of the HMDA dataset, which includes most originated mortgages, significantly enhances the likelihood of accurately identifying the correct loan when mapping to the ZCTA5 level. The second issue arises when loans from different census tracts, sharing identical matching variables, are grouped under the same ZCTA5.²⁵ By eliminating duplicates at the match variable level, we address this concern.

We report the match rate as well as summary statistics on the match in Table 21. Panel A displays the match rate for home purchase mortgages, refinance loans, and both types of loans combined. We find a corresponding HMDA mortgage for 43.02% of the mortgages in the credit bureau data with a slightly higher match rate for refinance loans as opposed to home purchase loans. Given that HMDA covers 95% of mortgages, 90% are unique at the census tract level, 95.5% of census tracts are merged to ZCTA5, and 67.24% of census tract level are unaffected by the ZCTA5 expansion, the conservative lower bound on the best achievable match rate is the product of these four numbers, approximately 54.9%.²⁶ In other words, our algorithm successfully found matches in roughly 78.4% ($0.4302/0.549$) of the potential cases.

The summary statistics in panels B and C of Table 21 assess whether our matched sample is representative of the original population for home purchase loans and refinance loans respectively. Panel B reveals that the successfully matched home purchase mortgage sample is largely representative of the initial population of credit bureau mortgages. Similarly, Panel C demonstrates that the matched sample of refinance loans accurately represents the starting sample from the credit bureau data.

Since our study focuses on the impact of race on debt collection judgments, it is essential to determine if minorities are underrepresented in the data or if a specific type of minority borrower, such as high or low-income, is underrepresented. We next investigate whether race plays a role in the likelihood of matching originated mortgages from the HMDA database to our sample of credit bureau records. The regressions in Table 22 assess the probability of successfully matching a loan based on borrower race. The coefficients on Black and Hispanic, and the interaction terms Black X log(Income) and Hispanic X log(Income) in column (1) (2) are insignificant. While there is a slight indication of selection bias through the combination

²⁵To view this from another angle, we assigned unique pseudo-IDs to each loan in the census tract-level HMDA data and then expanded these to the ZCTA5 level. We found that 67.24% of these pseudo-IDs had no conflicts at the ZCTA5 level. However, conflicts did arise: 23.57% of pseudo-IDs conflicted with one other, 6.62% with two others, and 2.58% with more than two. These conflicts affected 32.76% of HMDA observations at the census tract level, leading to the removal of 18.47% of pseudo-IDs. Our approach does not involve removing all conflicted pseudo-IDs; rather, we only eliminate those in ZIP codes with identified conflicts. This means a single census tract could span two ZCTAs, with conflicts present in one but not the other. We retain conflict-free pseudo-IDs. However, excluding ZIP codes with conflicts reduces the efficiency of our matching process if the correct ZIP code is among those removed.

²⁶Our estimated lower bound for the success rate of matches at the ZCTA5 level stands at 64.19% (0.955×0.6724) relative to the potential match rate achievable at the more precise census-tract level. Conversely, the upper bound for this success rate is 77.86% (0.955×0.8153). This lower bound is based on the assumption that in all cases of conflict, it was always the conflicting loans that would have been the correct matches. On the other hand, the upper bound is calculated under the assumption that the correct loans were never among those deleted, thereby allowing us to achieve a higher rate of successful matches.

of race and income for Black applicants seeking to refinance mortgages, we can control for variables from both databases, which mitigates concerns about selection bias.

Table 23 provides summary statistics for our individual-level panel data starting in 2013Q2 and ending in 2017Q2.²⁷ Similar to patterns identified in Butler, Mayer, and Weston (2022), Columns 1 and 2 indicate that individuals within the matched data set generally exhibit higher credit scores, are relatively younger, and have a higher likelihood of holding a mortgage compared to the average U.S. resident who has a credit history. These patterns are anticipated, as individuals must either obtain a new mortgage or refinance an existing one between 2010 and 2017 to qualify for inclusion in the matched panel. Columns 3–5 demonstrate that the White/Asian borrowers in the matched panel have higher credit scores and incomes in comparison to minority (Black/Hispanic) borrowers.

²⁷While a larger sample of data is used to match credit report and HMDA data, we limit our sample for our regression analysis as judgment data isn’t available in every quarter of the credit report data.

Table 21: Descriptive Statistics of Matching Data from Credit Bureau with HMDA

Panel A: Match rate					
	Credit bureau sample	Matched to HMDA	Match rate (%)		
Home Purchase Mortgages	31,411	12,761	40.63		
Refinance Loans	19,919	9,320	46.79		
All Loans	51,330	22,081	43.02		
Panel B: Home purchase mortgages					
	Credit bureau sample	Matched to HMDA	Unmatched	<u>Matched vs. Norm. diff.</u>	<u>Unmatched t-stat</u>
<u>Match criteria</u>					
Conventional	0.640	0.591	0.674	-0.17	-15.03
FHA-insured	0.285	0.327	0.257	0.16	13.62
VA-guaranteed	0.075	0.082	0.070	0.05	4.03
Fannie Mae	0.207	0.182	0.223	-0.10	-8.78
Freddie Mac	0.130	0.103	0.148	-0.14	-11.81
Loan amount	202,432	193,610	208,468	-0.08	-7.15
<u>Non-match characteristics</u>					
Credit score t-1	716	710	720	-0.15	-12.02
Age	40.7	39.4	41.5	-0.15	-12.26
Have mortgage t-1	0.099	0.064	0.123	-0.20	-17.17
Total debt t-1	29,933	26,043	32,472	-0.10	-8.08
Past due debt t-1	137	155	125	0.01	0.86
Observations	31,411	12,761	18,650		
Panel C: Refinance loans					
	Credit bureau sample	Matched to HMDA	Unmatched	<u>Matched vs. Norm. diff.</u>	<u>Unmatched t-stat</u>
<u>Match criteria</u>					
Conventional	0.861	0.842	0.877	-0.10	-7.17
FHA-insured	0.087	0.102	0.075	0.10	6.74
VA-guaranteed	0.052	0.056	0.048	0.04	2.60
Fannie Mae	0.202	0.213	0.192	0.05	3.70
Freddie Mac	0.131	0.128	0.134	-0.02	-1.26
Loan amount	218,933	202,233	233,619	-0.17	-12.33
<u>Non-match characteristics</u>					
Credit score t-1	739	739	739	-0.00	-0.16
Age	49.3	49.6	49.1	0.04	2.96
Have mortgage t-1	1	1	1	0	0
Total debt t-1	240,201	224,718	253,816	-0.15	-10.70
Past due debt t-1	1,039	540	1,477	-0.05	-3.51
Observations	19,919	9,320	10,599		

Notes: This table provides an overview of the credit bureau and the Home Mortgage Disclosure Act (HMDA) data matching. The initial sample of credit bureau mortgages comprises both home purchase mortgages and refinance loans issued between 2010 and 2017. Loan applicants must apply individually (excluding joint applications), reside within a metropolitan statistical area after loan origination in one of the 50 states (including D.C.), and have the mortgage as their sole first-lien mortgage. The matching process is described in Section 3.1. Panel A illustrates the success rate of the matching methodology. Panel B summarizes loan and borrower features for home purchase mortgages in the credit bureau data, the subset successfully matched to HMDA, and the unmatched loans. The final two columns display the normalized difference and the outcome of a t-test comparing the mean of the matched sample to the mean of the unmatched sample. Panel C offers comparable summary statistics for refinance loans.

Table 22: Does borrower race affect the Credit Bureau/HMDA match?

	Full sample (1)	Home purchase mortgages (2)	Refinance loans (3)
Match criteria			
FHA loan	0.0070*** (0.0023)	0.0369*** (0.0032)	-0.0567*** (0.0034)
VA loan	-0.0135*** (0.0031)	0.0110** (0.0044)	-0.0524*** (0.0044)
Purchased by Fannie Mae	-0.0231*** (0.0020)	0.0321*** (0.0036)	-0.0580*** (0.0025)
Purchased by Freddie Mac	-0.0316*** (0.0023)	0.0214*** (0.0044)	-0.0637*** (0.0028)
log(Loan amount)	-0.0173*** (0.0015)	-0.0366*** (0.0027)	-0.0036* (0.0018)
Non-match characteristics			
Black	-0.0024 (0.0070)	0.0124 (0.0536)	0.0109 (0.0072)
Hispanic	-0.0083 (0.0076)	0.0283 (0.0459)	-0.0036 (0.0077)
Black X log(Income)	-0.0003 (0.0007)	-0.0018 (0.0048)	-0.0019** (0.0007)
Hispanic X log(Income)	0.0010 (0.0007)	-0.0028 (0.0042)	0.0003 (0.0008)
log(Income)	0.0031*** (0.0003)	-0.0048*** (0.0012)	0.0006* (0.0003)
Census tract-by-year FE	Yes	Yes	Yes
R-squared	0.0027	0.0042	0.0092
Observations	20,250,348	9,706,546	10,543,802

Notes: This table presents regression analyses investigating the factors influencing the matching of a mortgage from the Home Mortgage Disclosure Act (HMDA) data to a credit bureau record. The sample includes all first-lien home purchase mortgages and refinance loans for owner-occupied properties in metropolitan statistical areas between 2010 and 2017 with reported income. Only single-applicant loans are considered, excluding joint applications. The dependent variable is a binary indicator representing a successful HMDA mortgage match to a credit bureau record, while the independent variables are loan and borrower attributes from the HMDA data. Columns 1, 2, and 3 display the results for the entire sample, home purchase mortgages, and refinance loans, respectively. The coefficients are reported in percentage point units. Standard errors are clustered by census tract-year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 23: Summary Statistics for the Credit Bureau/HMDA Matched Panel

	Full credit sample	Matched sample	White/Asian	Black	Hispanic
Age	50.86 (19.13)	44.94 (14.25)	45.21 (14.37)	45.00 (14.26)	42.68 (13.02)
Household income	74.80 (43.72)	93.25 (51.49)	96.43 (52.62)	77.45 (41.24)	79.28 (44.04)
Credit score _{t-1}	682.66 (103.49)	716.92 (81.23)	723.31 (79.03)	677.29 (87.61)	694.14 (81.52)
Have mortgage _{t-1}	0.268 (0.443)	0.687 (0.464)	0.699 (0.459)	0.641 (0.480)	0.627 (0.484)
Total debt _{t-1}	64,184 (151,233)	154,224 (169,753)	159,179 (175,425)	128,788 (122,437)	132,477 (148,105)
30 DPD debt _{t-1}	776.72 (13,062.49)	1,087.03 (14,750.96)	962.12 (14,439.15)	2,031.44 (17,405.45)	1,402.83 (15,066.73)
60 DPD debt _{t-1}	323.00 (9,031.04)	391.49 (8,324.76)	334.74 (7,912.57)	844.95 (11,289.45)	516.25 (8,957.47)
90+ DPD debt _{t-1}	1,892.81 (22,307.79)	1,205.09 (14,087.02)	1,073.66 (13,660.40)	2,577.32 (18,740.64)	1,246.79 (13,350.45)
90+ DPD on revolving debt _{t-1}	153.49 (4,963.97)	115.30 (2,016.56)	114.81 (2,095.42)	137.01 (1,800.13)	102.70 (1,413.69)
90+ DPD on installment debt _{t-1}	396.27 (6,653.93)	189.86 (3,419.63)	151.62 (3,141.07)	584.17 (5,929.77)	205.82 (2,962.45)
Collections debt _{t-1}	609.45 (3,955.72)	208.53 (1,720.72)	167.75 (1,627.30)	406.89 (1,673.76)	395.57 (2,368.18)
Public record bankruptcies _{t-1}	0.0489 (0.2354)	0.0527 (0.2344)	0.0486 (0.2222)	0.0857 (0.3202)	0.0618 (0.2531)
Public record judgments _{t-1}	0.0136 (0.1378)	0.0080 (0.0995)	0.0069 (0.0937)	0.0179 (0.1450)	0.0093 (0.1030)
Individuals	760,680	19,061	15,713	1,445	1,903

This table provides descriptive statistics for the Credit Bureau/HMDA matched panel. Column 1 outlines the sample means and standard deviations for the entire credit bureau dataset; column 2 portrays these statistics for the Credit Bureau/HMDA matched panel; and columns 3–6 display these statistics for the White/Asian, Black, and Hispanic borrowers within the matched dataset, respectively. The sample spans 2013Q2 to 2017Q2.

Match Validation

To verify the accuracy of the credit bureau data, we plot the evolution of our matching variables from 2010-2017. We then juxtaposed these results with data obtained from the Home Mortgage Disclosure Act (HMDA), presenting our findings in Figure 6. This figure comprises five panels: Panel (a) illustrates loan type trends, Panel (b) shows trends in purchaser types, Panel (c) details origination balances, Panel (d) depicts loan purpose trends, and Panel (e) focuses on match rates. In each panel, the left side represents credit bureau data, while the right side displays HMDA data.

Panels (a) through (c) reveal remarkably similar trends between the two data sources. In Panel (d), we observe a comparable trend with an increase in home purchase loans and a decrease in refinance loans. However, the credit bureau data shows a consistent 15-percentage point higher rate for purchase loans. This discrepancy is not a significant concern for two reasons. First, our ratio of home purchase to refinance mortgages aligns with that identified in Butler, Mayer, and Weston (2022). Considering their dataset is three times larger, our loan numbers are roughly one third in both categories. Second, to distinguish between transferred and refinanced first mortgages, we employed a criterion based on changes in the credit amount within the open first mortgage balance over the past two quarters. Importantly, refinancing does not always alter the loan amount; borrowers may refinance for better rates or terms without changing the principal, and we would not be able to capture those.

Panel (e) shows the percent of observations we matched in the Credit Bureau and HMDA data respectively. The match rate in the Credit Bureau data remains relatively steady over time, consistently around 40% until 2016-2017, when it increases to around 50%. Similarly, the HMDA data shows a stable match rate, averaging about 0.1% in most years, with a slight uptick to approximately 0.13-0.14% in the later years. These trends indicate a general stability in the match rate across both datasets over the analyzed period.

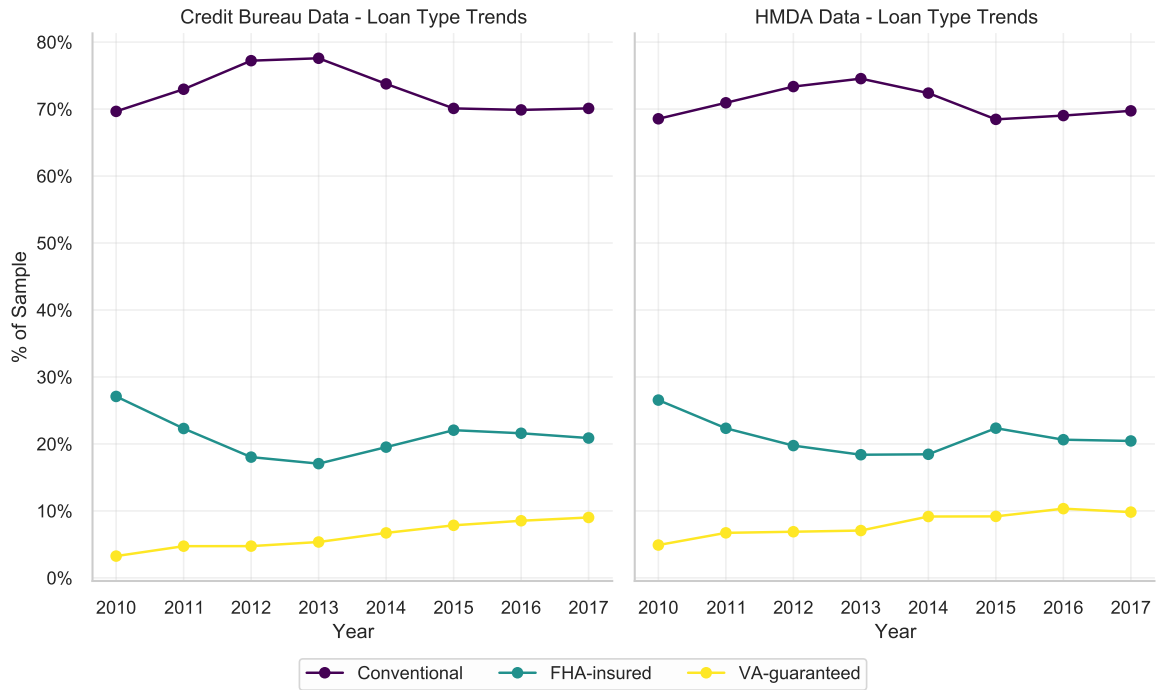
Credit Bureau Data Matching Variables

We matched mortgages in the credit bureau data to HMDA data based on six characteristics: ZCTA5, loan amount, loan purpose (purchase or refinancing), origination year, mortgage type (conventional, FHA, or VA), and if/to which quasi-government entity the loan is sold (purchaser type).

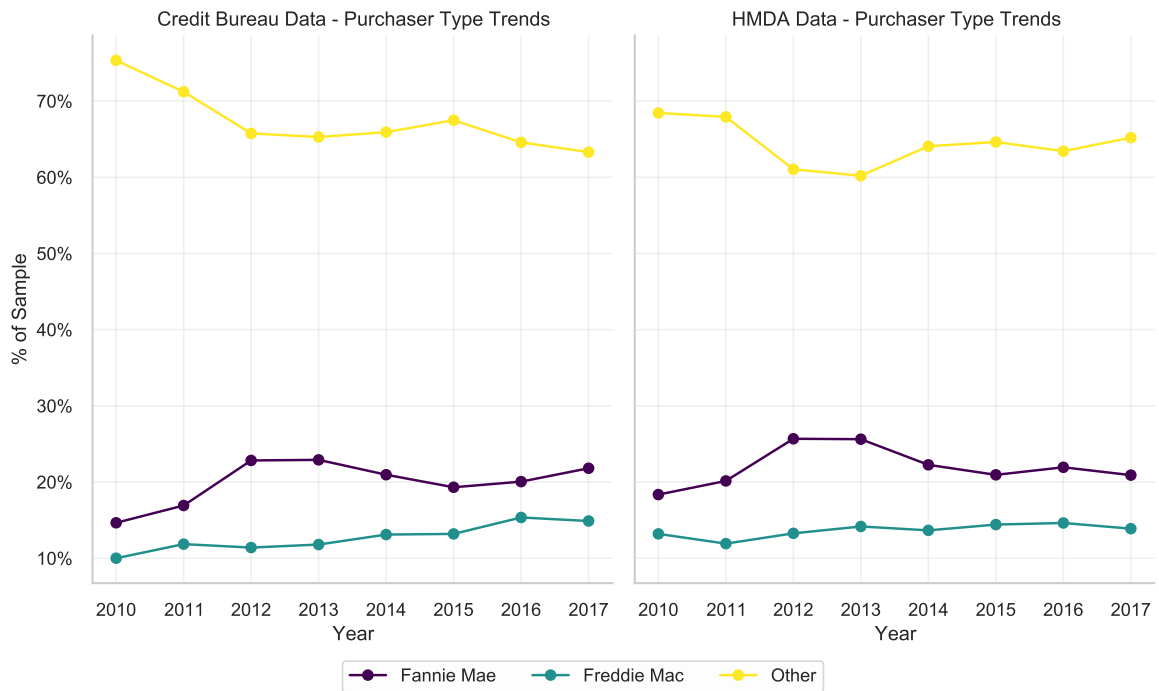
To enhance the accuracy of matching correct balances, especially in refinancing cases, our credit bureau sample is restricted to individuals with a single first mortgage trade in the current quarter and at most one in the past two quarters.²⁸ Utilizing the feature that records the total credit amount on open first-lien mortgages, we can accurately determine the loan amount. Additionally, with ZCTA5 directly observable in our data, two of the six matching variables required for our analysis are addressed.

For loan purpose determination, we analyze the total credit amount on open first mortgage trades, the time since most recent closure, transfer or refinancing of first-lien mortgages, the time elapsed since the most recent first-lien mortgage trade opening, and the number of outstanding first-lien mortgages. A mortgage is considered ongoing and not closed if there's

²⁸This restriction increases the likelihood that the properties in question are owner-occupied when analyzing purchase transactions.

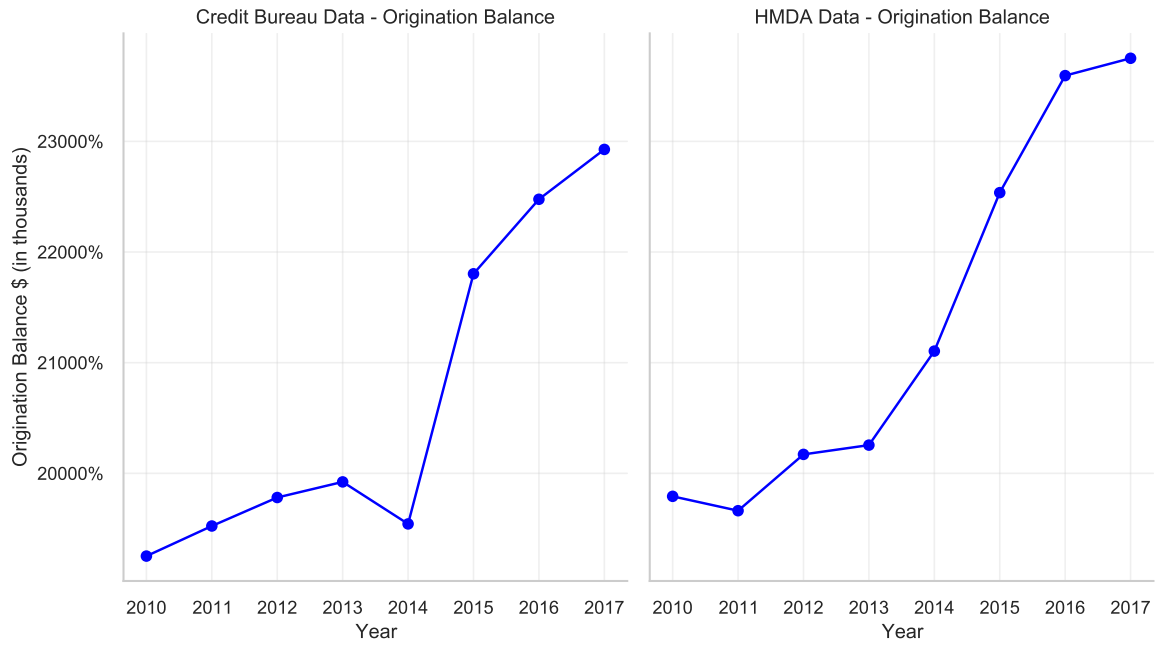


(a)

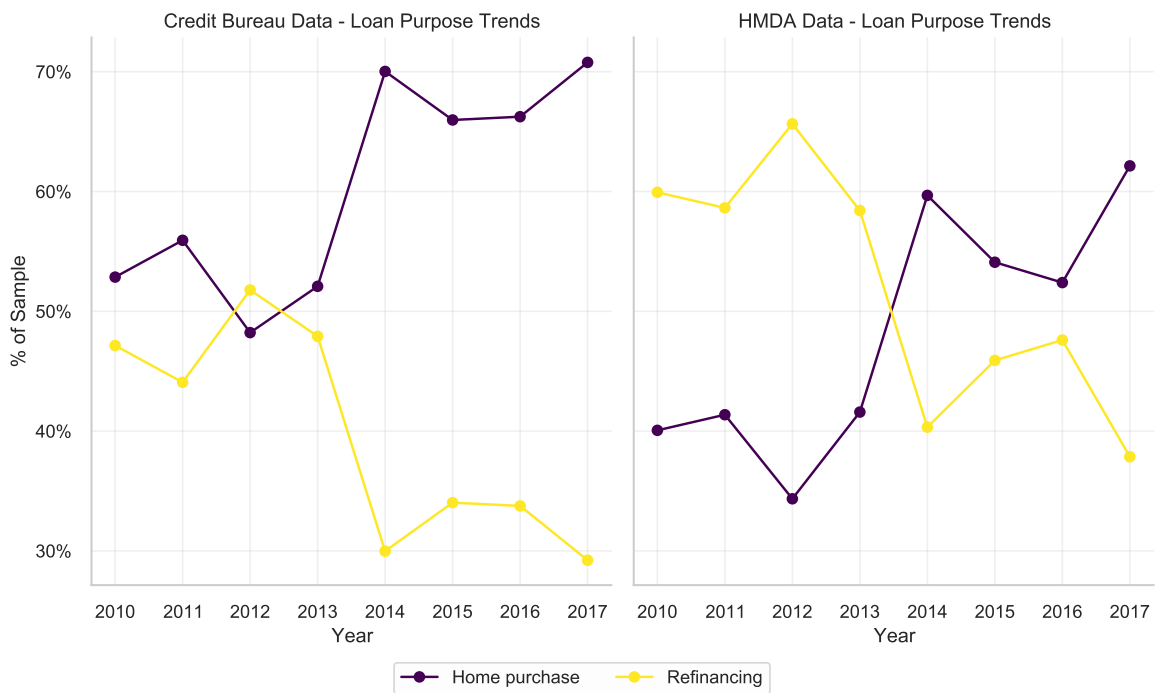


(b)

Figure 6: Credit Bureau & HMDA trends (part 1).

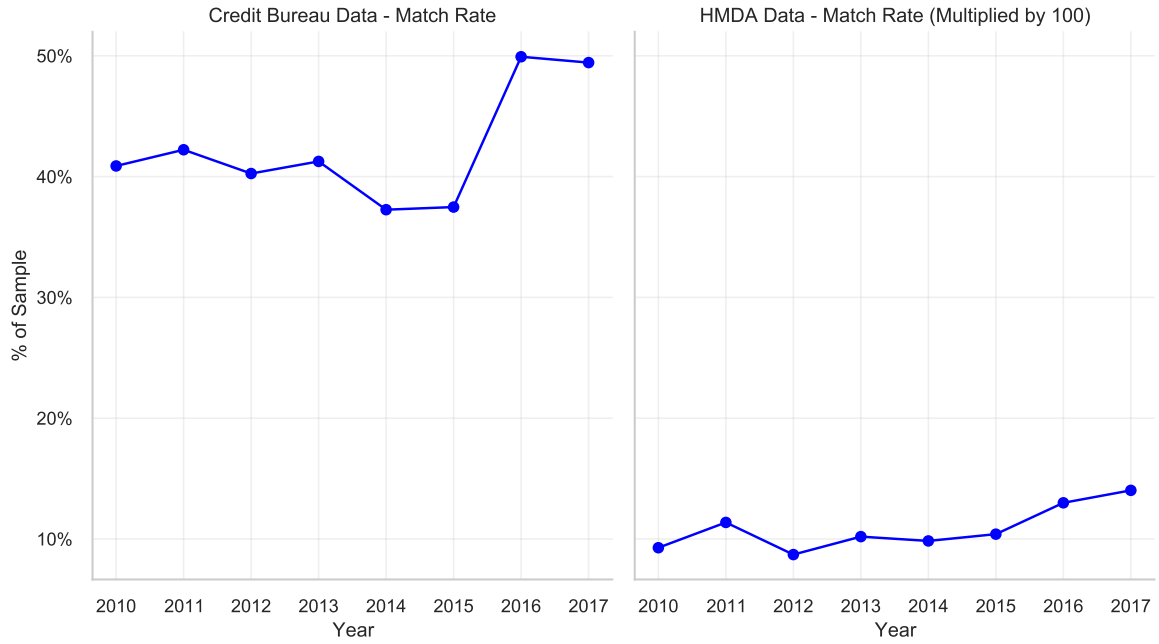


(c)



(d)

Figure 6: Credit Bureau & HMDA trends (part 2).



(e)

Figure 6: Credit Bureau & HMDA trends (part 3).

Notes: Figure 6 compares Credit Bureau and HMDA data from 2010-2017. Panels (a) to (e) display trends in loan types, purchaser types, origination balances, loan purposes, and match rates, respectively. In each panel, the left side represents credit bureau data, while the right side displays HMDA data. The HMDA match rate in Panel (e) is multiplied by 100 for clearer trend visualization. Both data sources exhibit similar trends, particularly in Panels (a) to (c). Panel (d) shows a consistently higher rate of purchase loans in Credit Bureau data, similar to Butler, Mayer, and Weston (2022).

an active credit balance in the current quarter, differing from the previous two quarters. This change in balance indicates that the transaction is a new financial activity, either a new purchase or a refinancing. For purchase identification, we look for loans opened within the past six months with no recorded open first-lien mortgages in the preceding two quarters. This confirms new purchases rather than extensions. For refinances, we identify mortgages with recent renegotiations or refinancing, evidenced by a mortgage closure, transfer, or refinancing in the last six months and an outstanding first-lien mortgage in the preceding two quarters.

The origination year for refinance mortgages is determined by subtracting the time since the mortgage's most recent closure, transfer, or refinancing from the current date. For purchase mortgages, it's calculated from the time elapsed since the opening of the latest first-lien mortgage trade.

Loan types are classified using a specific variable in the credit bureau dataset that categorizes the industry type of the most recently opened first mortgage in the past six months as VA, FHA, or conventional.

Purchaser type is determined by comparing the balances of open first mortgage trades for Fannie Mae and Freddie Mac over the last six months. A mortgage is classified based on which entity's balance is higher, falling into the 'Other' category if neither. Our data does not include cases where one balance is greater than the other while the lesser balance is not zero. This category also includes loans sold to private institutions, as our data does not distinguish these sales.

ReScorer: An Aggregation and Alignment Technique for Building Trust into LLM Reasons

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ABSTRACT

Large language models (LLMs) offer substantial potential for automating labeling tasks, showcasing robust zero-shot performance across diverse classification tasks. The LLM-generated reasons that accompany these classifications contain signals about the quality of the classifications. Estimates of quality of these reasons can, in essence, be used to detect potentially incorrect predictions. Conventional metrics for scoring reasons such as ROUGE-L and BLEU scores depend on ground truth reference reasons, which are challenging and expensive to acquire, and are not available at inference time for new examples. In this paper, we use a product classification dataset to evaluate two reasoning scoring strategies that do not rely on reference reasons: one involving an LLM-based scorer and another using recently proposed ROSCOE metrics. Our analysis reveals that LLM-based approaches are computationally intensive, while aligning ROSCOE metrics with human judgment presents challenges. Consequently, we propose an extension to the ROSCOE framework called ReScorer, which achieves 7% better alignment with human judgment compared to LLM-based evaluation and 59% better than ROSCOE, while being 89% cheaper compared to LLM-based scoring.

1 INTRODUCTION

Large Language Models (LLMs) such as Claude [1], GPT-3 [2] and PaLM [3] are capable of generating fluent and realistic responses to a wide variety of prompts. Recent work has also demonstrated the use of LLMs as annotators [4, 5]. Wei et al. [6] showed that LLMs are capable of providing a reason for their decision along with the end classification.

In classification tasks, the quality of a reason generated by an LLM can be a reliable indicator of the quality of the classification itself [6]. For instance, in e-commerce, an LLM may predict that a product does not violate a safety rule, and provide a corresponding reason. An instance of a safety rule is sale of Sky Lanterns in dry regions, which can be fire hazards. If a scoring algorithm assigns a low score to the reason generated by LLM allowing the sale of the product under the rule, the product can be flagged for human audit

before a safety violation happens. Hence, a metric measuring reason quality can help to identify potentially incorrect LLM classifications.

While traditional metrics like ROUGE-L [7] and BLEU [8] have been proposed for evaluating text generation models in summarization and machine translation, computing these metrics requires ground truth texts. This makes them unsuitable for inference time applications such as the Sky Lantern use case above. Consequently, there is a need for unsupervised metrics to evaluate the reasons generated by LLMs.

Zhang et al. [9] propose an unsupervised metric that uses models like BERT [10] to evaluate the output of text generation models. They show their BERT-based scoring mechanism better reflects human judgement than ROUGE-L or BLEU. Recent work by Manakul et al. [11] finds that BERT-based methods may have difficulty capturing subtle differences in reasons, leading to similar scores for right and wrong reasons. Hence Manakul et al. [11] propose using LLMs as verifier to evaluate language model reasons. The LLMs can be prompted multiple times to obtain consistency or confidence scores [12]. However, computing LLM-based metrics can be computationally expensive as they involve potentially costly LLM calls.

Recent work by Golovneva et al. [13] proposes evaluating language models using smaller embedding based models to compute metrics across different dimensions like semantic alignment, semantic similarity, logical inference, and grammatical correctness. This approach, named ROSCOE, is useful for identifying specific areas in which a reason is deficient. Its main limitation is that it produces 14 metrics rather than one, which makes it difficult to rank or filter reasons. There is also no guarantee that these metrics are all aligned with human preferences.

In this paper, we first assess two state-of-the-art evaluation methods for the language model output: LLM-based evaluation and ROSCOE. Our experiments use different task descriptions from a diverse set of programs within Amazon, with the goal of classifying products based on their adherence to the given task description. While LLM-based evaluation yields metrics that align more closely with human judgment, it is hindered by a significant computational cost, to the tune of \$2000 per million reason scores for our tasks [14]. On the other hand, ROSCOE metrics are inexpensive to compute at \$225 per million reasons evaluated on 1 GPU of p3.2xlarge. However, as we show in this paper, the plurality of ROSCOE metrics and their misalignment with human preferences renders them ill-suited for reason ranking and filtering. Motivated by these limitations, we propose Reasoning Scorer (ReScorer), a modification to ROSCOE

*Equal contribution

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that maps its 14 metrics to a single output. We accomplish the mapping with a lightweight logistic regression model trained on a small set of human judgments.

Our findings reveal that ReScorer outperforms both LLM-based and ROSCOE metrics in detecting low quality reasons (7% better than LLM-based scoring and 59% better than ROSCOE), underscoring the robustness of the ReScorer algorithm, while preserving a lower computational cost—89% reduction in computational cost compared to LLM-based scoring.

The remainder of this paper is structured as follows - Section 2 introduces various approaches to evaluate LLM reasons, including ReScorer. Section 3 outlines our experimental setup. We present key findings of the study in Section 4, followed by conclusions in Section 5.

2 REASON SCORING ALGORITHMS

In this section, we highlight different approaches we used to evaluate LLM reasons for product classification use cases. Section 2.1 introduces the LLM-based approach we used for evaluating reasons from LLMs; Section 2.2 introduces ROSCOE suite of metrics used for evaluating LLM output; and we propose ReScorer in Section 2.3.

2.1 LLM-based scoring

LLMs have recently shown to be effective in assessing information consistency between a document and its summary in zero-shot settings [11, 15]. We adopt a prompting strategy similar to Manakul et al. [11] for evaluating reasons produced by LLMs. In this work we use Claude-v2 [1] as a verifier LLM to evaluate reason output from Claude-v1. Our use case is product classification based on a task description (also called a *policy document*). We used the following prompt to score a previously generated LLM reason:

Given the following policy document:

{task_description}

Given this product to classify:

{product_features}

Given this response from the model classifying the product according to the policy:

{claude_response}

Please provide a score as either 0 or 1 about how good the response from the LLM is. Assign a score of 0 for a bad response and 1 for a good response.

To assess the likelihood that a provided reason is “good”, we employ a multi-prompt strategy. This involves prompting the model multiple times with different hyperparameters for the p in the top- p decoding strategy [16], resulting in various outputs. We then utilize these outputs to compute the probability of the reason being considered “good.” The computation of this probability is conducted using the following equation:

$$P(\text{Good Reason}) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(O_i = 1)$$

where N represents the number of times LLM is prompted, O_i represents i th output from LLM and $\mathbb{I}$ represents the indicator function defined as follow:

$$\mathbb{I}(O_i = 1) = \begin{cases} 1 & \text{if } O_i = 1 \\ 0 & \text{otherwise} \end{cases}$$

This approach is very computationally expensive as we have to prompt the verifier LLM multiple times.

2.2 ROSCOE

Golovneva et al. [13] propose a suite of metrics to evaluate language models output on different dimensions such as *semantic alignment*, *semantic similarity*, *logical inference*, and *grammatical correctness*.

The *semantic alignment* and *semantic similarity* metrics utilize an embedding model to generate embeddings for task descriptions, product details, and LLM reasons. These embeddings facilitate the computation of various scores, including faithfulness of the LLM output (reason) to the input text (task description + product details), and informativeness, which gauges the model’s ability to refer to the input text during output generation. In product classification tasks, it is crucial for the model to accurately refer to the task description. Higher informativeness scores indicate that the LLM output effectively refers to the task description, contributing to improved classification accuracy and good reasoning capability of the model.

Logical inference metrics gauge probability of contradiction between the source text and LLM output. Lower values of probability of contradiction are desirable, indicating minimal contradiction between the human-provided source text and the LLM output. Additionally, grammatical correctness assesses the coherence of the model output, with perplexity from GPT-2 utilized for this evaluation. Refer to Appendix B for comprehensive list of ROSCOE metrics.

Though ROSCOE is a powerful framework for unsupervised reasoning evaluation, there are two issues with using it to detect bad reasons. The first has to do with the fact that ROSCOE involves multiple scores as they evaluate LLM model capabilities across various dimensions. However, deriving decisions from multiple metrics poses a challenge, as it remains unclear how to prioritize different dimensions effectively. Additionally, ROSCOE metrics may not be fit for specific problem domains, being overly general in nature. In some scenarios, good reasons may score lower in certain metrics than bad ones. For example, reasons may exhibit poor grammar metrics because they quote product details or task descriptions, which are themselves grammatically incorrect. These limitations emphasize the need for an additional layer to *aggregate* and *align* metrics tailored to specific applications.

2.3 ReScorer

To address the aforementioned issues, we introduce a lightweight Reasoning Scorer—ReScorer (Figure 1). ReScorer uses a simple logistic regression model [17] to map ROSCOE metrics to a single score between 0 and 1. To ensure that ReScorer is aligned with human preferences, we fit the logistic regression model to a small number of human-assigned evaluations of LLM generated reasons.

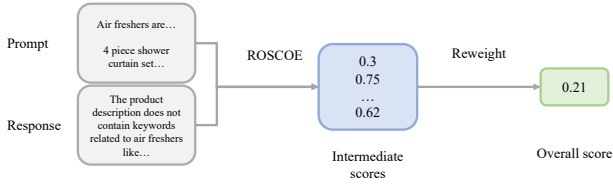


Figure 1: An overview of ReScorer. Given a prompt and corresponding LLM response, ReScorer first computes a set of unsupervised metrics via ROSCOE, then recombines them into a single score with a logistic regression model trained on human annotated data.

The benefits of ReScorer include its ability to align ROSCOE metrics with human judgements, its low computational cost, and the fact that it outputs a single summary score. If we want to use a score to rate LLM outputs, it is important for the score to adequately represent human preferences. Furthermore, when we need to assess a large number of LLM generated reasons, the low computational cost is critical. ReScorer achieves this by adding only a lightweight logistic regression model on top of ROSCOE. Consolidating performance into a single score, as opposed to utilizing a variety of metric, facilitates the decision making in the downstream tasks.

ReScorer’s main drawback is that it requires labeled data to learn the mapping component, even though the metric itself is unsupervised, that is it does not require reference reasons during scoring of reasons. Fortunately, the labeled data needed to train ReScorer is generally easier to obtain compared to ground truth generative output. Annotators need only assign LLM output with scores based on a target task, rather than composing text outputs (e.g. summaries or answers to questions). Another potential limitation is that, ROSCOE metrics that are the foundation for ReScorer may sometimes fail to capture relevant characteristics of LLM answers. However, it is easy to supplement the current ROSCOE metrics with additional metrics tailored for specific applications within the ReScorer framework.

3 EXPERIMENTAL SETUP

Dataset. We curated a corpus with each sample of the form <task description, product features, task label, reason>, with 764 total tasks. Figure 2 shows a sample of data in the corpus. The <task description, product features, task label> part of the corpus are human annotated. The remaining reason part of the corpus is derived by passing <task description, product features, task label> to an off-the-shelf LLM, Claude-v1 [1]. Table 1 records the statistics of our corpus. Human annotators were asked to score these reasons as low, medium and high quality. Refer to Appendix C for annotation guidelines provided to human annotators.

After annotation, there were 2,906 reasons with high scores, 4 medium scores, and 52 low scores, leaving us with a severely imbalanced dataset. Therefore we simplified the scores and added two types of synthetic bad reasons to the dataset. For simplicity, we mapped the original scores to binary scores by assigning 0 to low scores and 1 to both medium and high scores. We also added two types of synthetic negative examples: easy and hard perturbations, P_e and P_h . Both mismatch <task description, product

Task description → Defines the classification task
I am interested in classifying products into the following category: Personal Flotation Device. Below I have described the category: A personal flotation device is a flotation device in the form of a vest or suite that is donned and fastened to a user to prevent the wearer from drowning in a body of water. Products that satisfies the above description often contain keywords: boatcushion, lifejacket, lifepreserver, lifevest. Products that should be excluded from this classification task contain keywords: limetal, surgical, newborns, suitable from birth
Product features → Describes the query product
item_name:o'neill reactor teen life vest; bullet_point:flexible multidensity design closed cell pvc marine foam zipper safety tab quick release dual safety buckles mesh for drainage
Task label → Classification labels provided by human annotator
Yes
Reason → Claude generated classification reason based on <Task description, Product features, Task label>
This product contains keywords like life vest and is intended for safety on water, so it fits the category of Personal Flotation Devices.

Figure 2: Sample data from the dataset.

features> of a sample with <reason> from another sample in the original dataset. And we assign both scores of 0. We obtain easy perturbations by fixing a task and product then randomly selecting a reason from another example in the dataset. Such examples should be easy to detect since the perturbed reason is usually from a different task and hence easy to detect. We construct hard perturbations similarly, except we sample the new reason from another example with the *same task*. These negatives are more subtle because the reason references the correct task, but the wrong product. Further, we create a 80/20 train/test split, stratifying on the ground truth score. Only ReScorer uses the train set, and we evaluate all approaches on the test set.

Perturbations	# Samples	% Positives	Avg. Samples per Task
H	2962	98.2%	3.9
P_e	2962	0%	3.9
P_h	2962	0%	3.9
Overall	8886	32.7%	11.6

Table 1: Summary of dataset statistics. H is the set of human annotations, P_e is the set of easy perturbations, and P_h is the set of hard perturbations. “Overall” represents the full dataset. All perturbations include all 764 tasks.

ROSCOE. As baselines, we test two straightforward techniques for mapping ROSCOE metrics to a single score. **ROSCOE avg.** computes the mean of the 14 ROSCOE metrics. And **ROSCOE one-hot** uses the metric which has the best correlation with the target score. We specifically use Somers’ D correlation [18], elaborated at the end of this section. In this case, the selected metric was repetition_step (refer Appendix B for details on ROSCOE metrics). Both methods generate unsupervised metrics. To calculate semantic alignment and semantic similarity metrics, we employ the off-the-shelf embedding model from Hugging Face, specifically using all-MiniLM-L6-v2 [19]. For logical inference metrics, we adopt the model recommended by Golovneva et al. 13: DeBERTa-v3-large-mnli-fever-anli-ling-wanli. This model, based on the DeBERTa-large architecture [20], is fine-tuned on the MNLI, ANLI, and WNLI datasets [21]. To assess grammatical correctness, we use perplexity output from GPT-2 [22].

ReScorer training. We trained ReScorer’s supervised module (logistic regression) on the train set with hyperparameter optimization and 3-fold cross-validation using scikit-learn [17] (see Appendix A for details). We obtained the best overall performance by training on human annotations plus hard perturbations (see Section 4.2). Unless otherwise noted, the results in the following section all use this model.

Metric. Similar to Golovneva et al. [13], we use Somers’ D¹ [18] to measure the correlations between the scorers and human judgements. The scorers output scores between 0 and 1, and human judgements are either 0 (for bad reasons) or 1 (for good reasons). Somers’ D is a way to measure the correspondence between two rankings. Somers’ D closer to 1 represent strong agreement, while closer to -1 indicates strong disagreement. To better understand how well the methods detect different types of negative examples, we evaluate each on different subsets of the test data: human annotations H , human annotations + easy perturbations (H, P_e), human annotations + hard perturbations (H, P_h), and the full test set—human annotations + easy perturbations + hard perturbations (H, P_e, P_h).

4 RESULTS

Our results are organized into four sections. Section 4.1 gives the main results: a comparison of the methods of Section 2 on our reasoning dataset. Section 4.2 explores the effects on ReScorer of adding perturbations to the training data. We consider specific examples of where ReScorer correctly identifies bad reasons in Section 4.3. And Section 4.4 further discusses why ROSCOE metrics alone are insufficient for scoring reasons for specific applications.

4.1 Main results

Method	Somers’ D			
	H	(H, P_e)	(H, P_h)	(H, P_e, P_h)
Random	-0.3081	0.0159	0.0356	0.0278
LLM-based [1]	0.2294	<u>0.8485</u>	<u>0.2629</u>	<u>0.5470</u>
ROSCOE avg. [13]	<u>0.2138</u>	-0.0165	0.0651	0.0213
ROSCOE one-hot [13]	0.1242	-0.0032	0.0686	0.0310
ReScorer	-0.1111	0.9616	0.2934	0.6195

Table 2: Somers’ D between ground truth scores and scores assigned by reason evaluation methods. We evaluated on subsets of the test set with various types of perturbations: human annotations H , easy perturbations P_e , and hard perturbations P_h . The best results on each subset are bold and the second best are underlined. We compare our method, ReScorer, against four baselines: Random—randomly assigned scores (selected uniformly from $\{0, 1\}$), LLM-based—scores assigned by Claude, ROSCOE avg.—the mean of ROSCOE metrics, and ROSCOE one-hot—the best-performing single ROSCOE metric.

Table 2 shows the Somers’ D values for all reason evaluation methods on different types of perturbations. ReScorer demonstrates strong performance in distinguishing good and bad reasons when we include any or all perturbation types, improving significantly over LLM-based scoring by 7% (Somers’ D increase from 0.54 to 0.61) and ROSCOE by 59% (Somers’ D increase from 0.03 to 0.62). However, it struggles on the severely imbalanced human annotated data, many of which it confuses for negatives. LLM-based scoring provides a reasonable level of performance across different strata. Whereas the ROSCOE approaches do poorly at detecting perturbations, suggesting that ReScorer is needed to re-weight the unsupervised metrics. Interestingly, we find that a simple average of ROSCOE metrics performs reasonably well on human annotations H , but this may be because this set is over 98% good reasons.

The Somers’ D values reported here are generally lower than the values in [13], which suggests that our task may be more challenging. Indeed, the bad reasons from the human annotations and the hard perturbations are often wrong in subtle ways. For example, it may be hard to classify as good or bad a reason that is trying to determine whether a vegetable-derived oil is meant for cooking or personal use. Both uses have similar keywords such as “organic”, “vegan”, and “cold-pressed”. To determine whether the reason is good, one must essentially solve the underlying classification problem. It is not reasonable to expect methods based on unsupervised metrics, like ROSCOE and ReScorer, to consistently solve such tasks.

4.2 Training data and ReScorer

We explore the effects of supplementing the human-annotated training data by varying the types of perturbations added to the training data and evaluating ReScorer’s performance on different subsets of the test set. Table 3 shows the results. Unsurprisingly, we find that adding perturbed examples improves ReScorer’s ability to detect perturbations, i.e. performance on (H, P_e, P_h) —Somers’ D 0.6195 in Table 3. Adding only hard negatives/perturbations improves performance the most, but the larger message is this: the biggest gains come from adding *any* extra negative examples. As with any supervised model, ReScorer performs best when its train and test distributions are aligned. Therefore, on human annotated test data H , ReScorer does best when no perturbations are added to the training data, in which case it is comparable to the LLM baseline (0.2294 from Table 2). To optimize performance on such data, it may be best to omit added perturbations. However, given that our annotated dataset is highly imbalanced, it is still important to add perturbations to ensure robustness against some reasoning errors. Since the perturbations are essentially free, the main cost incurred is a minor increase in training time.

4.3 Anecdotes

It is informative to consider specific examples where ReScorer detects bad reasons (Table 4). In each example, the reason contains relevant information along with one or more hallucinations, and in each case ReScorer correctly assigns the reason a low score, i.e. below 0.5. It is promising that ReScorer is able to detect such nuances in LLM-based reasons.

¹We use the SciPy implementation [23].

Train perturbations	Test perturbations			
	H	(H, P _e)	(H, P _h)	(H, P _e , P _h)
<i>H</i>	0.2188	0.4929	-0.0919	0.1902
(<i>H</i> , <i>P_e</i>)	0.1243	0.9820	0.0643	0.5090
(<i>H</i> , <i>P_h</i>)	-0.1111	0.9616	0.2934	0.6195
(<i>H</i> , <i>P_e</i> , <i>P_h</i>)	-0.0183	0.9834	0.2593	0.6119

Table 3: Somers’ D for ReScorer models with different types of perturbations included in the training data (Train perturbations). We report results on different segments of the test set (Test perturbations).

Class	Product features	LLM reason
Cooking thermometers	gomesme - dual channels wireless waterproof bbq food thermometer thermometer digital cooking meat food oven grilling with timer...	This product mentions wireless and waterproof features but does not explicitly mention cooking or food thermometers.
Memory cards	new ssk usb 2.0 card reader for ms, xd, cf, mmc, sd, microsd, m2 card. guarantee the quality of products delivery time 20-30 days if you have any questions, please feel free to contact us	This product contains keywords like " 64gb ", "microsd", " memory card ", and mentions it is for a Samsung tablet . Based on the SOP, this satisfies the description of a Memory Card.
Apple laptops	letter h laptop carrying bag sleeve, neoprene sleeve case/ancient, jade wooden h symbol from alphabet uppercase scratched worn design print decorative/for apple macbook air samsung google acer hp...	This product mentions Apple Macbook Air and includes specifications like processor speed and graphics that match the SOP criteria for Apple Laptops.

Table 4: Examples of bad reasons that ReScorer detects. For brevity we show the class names in place of the full class descriptions and truncate long product features at the item name. Hallucinations are shown in bold. Note that “SOP” is synonymous with “class description.”

4.4 Why can’t we just use ROSCOE?

There are two reasons behind our decision to put a supervised fine-tuning layer on top of ROSCOE metrics instead of using them directly. The first is simply the issue of deriving a decision from multiple metrics. For example, how should we decide which of two reasons is better when reason *A* scores (0.9, 0.1) and reason *B* scores (0.7, 0.7)? Two obvious approaches are to pick one dimension/metric or to average across multiple dimensions. But we showed that these two strategies perform worse than ReScorer (Table 2).

The second issue is that the ROSCOE metrics are sometimes too general-purpose for specific problem domains. In some applications good reasons can have score lower in some metrics than bad reasons. For example, `repetition_word` and `repetition_step` measure the repetitiveness of a text. Usually it is bad for text to be too repetitive, but in our classification use-case, it is sometimes desirable. The following reason was assigned low repetition metrics despite being satisfactory:

This product does not mention a glue gun, only glue bars. The SOP defines Glue Guns as devices that dispense hot melt glue, while this item only includes glue bars.

Similarly, we found that some good reasons had low `grammar_step` and `grammar_step_max` metrics because they quoted product details or class descriptions, which were themselves grammatically incorrect. These examples show the need for an extra step on top of ROSCOE to align and aggregate the metrics for the application at hand.

5 CONCLUSION

We explored the *evaluation* of LLM generated explanations alongside classification decisions. Using a e-commerce product classification dataset, we assessed two reasoning evaluation strategies: one using an LLM-based scorer and the other using ROSCOE metrics. Our analysis highlighted the computational intensity of LLM-based approaches and the difficulties in aligning ROSCOE metrics with human judgment. Consequently, we proposed ReScorer, an extension to the ROSCOE framework, achieving a 7% improvement over LLM-based scoring and a 59% improvement over ROSCOE, while reducing computational costs by 89% compared to LLM-based scoring. These findings emphasize the importance of efficient evaluation techniques and highlight ReScorer’s potential to enhance the reliability of language model predictions in critical applications.

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A RESCORER HYPERPARAMETER OPTIMIZATION

When training ReScorer, we used the scikit-learn logistic regression implementation and randomized 3-fold cross-validation search over the following hyperparameters with random state 42 and 128 iterations:

```
"penalty": [None, "l1", "l2"],
"C": [1e-3, 1e-1, 5e-1, 1, 5, 10, 100],
"solver": ["liblinear"],
"tol": [1e-5, 5e-5, 1e-4, 5e-4, 1e-3],
"class_weight": [None, "balanced"],
"max_iter": [100, 200, 500]
```

B LIST OF ROSCOE METRICS

ROSCOE [13] semantic alignment metrics is the reasoning alignment vector from the N -step hypothesis h (LLM response) to the source s (task description + product details) of length T :

$$r-align(h \rightarrow s) = \{\alpha_1, \alpha_2, \dots, \alpha_N\},$$

where each alignment value

$$\alpha_i = r-align(h_i \rightarrow s) = \frac{1}{2} \left[1 + \max_{j=1}^T (\cos(h_i, s_j)) \right] \in [0, 1]$$

is the normalized cosine similarity between the hypothesis step and the most similar sentence in a context. It explicitly measures the grounding of the step-wise reasoning with respect to the source text. The ROSCOE metrics are summarized in Table 5

C ANNOTATION GUIDELINES

Following are the annotation guidelines provided to human reviewers for evaluating reasoning output from LLM:

- Provide score 1, 2 or 3 depending on quality of reason in <task description, product features, task label, reason>
- A low score of 1 should be given when there are outright hallucinations, such as task asks for bicycles or scooters, product is a scooter, but reason provided is that product is a bicycle.
- A medium score of 2 should be given when the reason is correct but contains general knowledge not appears in product attributes - this can happen since LLMs are pre-trained on billions of internet-scale data and hence could bring general knowledge during reasoning.
- A high score of 3 is given when none of the above are true.

Metric	Description
Faithfulness-Step ($h \rightarrow s$)	This step-level score is based on the alignment from the hypothesis steps to the source sentences, and is calculated as the mean reasoning alignment score over the steps of reasoning (see illustration in Appendix D, Figure 3): $\frac{1}{N} \sum_{i=1}^N r-align(h_i \rightarrow s)$. Faithfulness measures if the model misinterpreted the problem statement, or the reasoning chain is too vague, irrelevant, or misuses information.
Faithfulness-Token ($h \rightarrow s$)	We extend step-level embeddings of the Faithfulness-Step by measuring similarities between the token embeddings: $\frac{1}{N+M} \sum_{i=1}^N [r-align(h_i \rightarrow s) + \sum_{j=1}^{M_i} r-align_{token}(h_{i,j} \rightarrow s)]$, as shown in App. D, Fig. 3. M_i is the number of tokens in step h_i , $M = \sum_{i=1}^N M_i$ is the total number of tokens in the reasoning chain, $h_{i,j}$ is the j th token in the i th step, and $r-align_{token}$ is the alignment vector from tokens in step h_i to all tokens in s .
Informativeness-Step (Info-Step) ($h \leftrightarrow s$)	Measures how well information present in the source is used in the reasoning steps: $[\frac{1}{T} \sum_{t=1}^T r-align(s_t \rightarrow h) + \frac{1}{N} \sum_{i=1}^N r-align(h_i \rightarrow s)] / 2$. Info-step gives a higher score to reasoning steps that are well-grounded with respect to the source, and identifies the degree of information from source that is covered by the generated hypothesis. A lower Info-Step score corresponds to the reasoning steps that are not related to the source sentences or have missed information provided in the context.
Repetition-Token ($h_i \rightarrow h_j$)	To identify repeated, or paraphrased steps, we look at the token alignment scores between all steps in the hypothesis chain: $1 - \max_{i=2..N} \max_{j=1..i-1} [\frac{1}{M_i} \sum_{l=1}^{M_i} r-align_{token}(h_{i,l} \rightarrow h_j)]$. For each pair of sentences, we look at the mean token alignment, and find those sentences that maximize this alignment score. In other words, Repetition-Token will punish chains where there are at least two steps with high overlap in token embeddings.
Informativeness-Chain (Info-Chain) ($h \rightarrow s$)	Similar to Info-Step, this metric quantifies the degree of agreement between the hypothesis chain and the source and is calculated as $[1 + \cos(h, s)] / 2$. We embed reasoning chain and source context as a whole, as opposed to using step-wise embeddings.
Repetition-Step ($h_i \leftrightarrow h_j$)	Measures repetition-related errors on the step level by checking if it paraphrases information already mentioned in the previous steps: $(1 - \max_{i=2..N} \max_{j=1..i-1} [\cos(h_i, h_j)]) / 2$. Unlike Repetition-Token, which is orderless and compares individual tokens in pairs of steps, Repetition-Step considers step embeddings similarity and is more robust to changing contexts.
Semantic Coverage-Chain ($r \leftrightarrow h$)	Reflects the overall degree of similarity between the reference and hypothesis chains, comparing reference and hypothesis embeddings as a whole: $[1 + \cos(r, h)] / 2$.
Self-Consistency ($h_i \leftrightarrow h_j$)	Measures logical entailment errors within the reasoning steps: $1 - \max_{i=2..N} \max_{j < i} p_{contr}(h_i, h_j)$. This metric will punish chains where there is a pair of steps that are likely to contradict each other.
Source-Consistency ($h \leftrightarrow s$)	Measures logical entailment errors between any generated reasoning h and the source context s : $1 - \max_{j=1..N} \max_{j=1..T} p_{contr}(h_i, s_j)$. Specifically, for each reasoning step we measure the probability that it contradicts any sentence in the context. We take the maximum probability of contradiction over all steps, following the logic that a contradiction anywhere in the reasoning chain signals a failure of the overall argument.
Perplexity-Chain (h)	Average perplexity of all tokens in the generated reasoning steps: $1/PPL(h)$. The context used to score each token is the previous tokens in the current and all previous steps. Steps are joined with a space character. To keep the range and orientation consistent with the other scores, we invert the perplexity.
Perplexity-Step (h_i)	Average perplexity of all tokens in the generated reasoning steps, where the context used to score each token is only the previous tokens within the current step: $1/[(1/N) \sum_{i=0} PPL(h_i)]$. To keep the range and orientation consistent with the other scores, we invert the perplexity.
Grammar (h_i)	Probability of grammatical acceptability of each step, averaged over all steps: $(1/N) \sum_{i=0} p_{gram}(h_i)$.

Table 5: Description of ROSCOE metrics